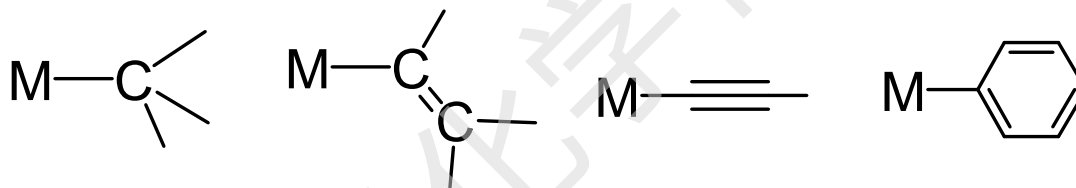


Chapter 2. Complexes with metal-carbon σ bonds



Related compounds:

$M-H$, $M-OR$, $M-NR_2$, $M-F$, $M-Cl$, etc.



Outline

2.1 M-C bond strength

2.2 Preparation of complexes with metal-carbon σ bonds

2.3 Chemical properties of complexes with metal-carbon σ bonds

2.4 Complexes with related σ -bonded ligands

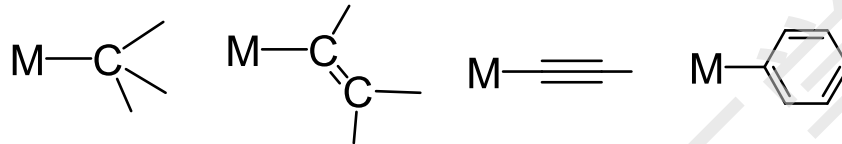


2.5 Metal hydrides



2.1 M-C bond strength

The stability of transition metal-carbon σ bonds



Related compounds:

M-H, M-OR, M-NR₂, M-F, M-Cl, etc.

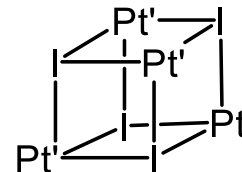
Formation and breaking of metal-carbon σ -bonds is the central event in organometallic reactions and in many organic reactions catalyzed by transition metal complexes.

Pope, Peachey, in 1909:



$[\text{Me}_3\text{PtI}]_4$:

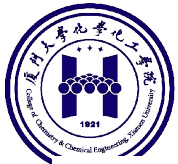
$\text{Pt}' = \text{PtMe}_3$



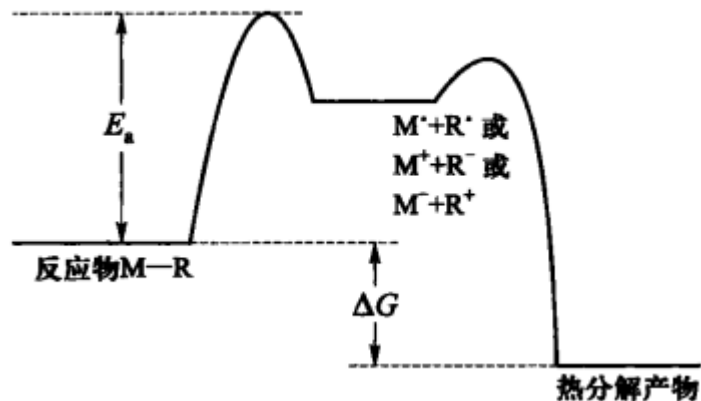
18e ?

Some questions:

- Any difference in the bond strengths of transition metal-carbon σ bonds and main group metal-carbon σ bonds?
- How strong are transition metal carbon σ bonds?
- What affects transition metal carbon M-C σ bond strength?
- Relative bond strengths of M-C and M-H, M-X bonds?



Bond dissociation energy (BDE)

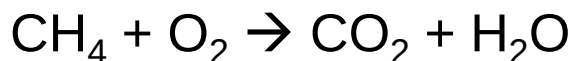


Stability

Thermodynamic **Kinetic**

$$G(T,p) = H - TS$$

$\Delta G < 0$, thermally unstable
Kinetically stable



The bond strength can be related to **bond dissociation energy**.



If BDE (M-X) is larger, M-X is stronger.

Main group metals:

BDE(M-C): 89-280 kJ/mol [Except B-C]

Transition metals:

BDE (M-alkyl) : 150 - 260 BDE (M-aryl): 250 - 350

Normally, BDE (M-alkyl) < BDE (M-aryl)



Range of M-C bond dissociation energy (BDE)

BDE (M-C) values in MR_n complexes (KJ/mol)

<i>Non-transition metals</i>		<i>Transition metals</i>		
M-C bond	BDE	Compound	M-C bond	BDE
Li-Et	209	$Ti(CH_2C(CH_3)_4)_4$	Ti-CH ₂ R	170
Li-Bu	248	$Ti(CH_2Ph)_4$	Ti-CH ₂ R	240
Zn-Me	176	$Ti(CH_2Si(CH_3)_4)_4$	Ti-CH ₂ R	250
Zn-Et	145	$Cp_2Ti(CH_3)_2$	Ti-CH ₃	250
Cd-Me	139	Cp_2TiPh_2	Ti-Ph	350
Hg-Me	122	$(CO)_5MnCH_3$	Mn-CH ₃	150
Hg-Et	101	$(CO)_5ReCH_3$	Re-CH ₃	220
Hg-i-Pr	89	$Zr(CH_2C(CH_3)_4)_4$	Zr-CH ₂ R	220
Hg-Ph	136	$Zr(CH_2Ph)_4$	Zr-CH ₂ R	380
B-Me	363	$Zr(CH_2Si(CH_3)_4)_4$	Zr-CH ₂ R	225
B-Et	342	$CpPt(CH_3)_3$	Pt-CH ₃	165
Al-Me	276	$(Et_3P)_2PtPh_2$	Pt-Ph	250
Al-Et	242	$Ta(CH_3)_5$	Ta-CH ₃	260
Ga-Et	237	$W(CH_3)_6$	W-CH ₃	160

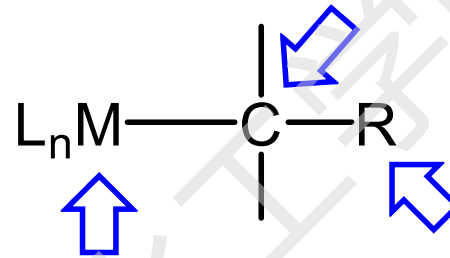
Conclusion: Transition metal-carbon bond energies are not so different from those of main group metals.



Factors affecting M-C bond strength

Affecting Factors :

- *M effect*;
- *R effect*;
- *C-(sp^3 , sp^2 , sp) effect*.



a) Metal effect

BDE (M-CH₃), kJ/mol

For main group metals,

B-Me	363	Zn-Me	176
Al-Me	276	Cd-Me	136
Ga-Me	247	Hg-Me	122

BDE(M-C) decreases when going down a group.

For transition metals,

(CO) ₅ Mn-CH ₃	150	(PhCH ₂) ₃ Ti-CH ₂ Ph	240
(CO) ₅ Re-CH ₃	220	(PhCH ₂) ₃ Zr-CH ₂ Ph	380

BDE(M-C) increases when going down a group.

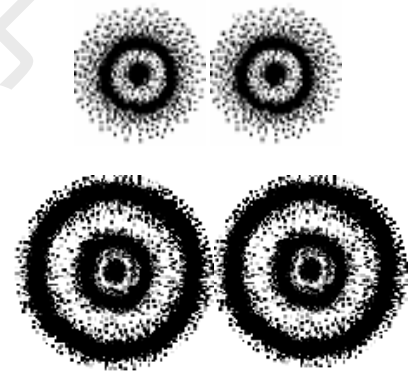


Why?

Possible explanation:

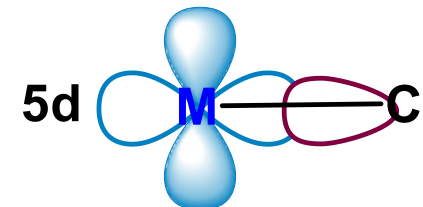
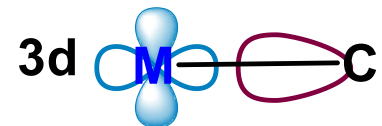
- Main group metals use s , p orbitals to form M-C bonds.

Down a group, the orbitals become more diffuse $\rightarrow$ less effective overlap $\rightarrow$ weaker bond.



- Transition metals may use d orbitals to form M-C bonds.

Down a group, d orbitals become larger $\rightarrow$ more effective overlap $\rightarrow$ stronger bond.





b) R effect (M-R)

Bond dissociation energies $\Delta[R-Mn(CO)_5]$

Complex	BDE[R-MnCO] ₅ (kJ/mol)
[Mn(CO) ₅ CF ₃]	172
[Mn(CO) ₅ C ₆ H ₅]	170
[Mn(CO) ₅ CH ₃]	153
[Mn(CO) ₅ COCH ₃]	129
[Mn(CO) ₅ COC ₆ H ₅]	89
[Mn(CO) ₅ COCF ₃]	147
[Mn(CO) ₅ H]	213
[Mn(CO) ₅ I]	195
[Mn(CO) ₅ Br]	252
[Mn(CO) ₅ Cl]	294

Relative bond strength: (M-Cl > M-Br > M-H = or > M-I)

M-H > M-CF₃ > M-Ar > M-CH₃ > M-COCH₃

M-C-R: stability increases with more electron-withdrawing R.



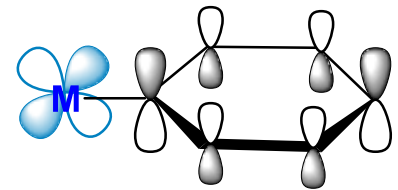
Questions

Why M-H is stronger than M-CR₃?

A: M-H bond is shorter than M-C, 1s orbital of H is less diffuse than 2s and 2p orbitals of C. Steric effect may also cause M-C bond to be weaker.

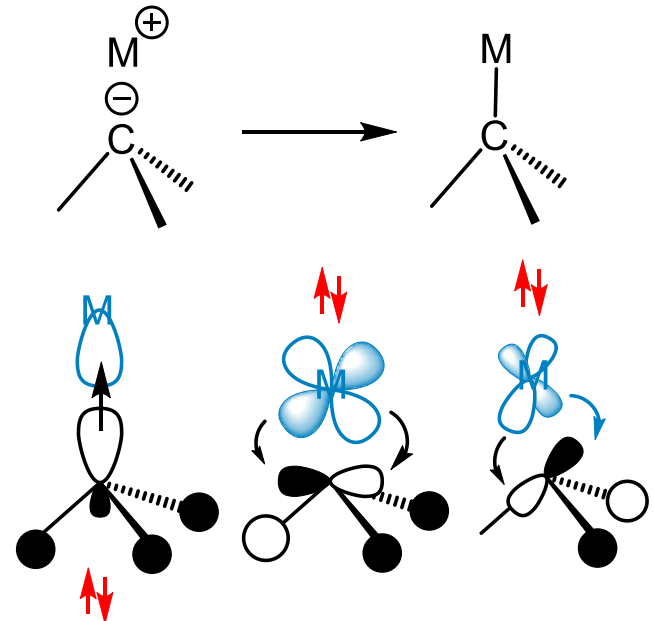
Why M-Ar is stronger than M-CR₃?

A: Ar can have d-p π interaction with M. The interaction makes M-Ar bond to have some multiple character.



Why M-CH₃ is weaker than M-CF₃?

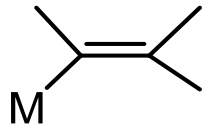
A: Electron-withdrawing group encourage backdonation, making M-C bond to have some multiple character.



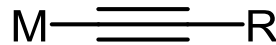


C) C-(sp³, sp², sp) effect

Would you expect M-CH₃ bond be stronger or weaker than



alkenyl
(vinyl complexes)



alkynyl

(acetylide complexes)

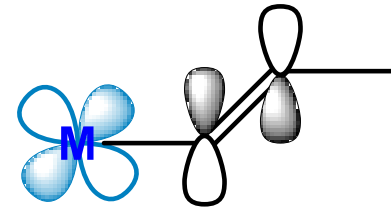
Why?

(answer: $M-CH_3 < M-CH=CH_2 < M-C\equiv CH$)

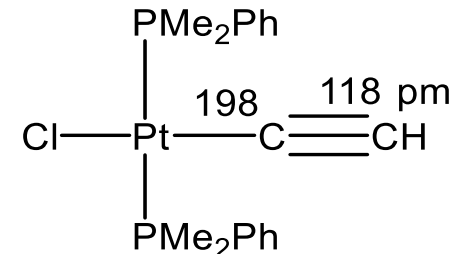
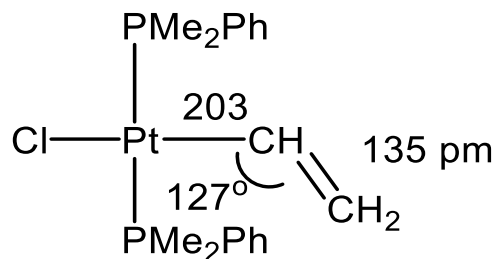
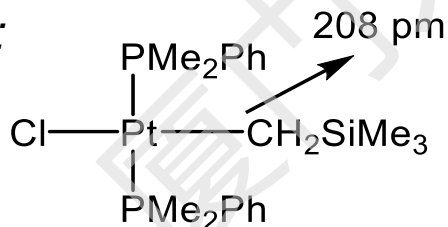
M-CH₃: no dπ-pπ interaction;

M-CH=CH₂: one dπ-pπ interactions;

M-C≡CH: two dπ-pπ interactions



Evidence:



Increasing s character of the M-C bond, increasing M-C bond strength.



Summary

- The M-C bond in transition metal alkyl complexes is not as weak as was thought previously.
- There is no inherent instability of transition metal alkyls.
- The BDE(M-R) value increases as the atomic number increases among the congeners in the same group, in an opposite trend to non-transition metal alkyls.

The earlier failure in obtaining TM alkyls is not due to thermodynamic reasons!

Differences between metals

- Electronegativity, size: For a given group, 3d metals are quite different from 4d and 5d metals, 4d and 5d metals are similar.
- The energy of valence d electrons in an metal are affected by charge, and locations in the periodic table.

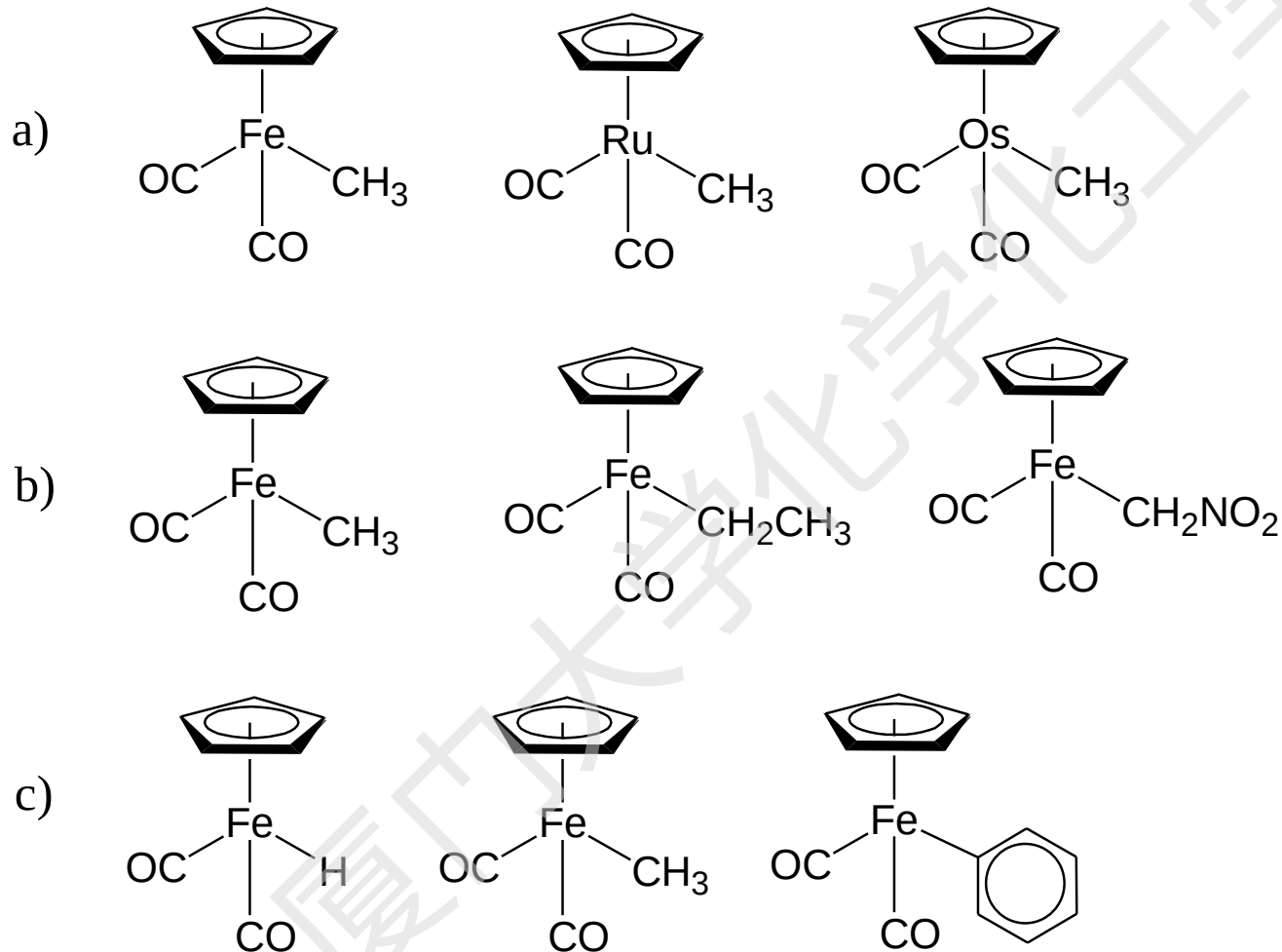
R. H. Crabtree:

“This example of the historical evolution of our ideas implies that just as some of the early assumptions in this area proved to be wrong, some of our ideas today will probably turn out to be wrong too – the problem is we do not know which ones!”



Practice Problems

Which one has the strongest stability of the following complexes ?





Origin of earlier failure in obtaining TM alkyls

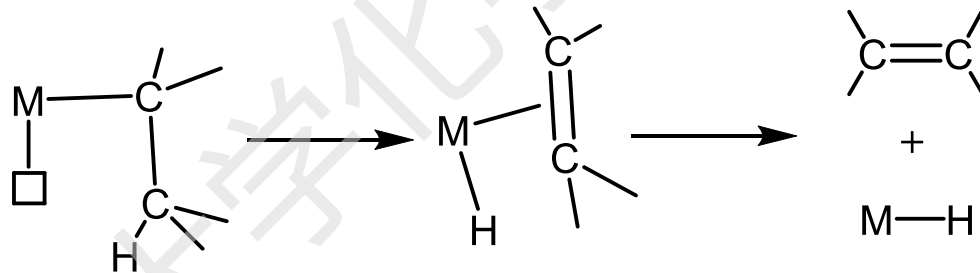
(Kinetic consideration)

Q1. Why could many TM alkyl complexes not be isolated?

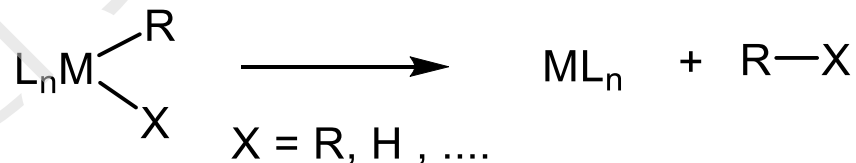
There are many pathways that TM alkyl complexes can be decomposed.

Q2. What are the common decomposing pathways for transition metal alkyls?

✓ β -elimination (The most common route!)



✓ Reductive elimination



✓ Others, e.g. α/γ -H eliminations, intramolecular reactions

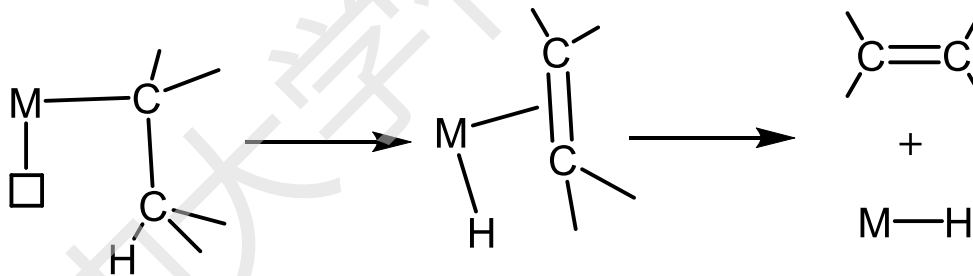


How can we obtain stable M-alkyl complexes?

Stable M-alkyl complexes could be obtained if the above decomposing pathways can be prevented.

α) β -elimination.

The most common decomposition pathway for alkyls is β -H elimination, which converts a metal-alkyl into a **hydrido-olefin complex**.



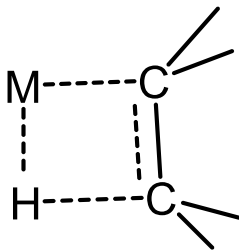
hydrido-olefin complex



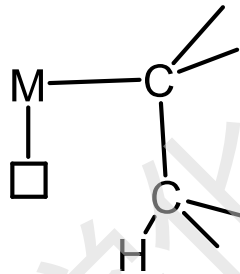
What is the condition for β -H elimination?

A. There must be a β -H.

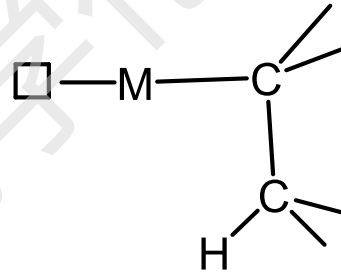
B. The complex should be able to form four-membered co-planar transition state (see (A)) Normally 16e species



(A)



(B)



(C)

C. There is a vacant site *cis* to the alkyl (e.g. (B). but not (C)).

(need an empty orbital to accept C-H electron.)

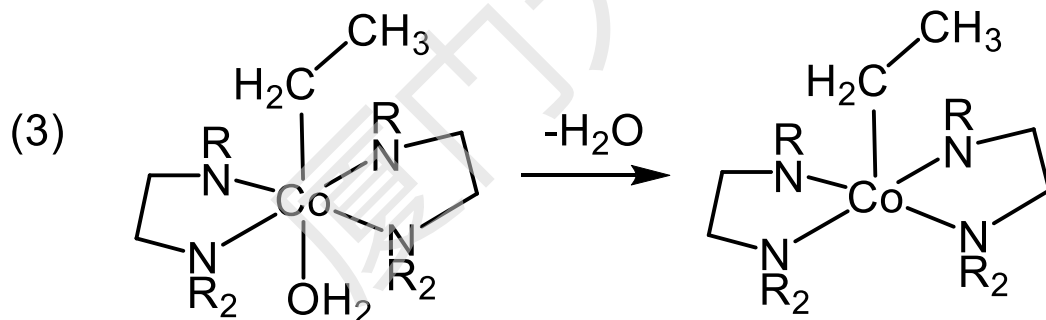
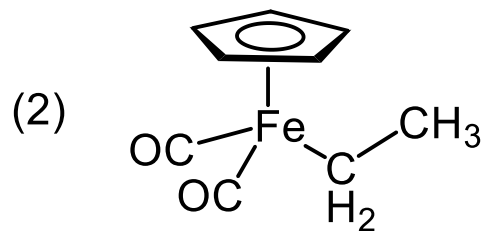
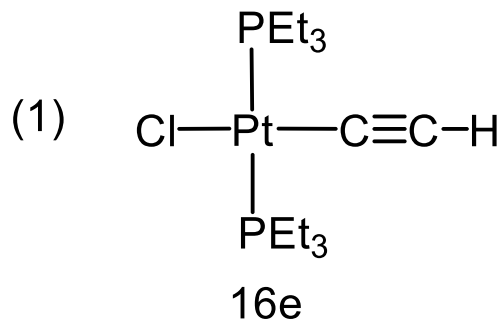
D. The olefin formed is stable.

E. The metal has e^- to go to $\sigma^*(\text{C-H})$.



Practice Problems

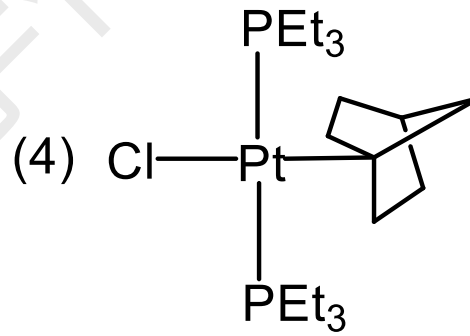
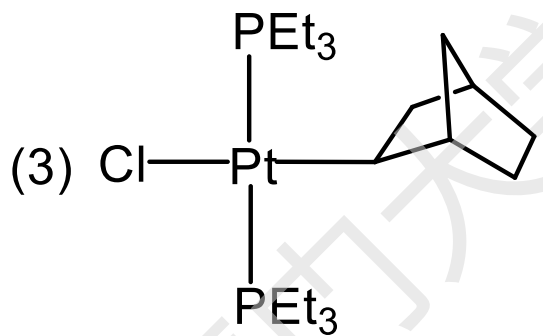
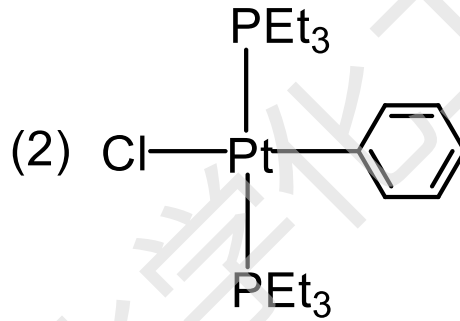
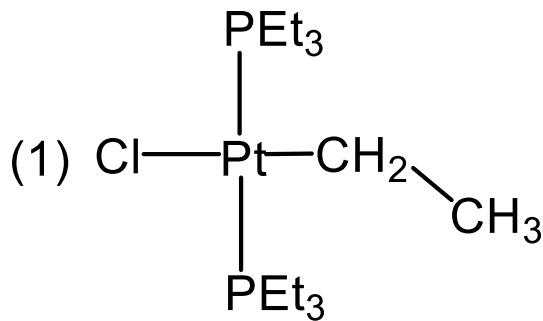
It is difficult for the following compounds to undergo β -hydrogen eliminations. Suggest main reasons.





Practice Problems

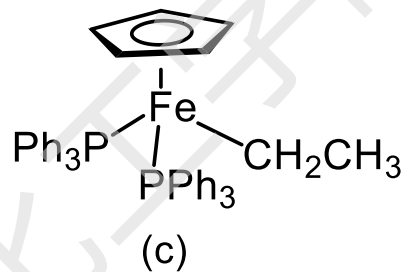
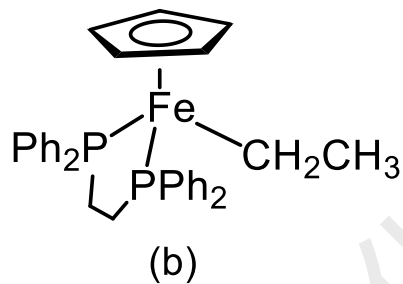
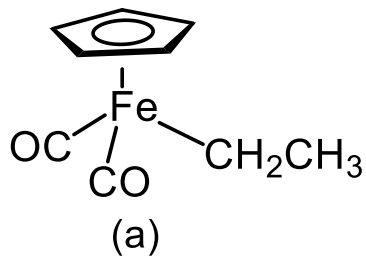
Would you expect the following complexes to undergo β -H elimination easily?





Practice Problems

Which of the following complexes would you expect to be least stable?

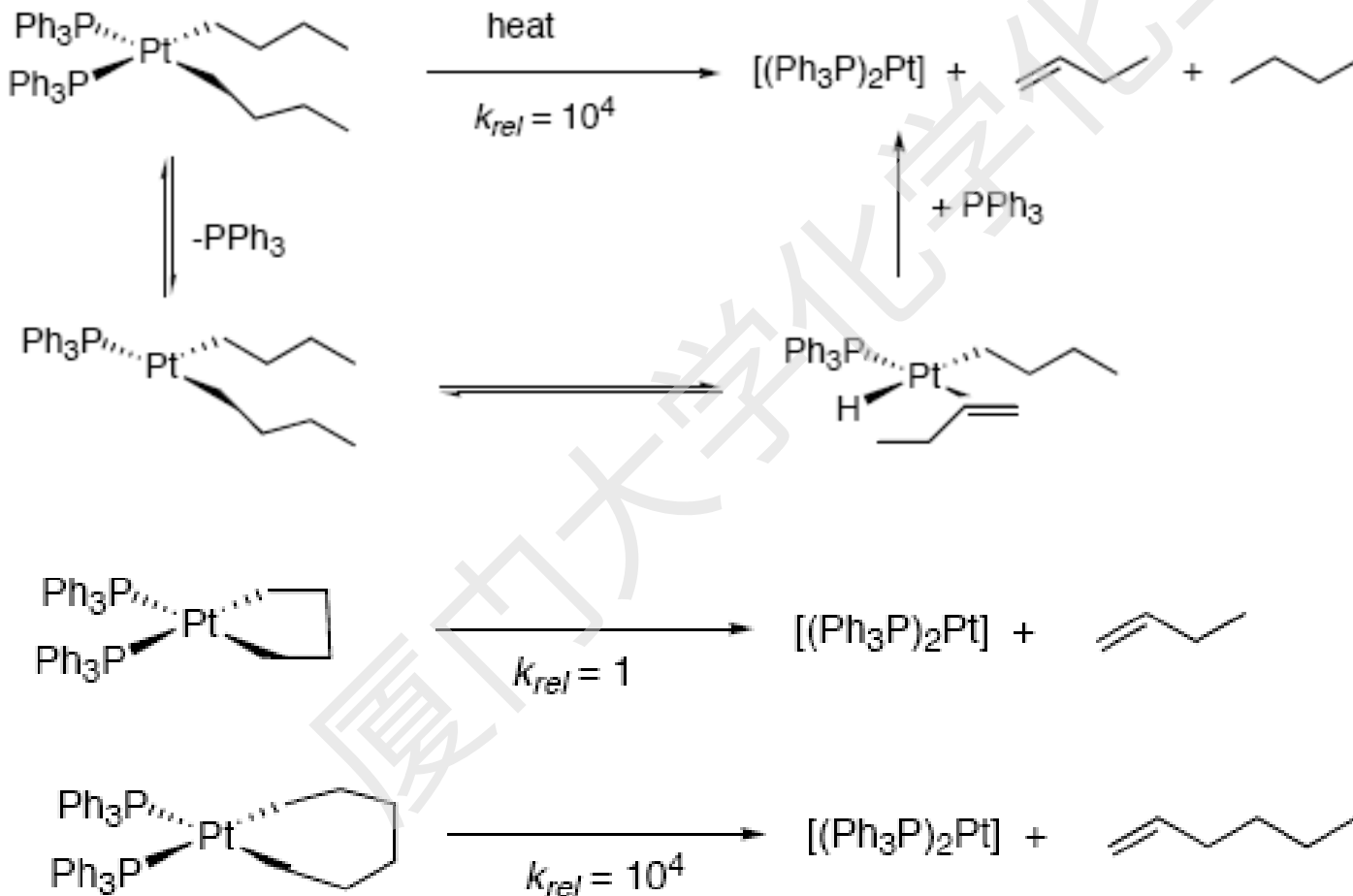




For 16 electron complexes, prior ligand dissociation is not required

Exception is observed for d^8 Pd(II) and particularly Pt(II), which may involve 14e species.

Thus rate of decomposition of $(\text{Ph}_3\text{P})_2\text{PtBu}_2$ to give butane and butene is inversely dependent upon $[\text{PPh}_3]$.



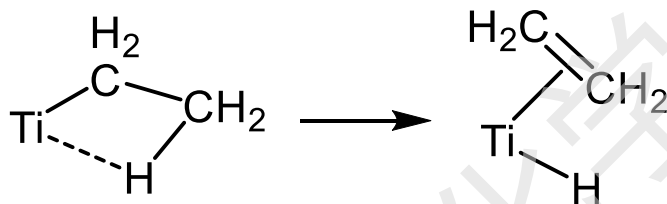
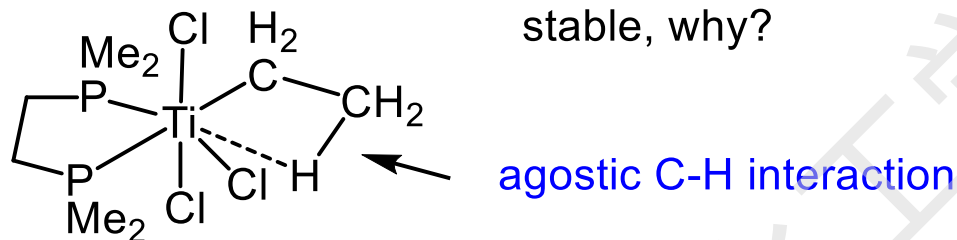
The platina-cyclopentane analog undergoes decomposition 10,000 times more slowly.

However larger platina-cycles undergo decomposition at a similar rate to the dibutyl complex and by the same mechanism.



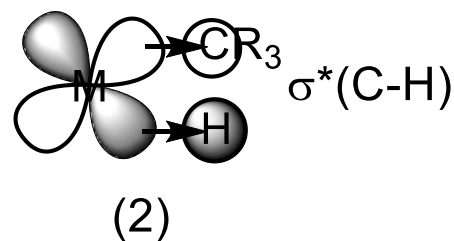
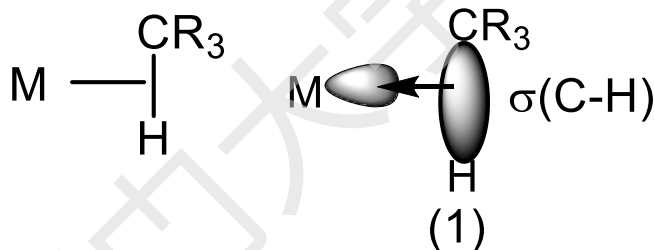
Further notes on β -H elimination

(i) β -H elimination on d^0 (electron deficient) metal complexes may not occur.



A C-H bond is broken!

Bonding in $M(\eta^2\text{-C-H})$



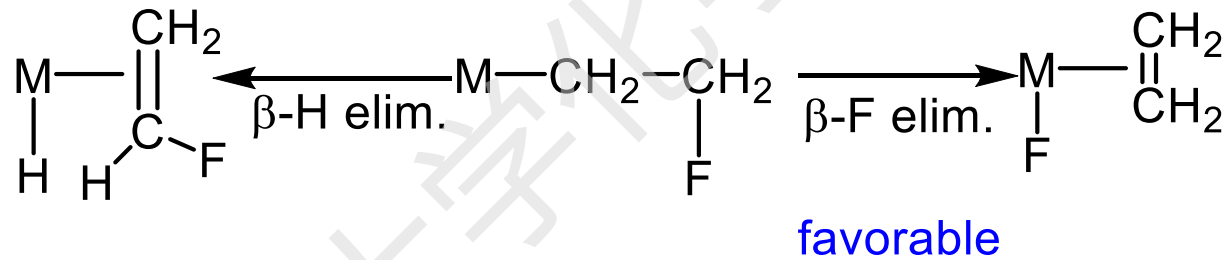
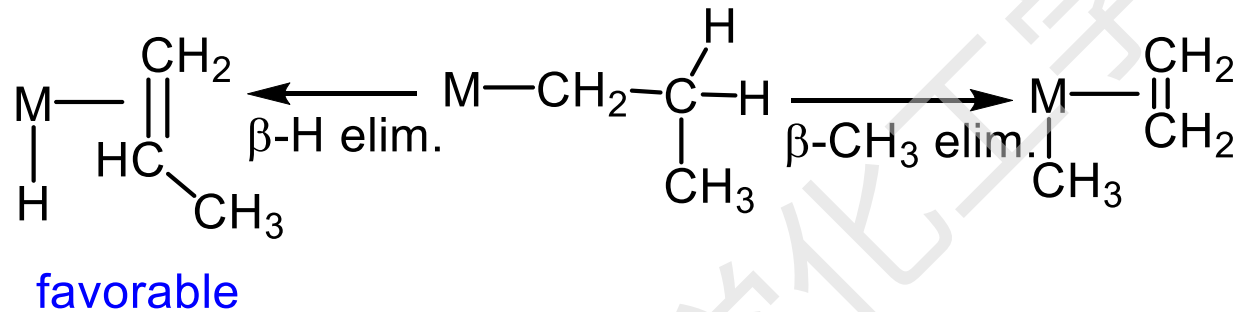
The complex has no d electron for backdonation to $\sigma^*(\text{C-H})$, so it is difficult to undergo β -H elimination.

In general, d^0 metal-alkyls has a lower tendency to undergo β -H elimination than non- d^0 alkyls.



Further notes on β -elimination

(ii) Other β -elimination is also possible



because: $\text{M}-\text{H} > \text{M}-\text{CH}_3$ (by 15-25 kcal/mol); $\text{M}-\text{F} > \text{M}-\text{H}$



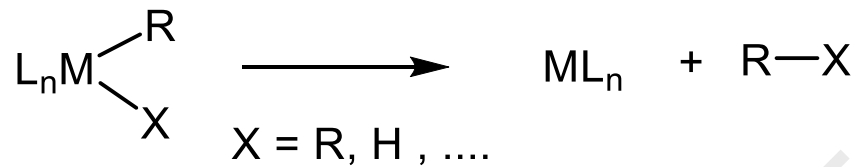
Summary

Metal-alkyls that are stable to β -hydrogen elimination

1. Alkyls that do not have β -hydrogens
2. Alkyls for which the β -hydrogen cannot approach the metal center (as in cyclic alkyls or as a result of steric crowding)
3. Alkyls in which the M-C-C-H unit cannot become coplanar
4. Olefin generated is unstable
5. 18-electron species that cannot dissociate a ligand
6. Some d^0 - alkyls

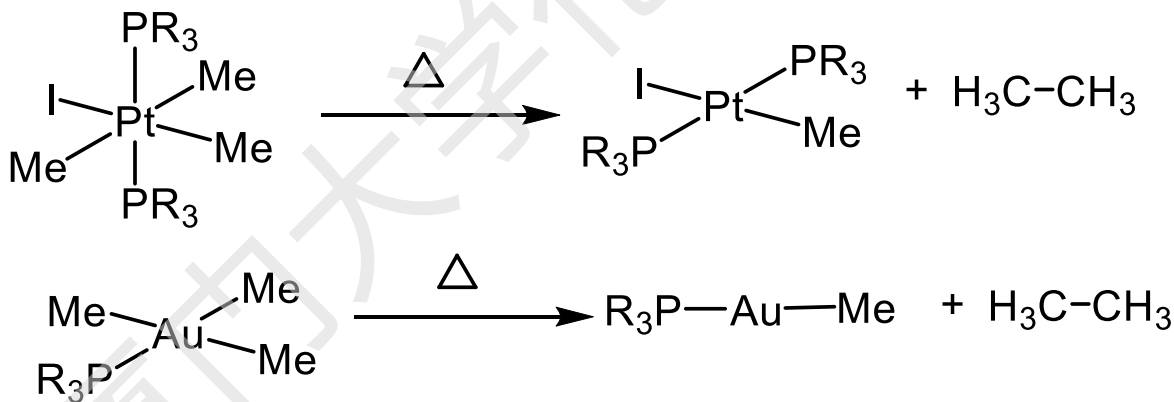


b) Reductive elimination



- ✓ Coordination number decreases by 2.
- ✓ Oxidation number decreases by 2.

The process is the second most common decomposition pathway for metal alkyls. e.g.



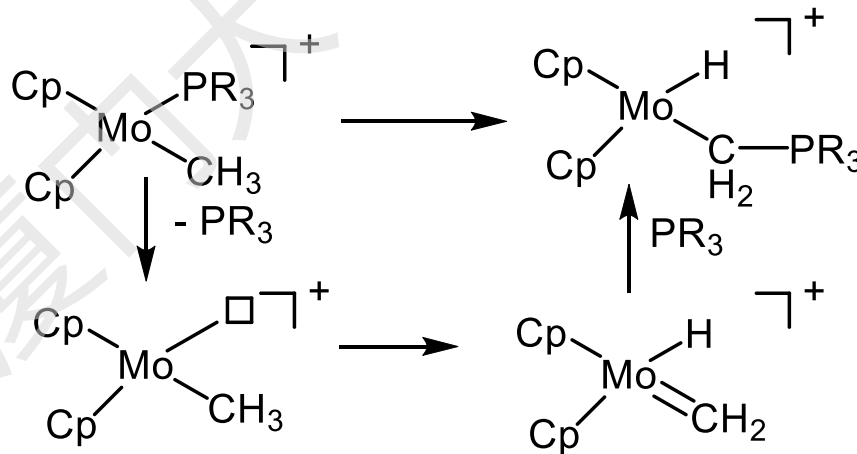
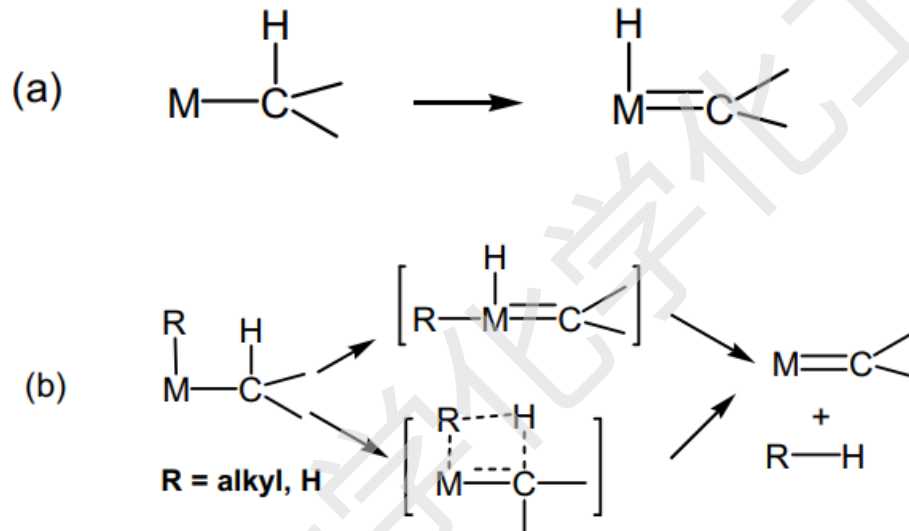
More detailed discussion will be given later.



c) Other pathways

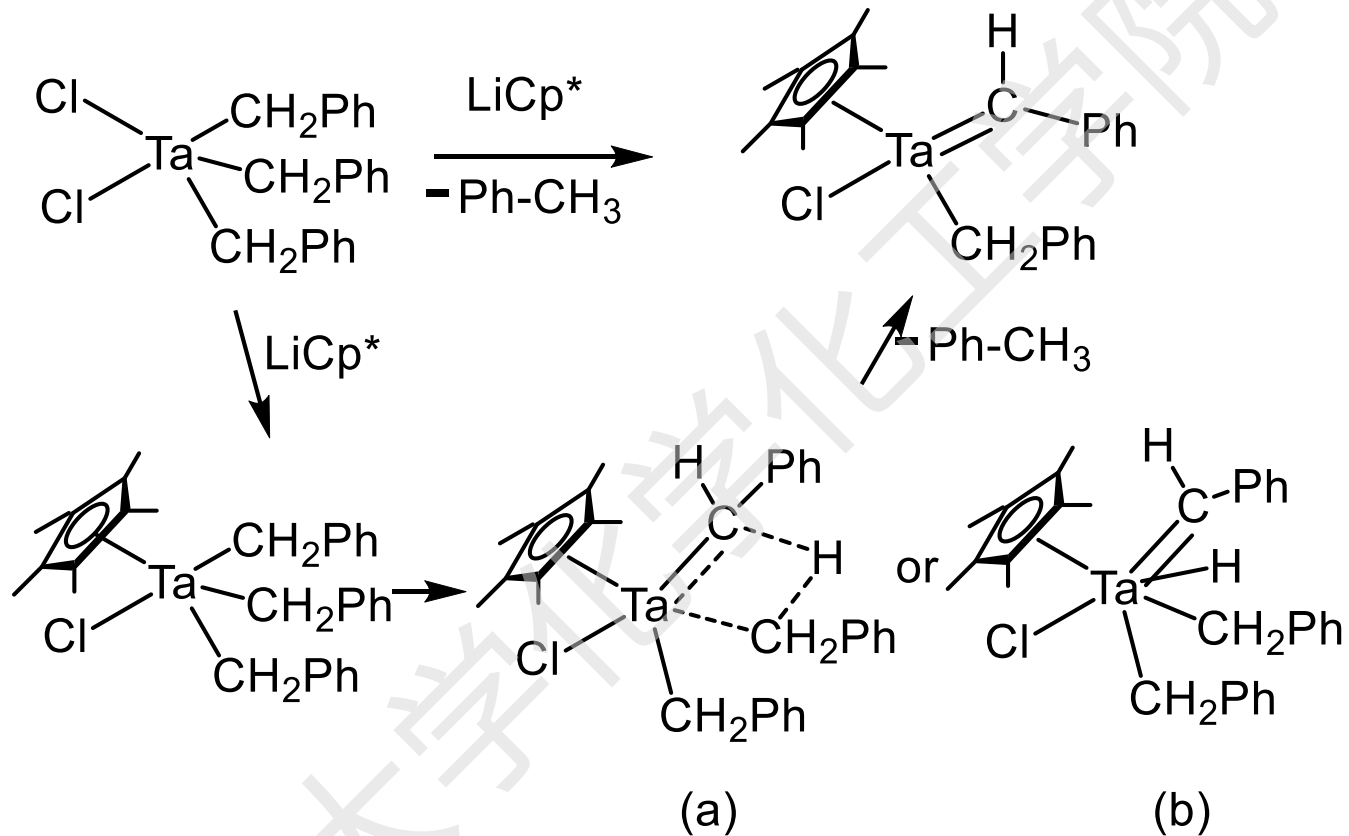
α -H elimination

For early transition metal alkyls, α -H elimination can be an important decomposition pathway.





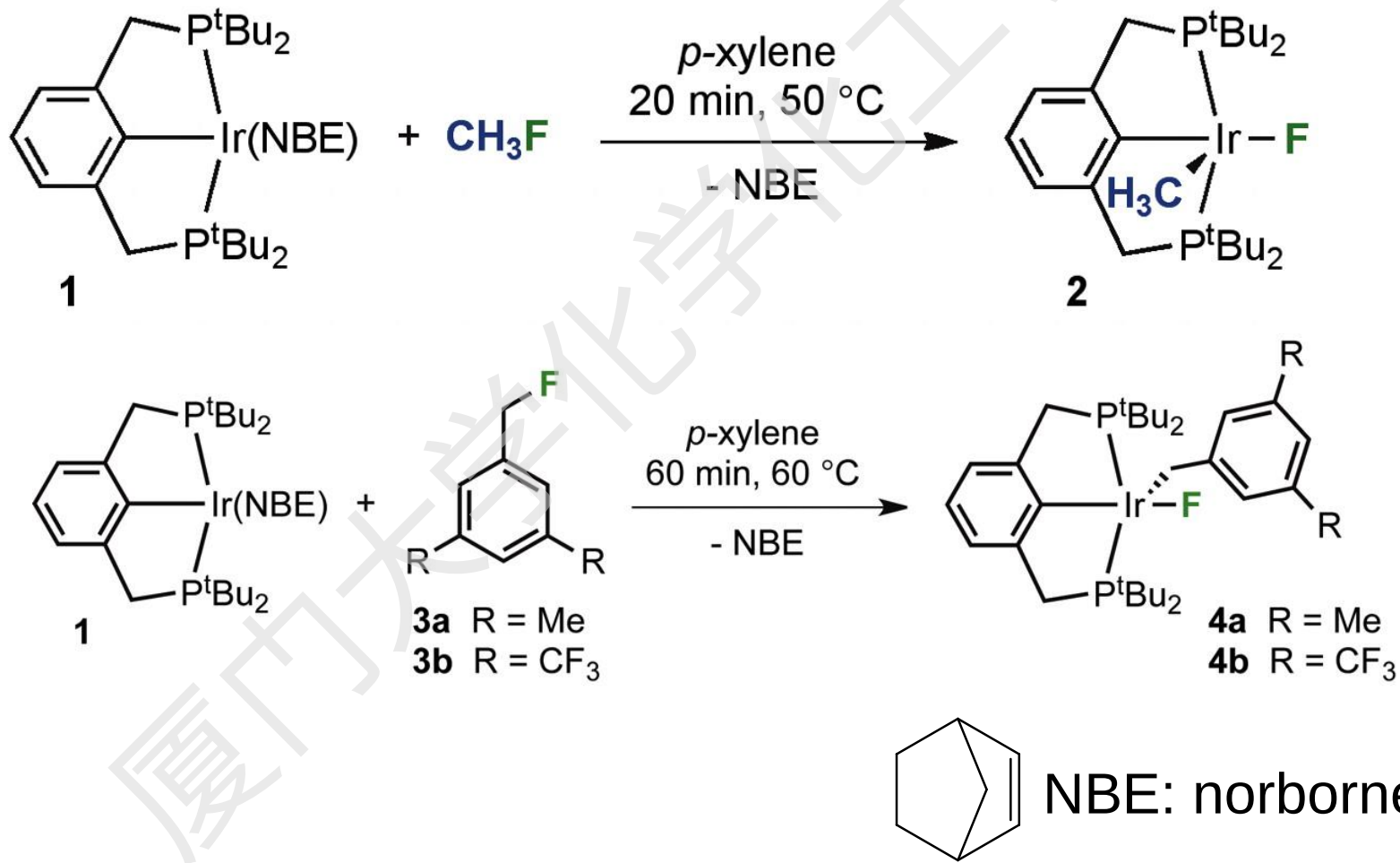
Another example



Which intermediate is more likely?



Net Oxidative Addition of C(sp³)-F Bonds to Iridium via Initial C-H Bond Activation



Net Oxidative Addition of C(sp³)-F Bonds to Iridium via Initial C-H Bond Activation

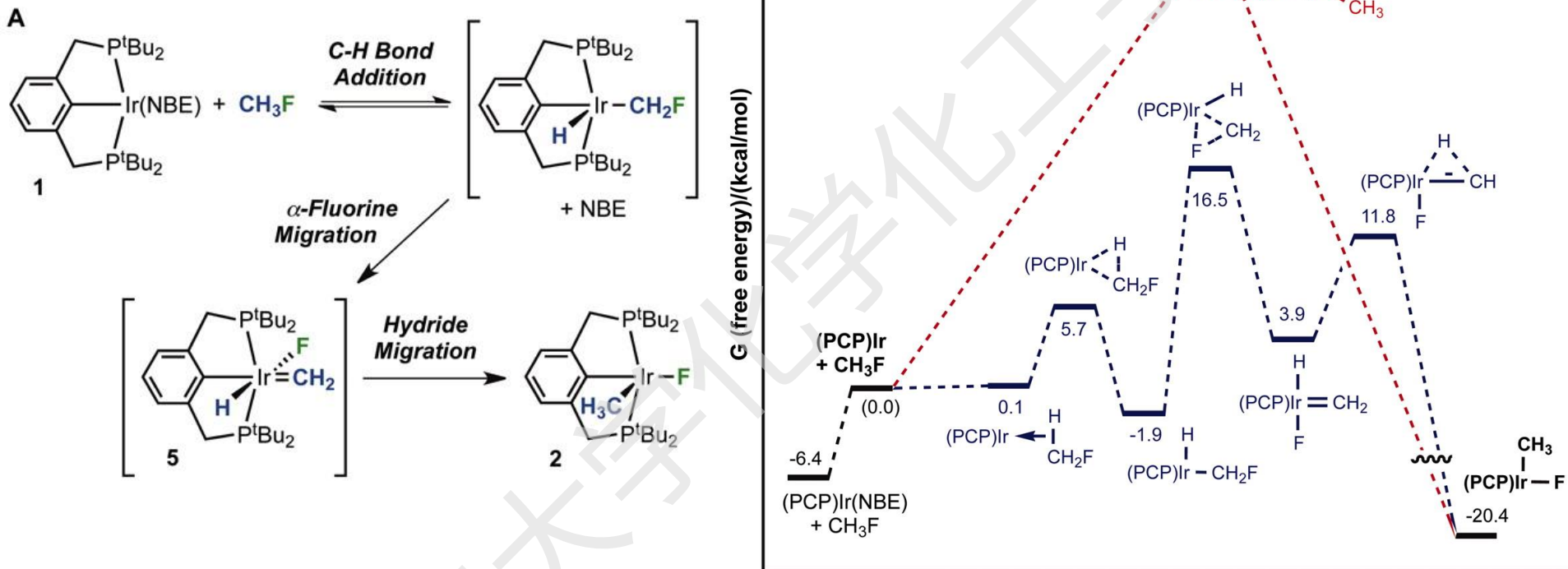
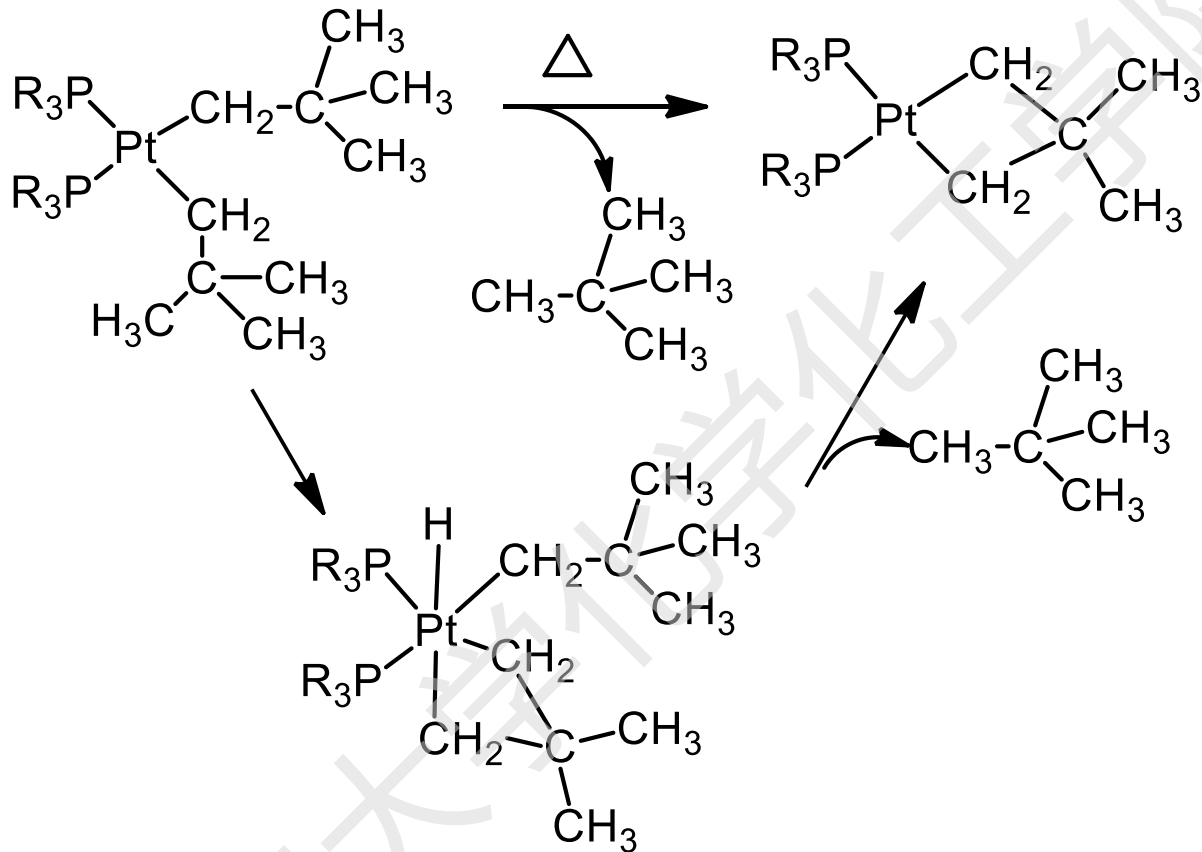


Fig. 4 (A) Proposed mechanism for oxidative addition of CH₃-F to (PCP)Ir. (B) Free energy diagrams pertaining to the proposed pathway for the reaction of (PCP)Ir with CH₃F (blue) and to direct C-F addition (red).



γ -H elimination



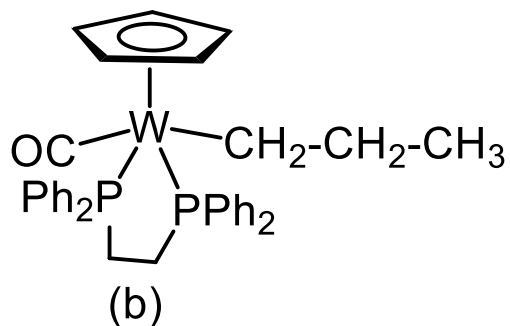
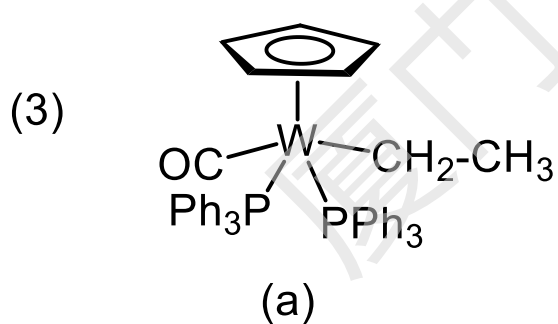
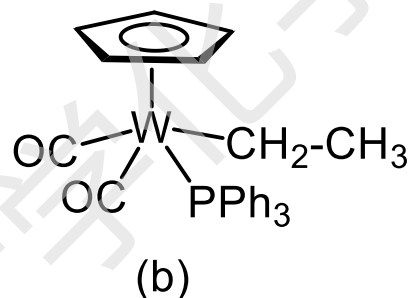
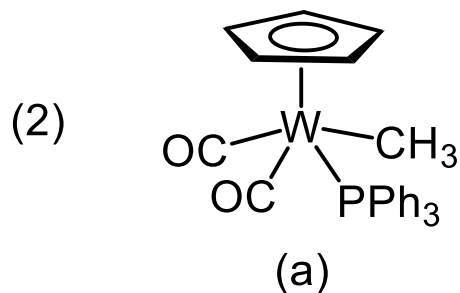
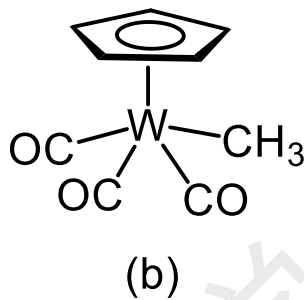
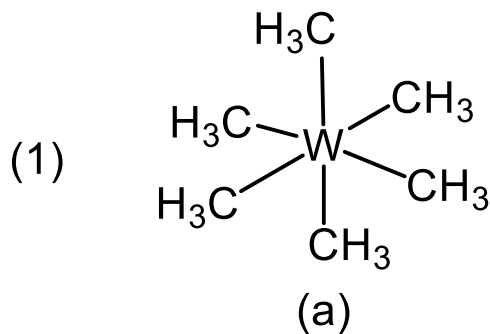
Summary: Common pathways of decomposing M-alkyls:

- 1) β -H elimination
- 2) Reductive elimination
- 3) Others.



Practice Problems

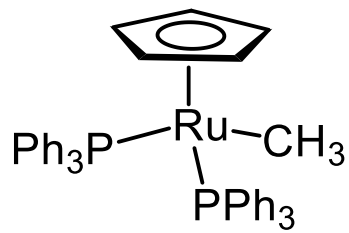
1) For each pair of complexes listed below, which one is less stable?



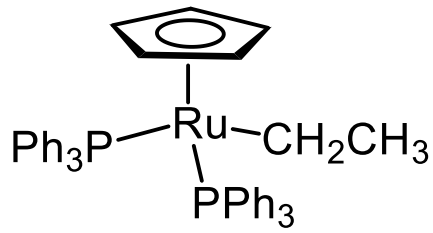


Practice Problems

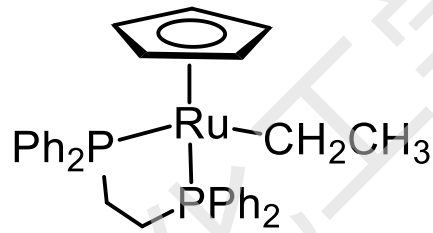
2) Which one is least stable ?



(a)

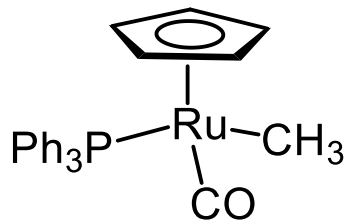


(b)

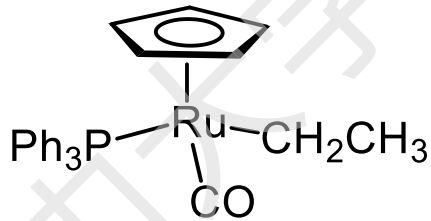


(c)

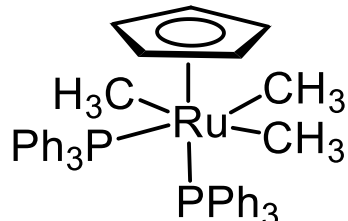
3) which one is most stable ?



(a)



(b)



(c)



2.2 Preparation of complexes with metal-carbon σ bonds

A. Synthesis by alkyl transfer reactions.



B. Synthesis from anionic transition metal complexes.



C. Synthesis by oxidative addition reactions.



D. Synthesis involving insertion Reactions.



E. Synthesis involving elimination reactions.



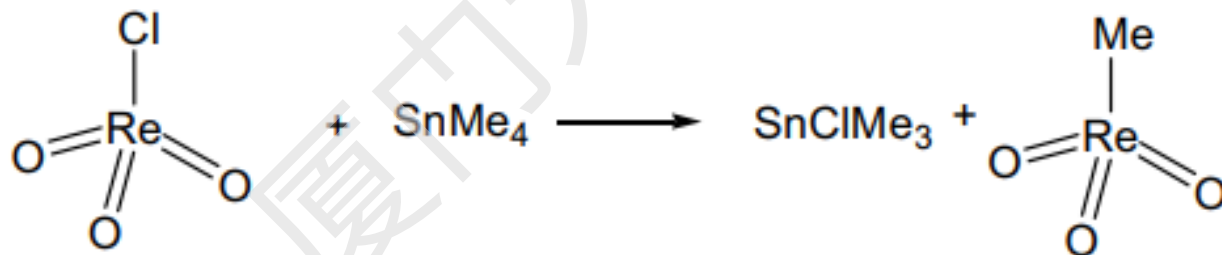
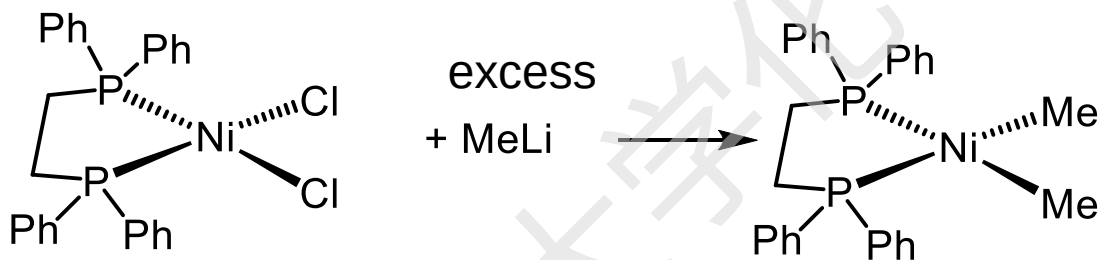
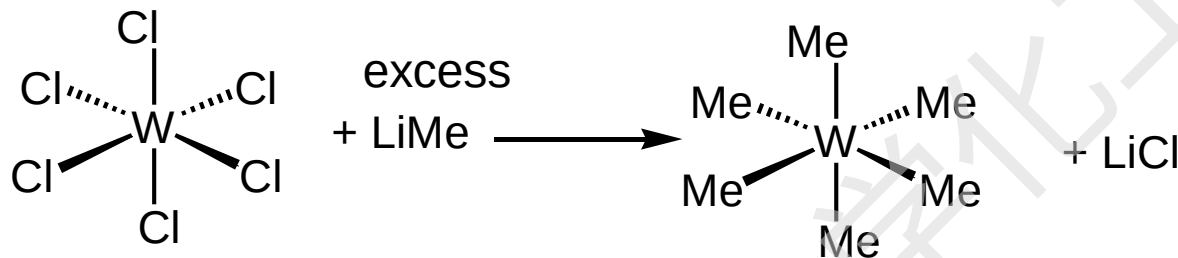
F. Other routes



A. Synthesis by alkyl transfer reactions



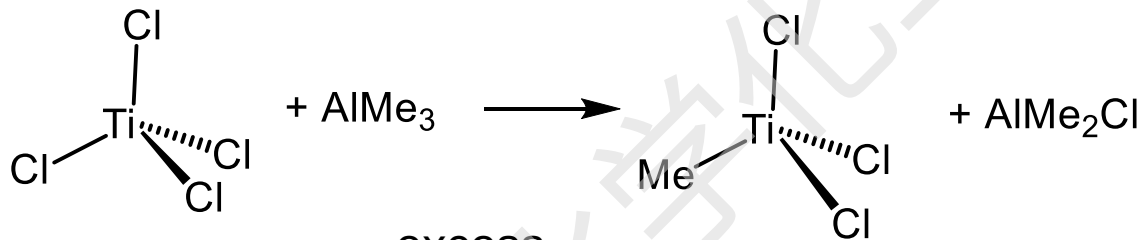
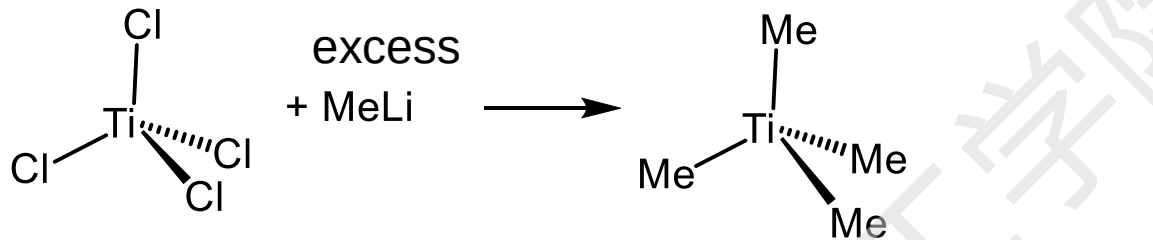
Typical R^- reagents: RLi , $RMgX$, AlR_3 , AlR_2X , ZnR_2 , etc.



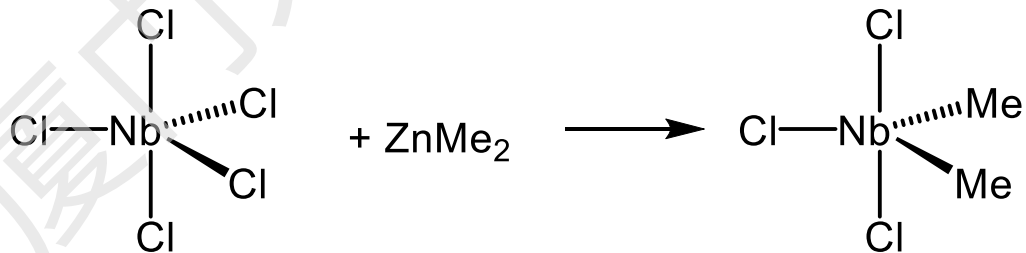
Methylrhenium trioxide (MTO) is a catalyst for a variety of transformations.



Note: different alkylating agents may have different reactivity



ZnR_2 is also a weak alkylating reagents. e.g.

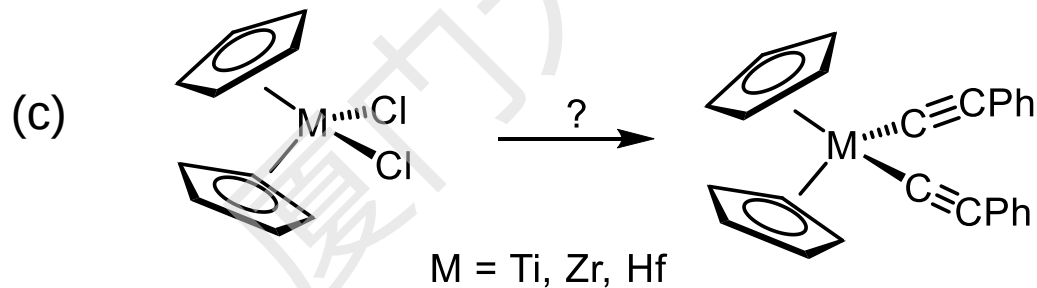
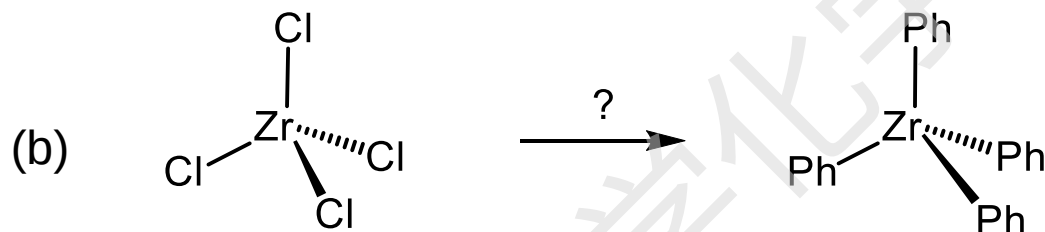
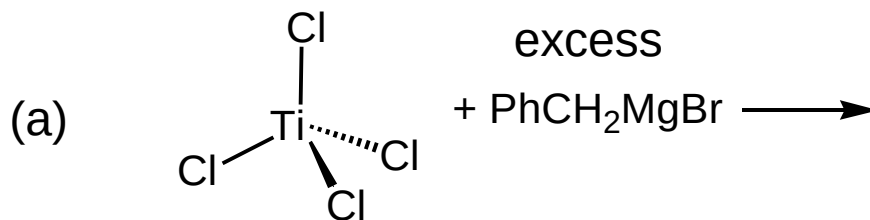


Alkylating strength: $\text{RLi} > \text{RMgX} > \text{AlR}_3 > \text{AlR}_2\text{X} \approx \text{ZnR}_2 > \text{SnR}_4$



Practice Problems

Suggest products/reagents for the following preparations.

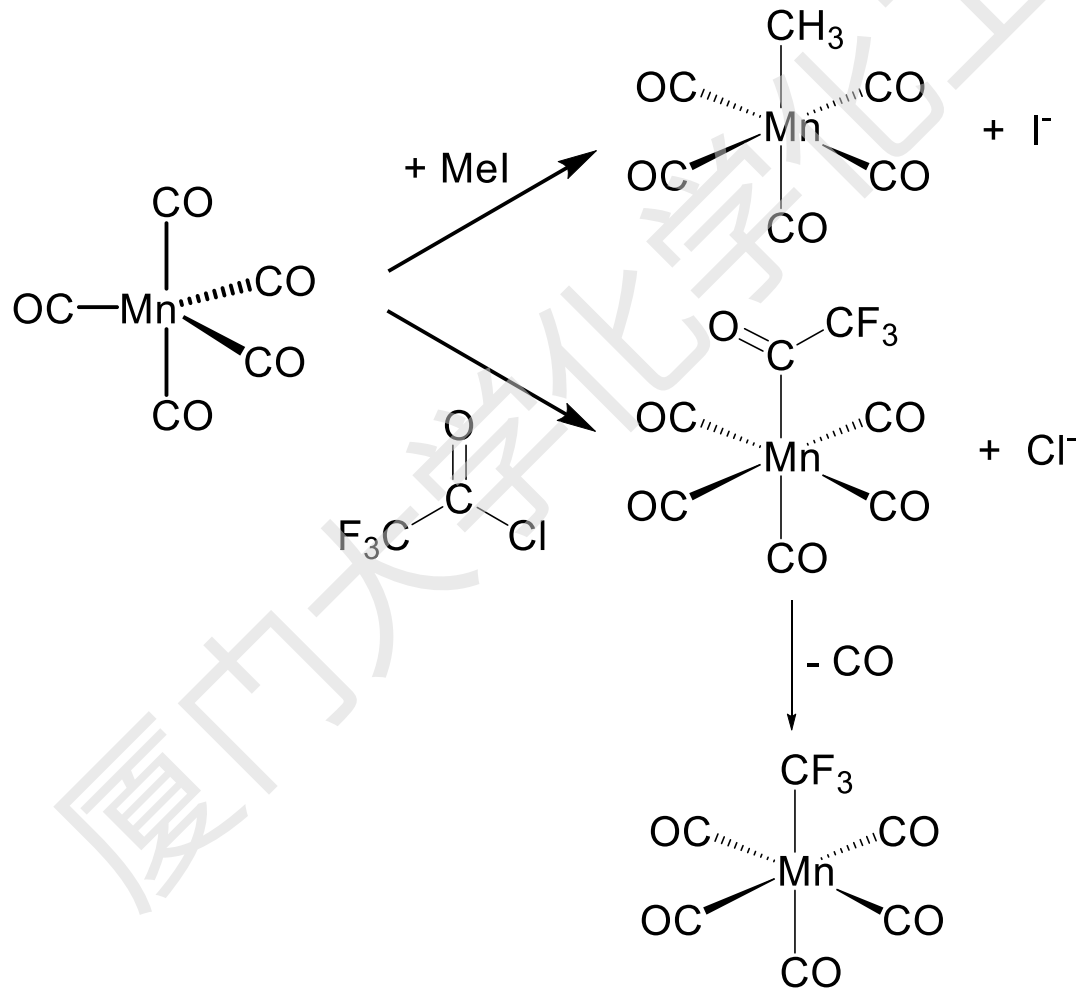




B. Synthesis from anionic transition metal complexes.

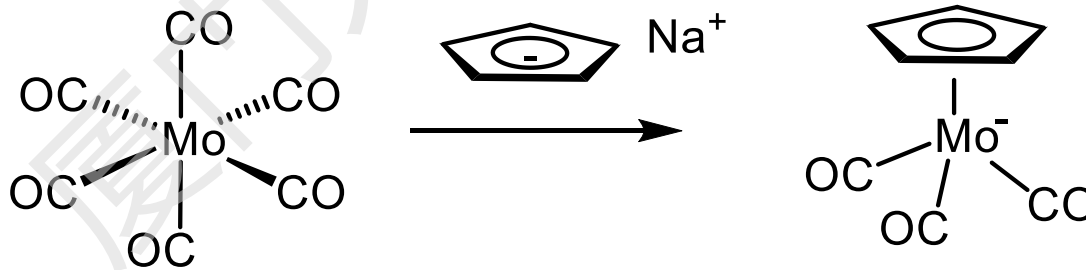
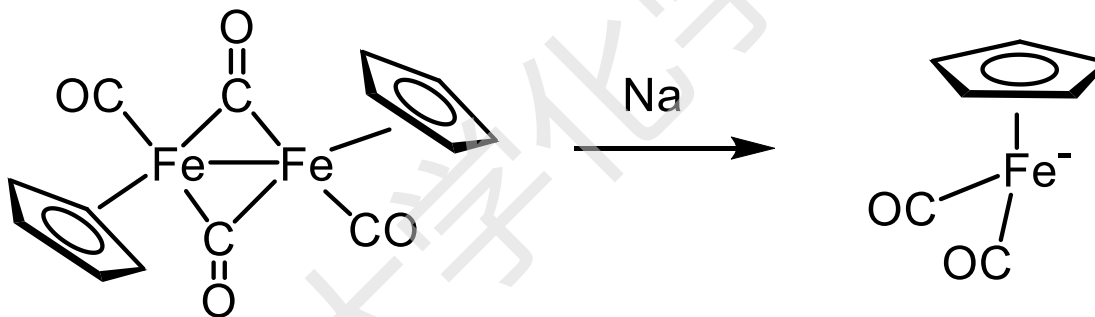
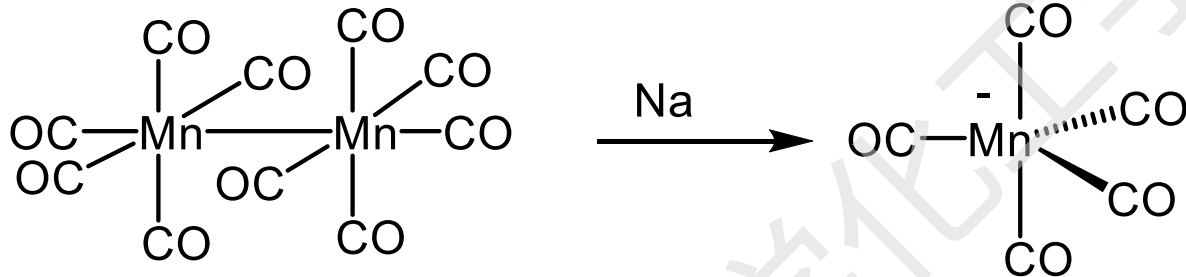
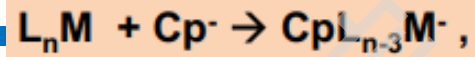
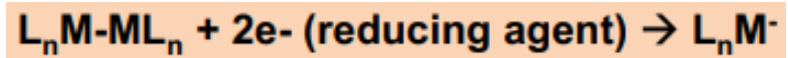


a sufficiently nucleophilic metal can undergo electrophilic attack





Generation of L_nM^-





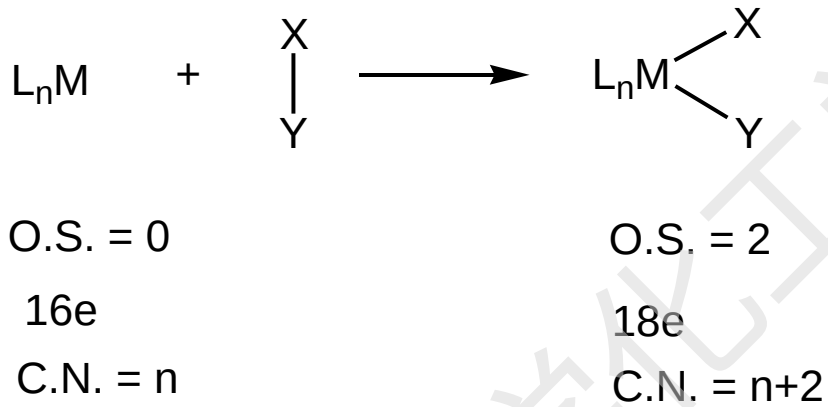
(a)





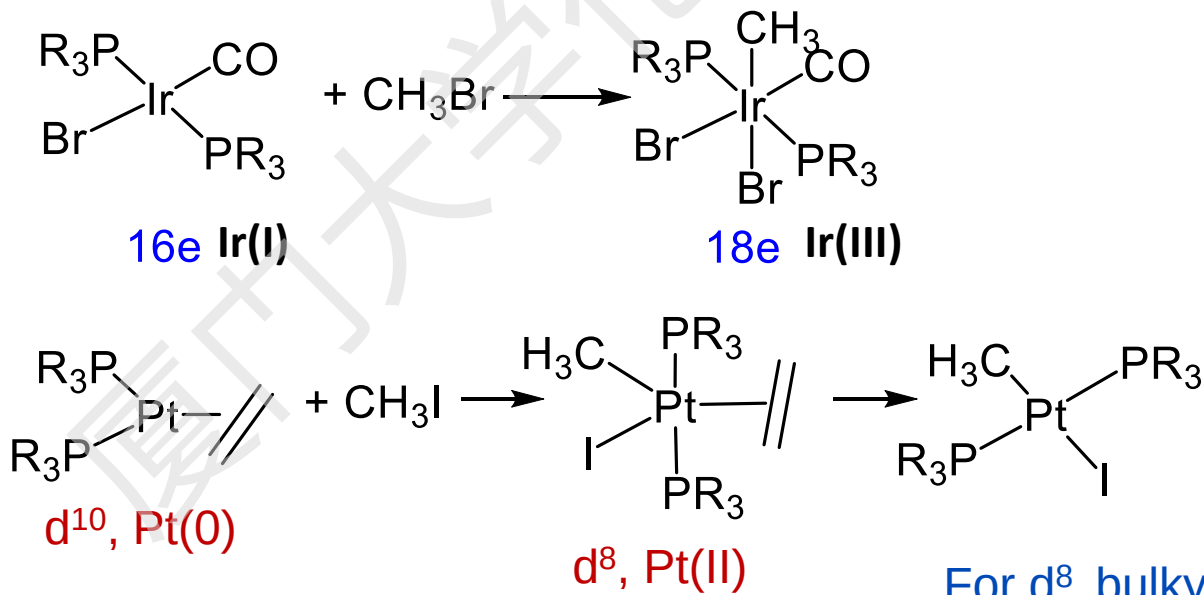
C. Synthesis by oxidative addition reactions.

Typical oxidative addition reactions:



The reactions are common for RX and ArX (X = halides, OTf).

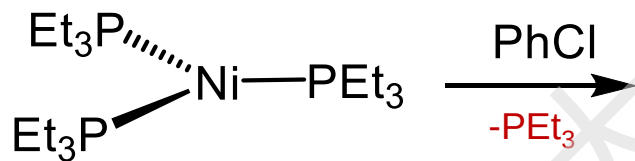
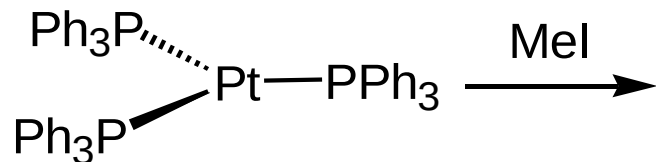
Works for many low valent complexes, e.g. $\text{Pd}(0)$, $\text{Pt}(0)$, $\text{Rh}(\text{I})$, $\text{Ir}(\text{I})$, $\text{Re}(\text{I})$





Practice Problems

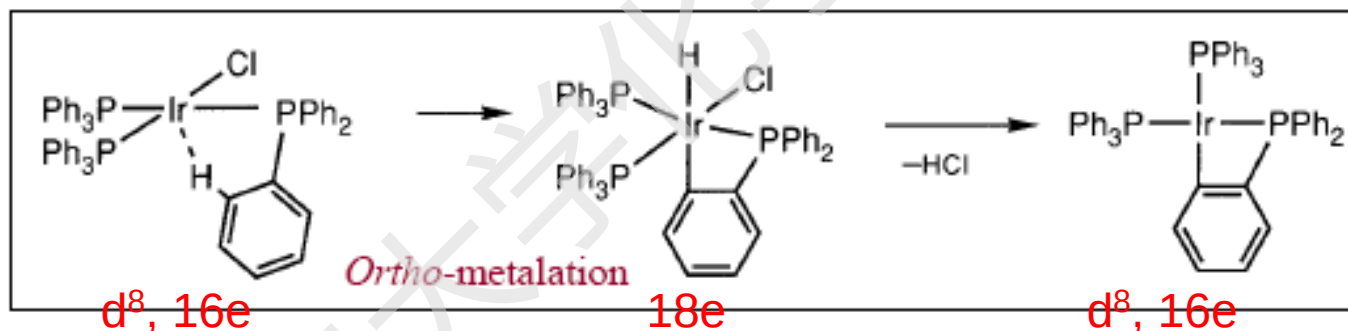
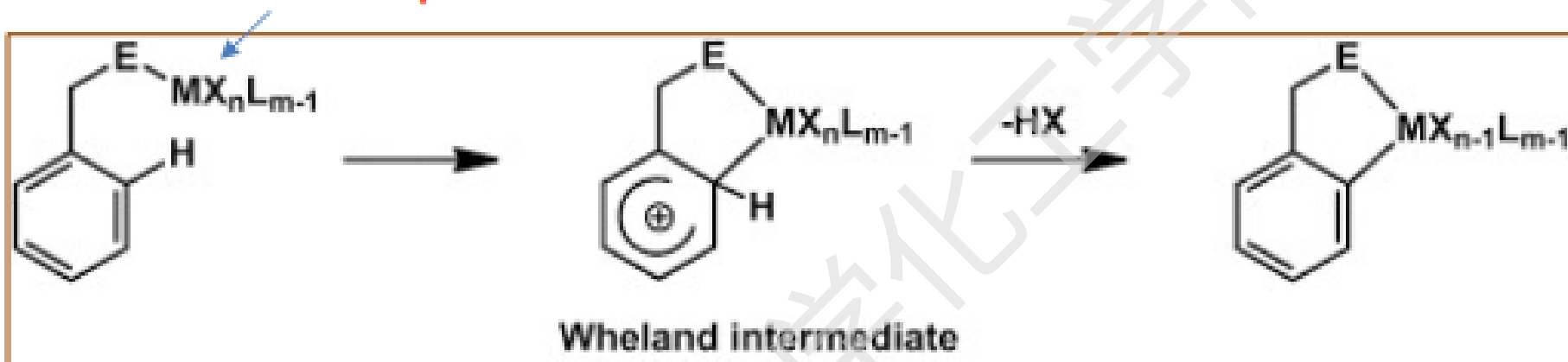
Give the product of the following reactions.





Synthesis by oxidative addition of C-H bonds.

A general mechanism for cyclometallation of aryl groups
electrophilic



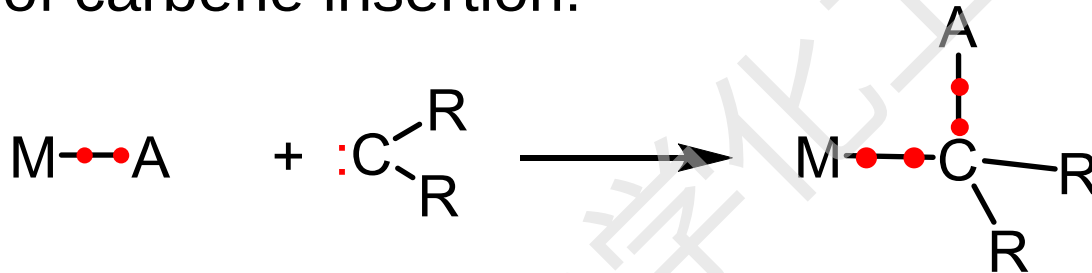
Intermolecular oxidative addition of C-H bonds is rarely observed. Note that this requires ca. 95 kcal/mol, which is close to the combined energies of M-H (~60 kcal/mol) and M-C (30-50 kcal/mol) bonds and is entropically unfavorable.



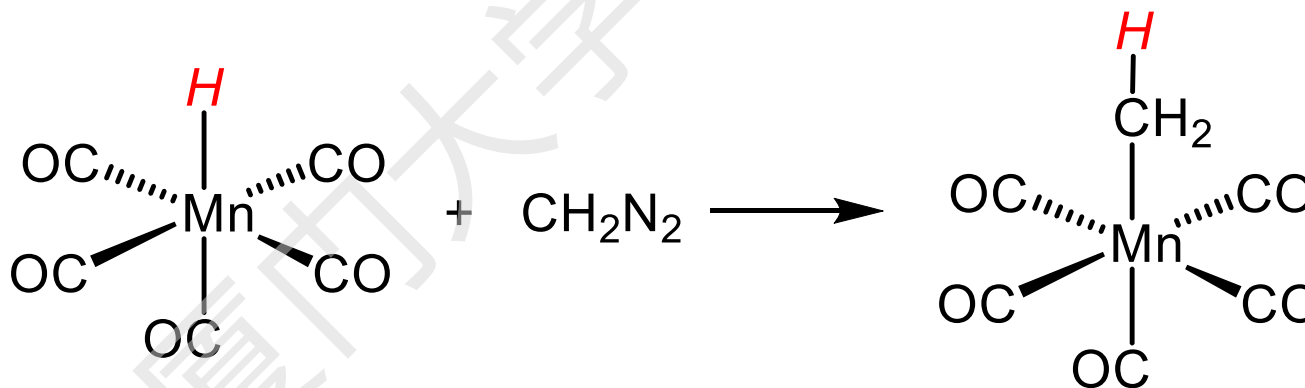
D. Synthesis involving insertion reactions.



Examples of carbene insertion:

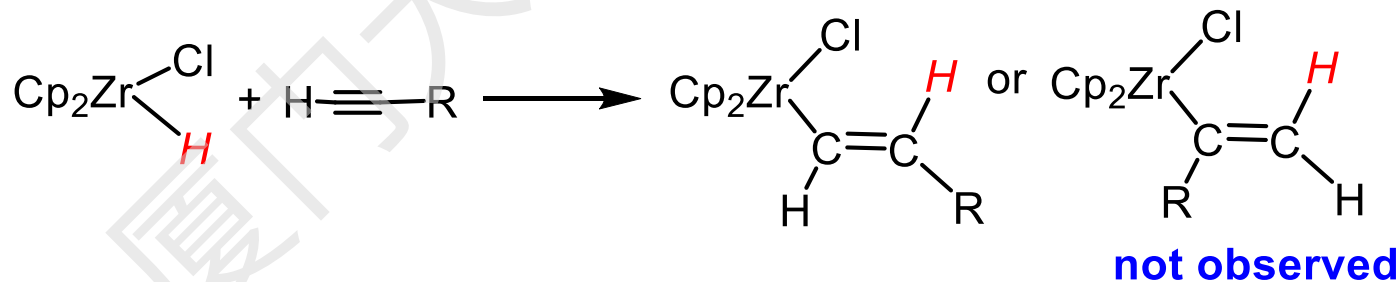
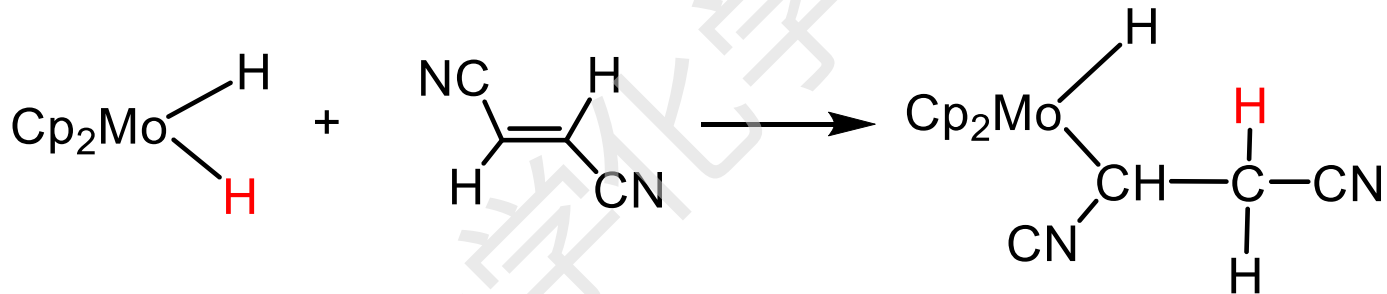
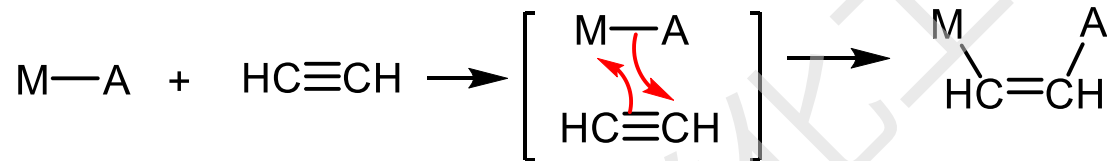
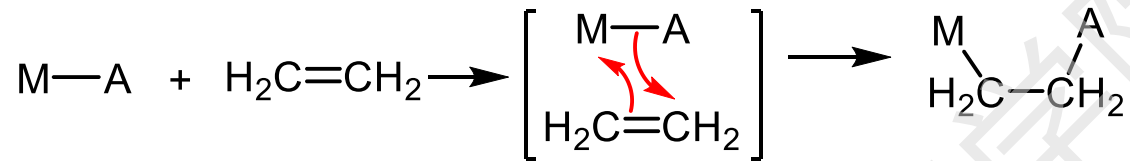


A can be H, alkyl, halide





Examples of olefin and alkyne insertion

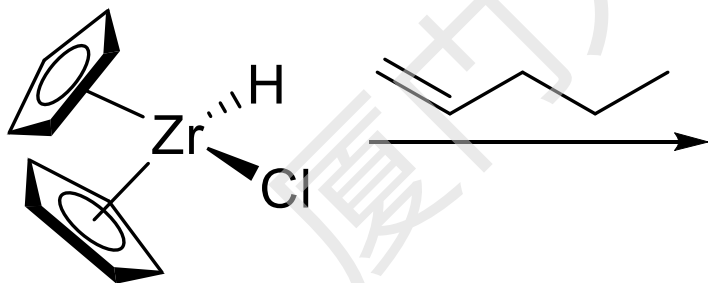
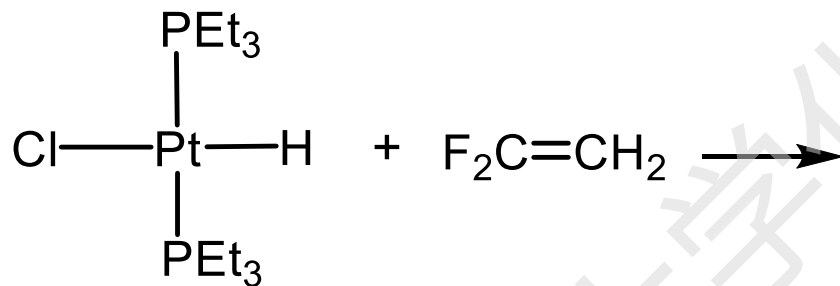
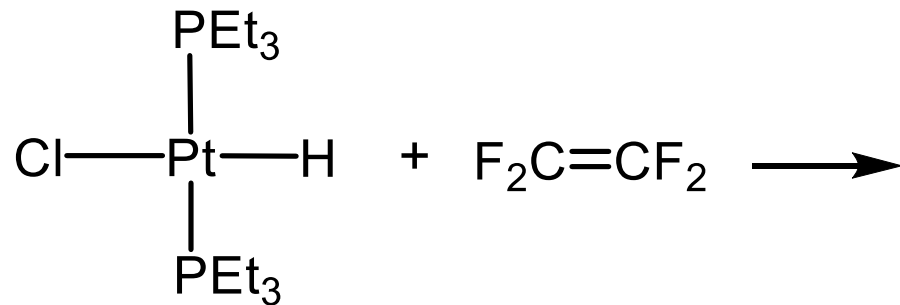


For steric reason



Practice Problems

Give the product of the following reactions.



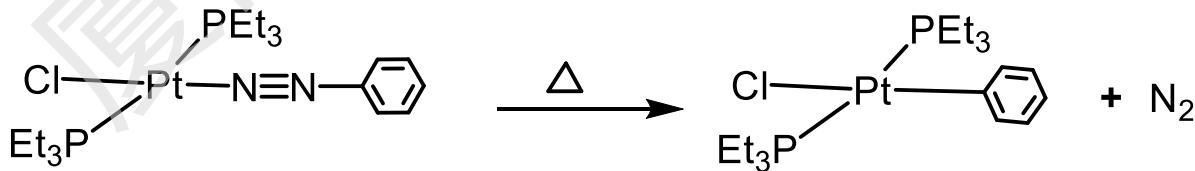
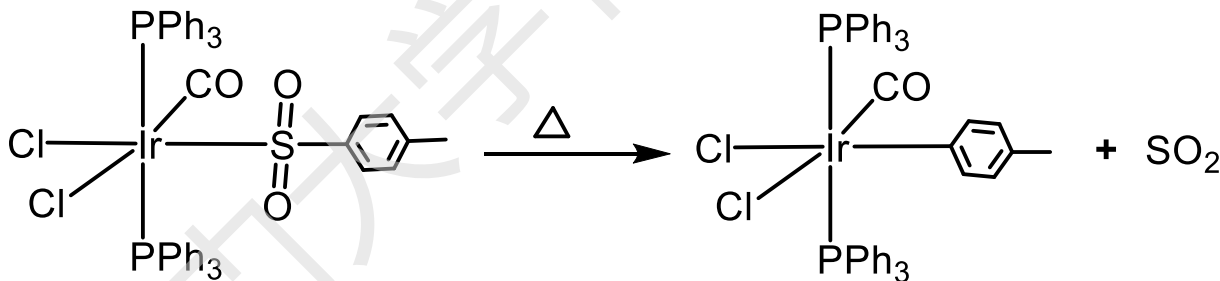
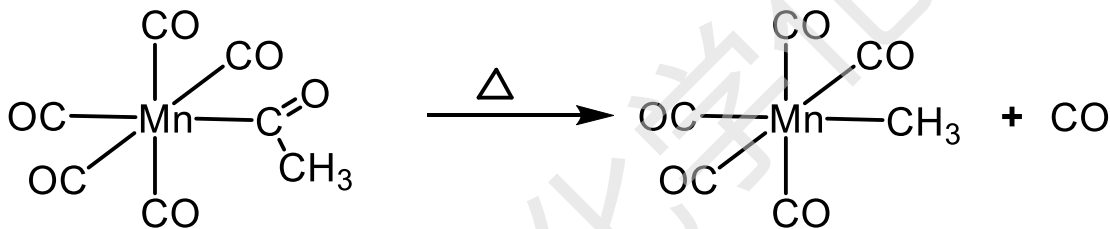


E. Synthesis involving elimination reactions.



A = stable species, such as CO, CO₂, SO₂, N₂.

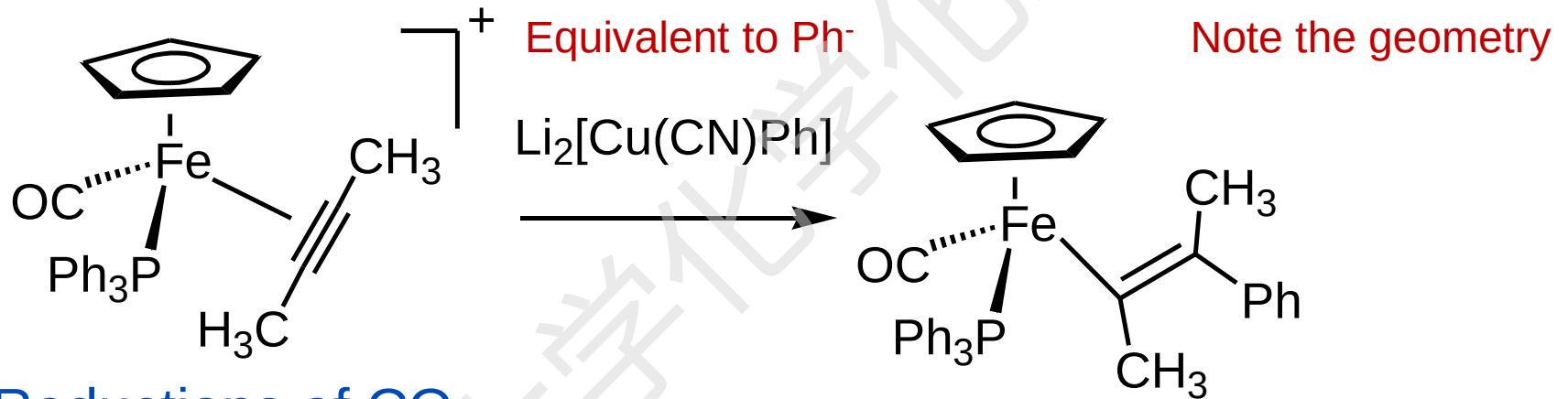
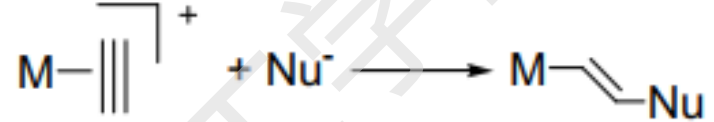
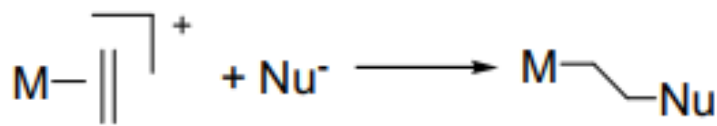
Examples of preparation of complexes.



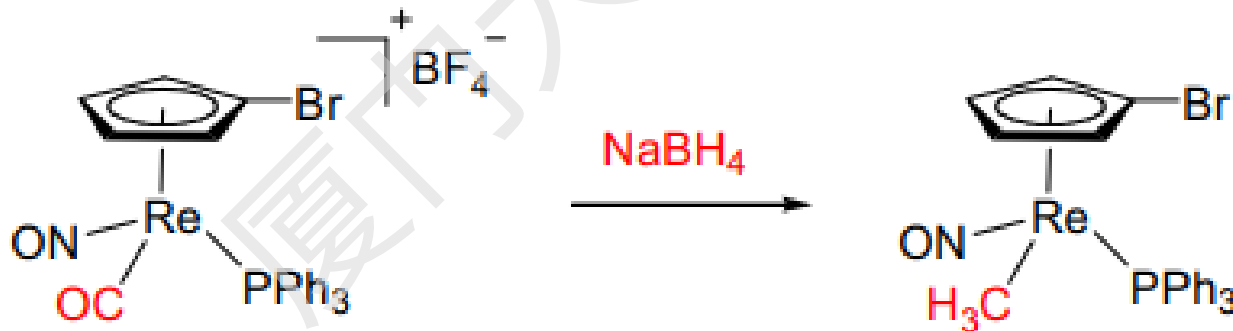


F. Other routes

- Nucleophilic addition to coordinated olefin or alkynes



- Reductions of CO

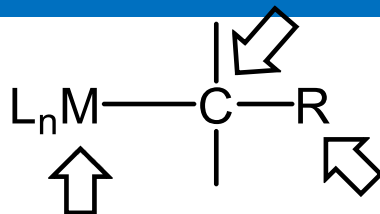


We will go to more details later. ⁴⁵



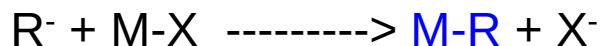
Summary

-M-C bond strenths

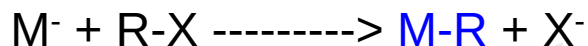


- Preparation of complexes with metal-carbon σ bonds.

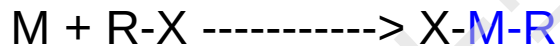
A. Synthesis by alkyl transfer reactions.



B. Synthesis from anionic transition metal complexes.



C. Synthesis by oxidative addition reactions.



D. Synthesis involving insertion Reactions.



E. Synthesis involving elimination reactions.

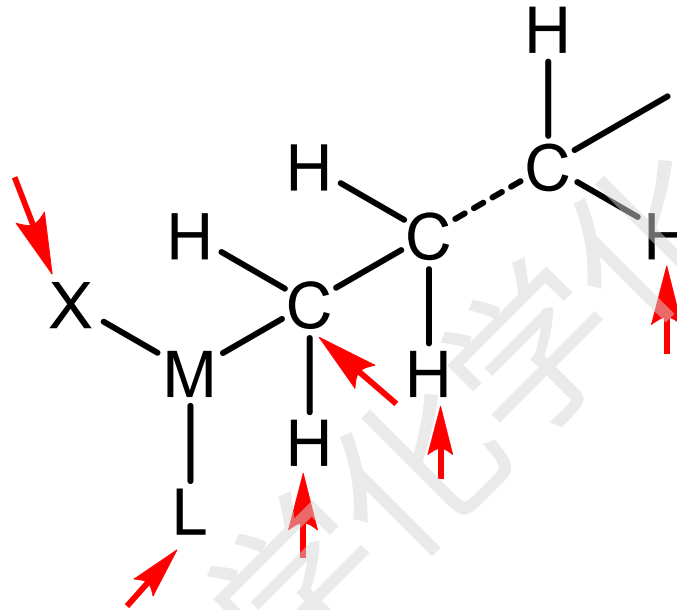


F. Synthesis by attack on coordinated ligands



2.3 Chemical properties of complexes with metal-carbon α bonds

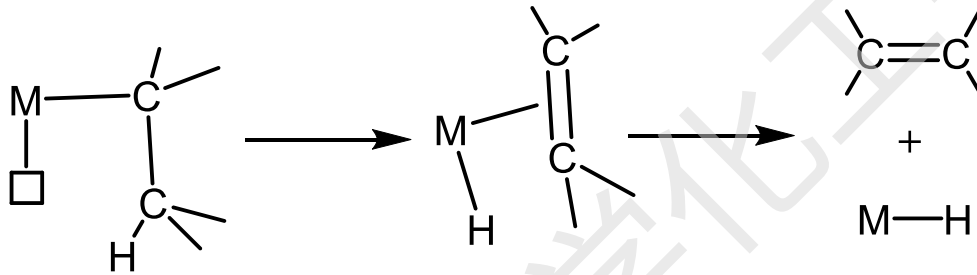
Possible reaction sites:



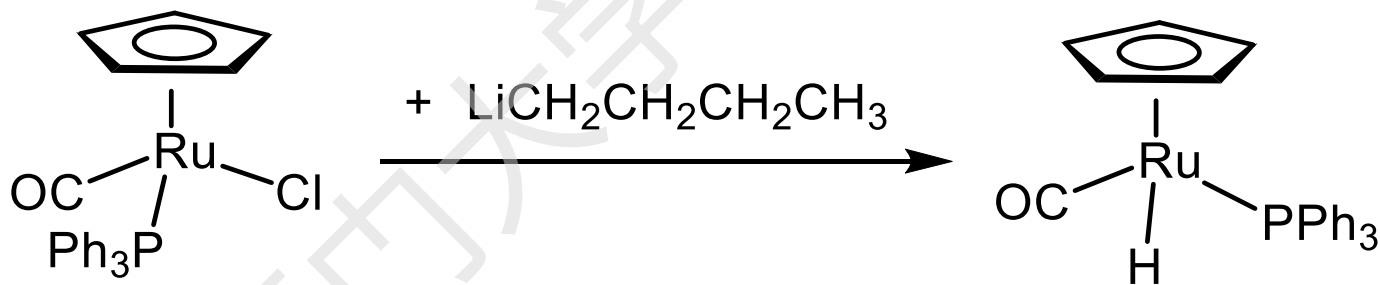


A. β -H elimination

It is the most common decomposition pathway for alkyls, which converts a metal-alkyl into a hydrido-olefin complex.



hydrido-olefin complex

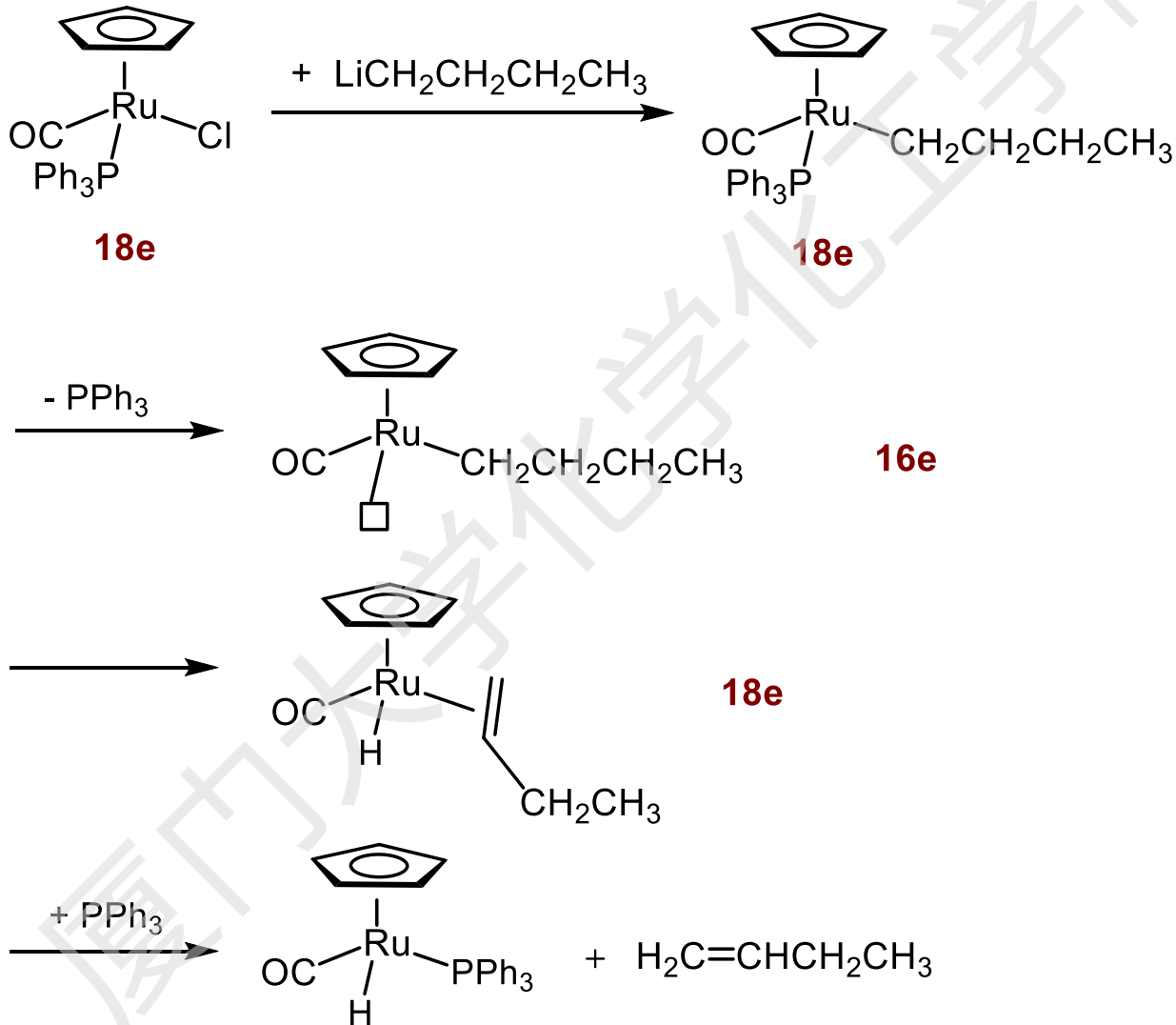


How?



A. β -H elimination

Note: $18 \leftrightarrow 16 \leftrightarrow 18 \dots$

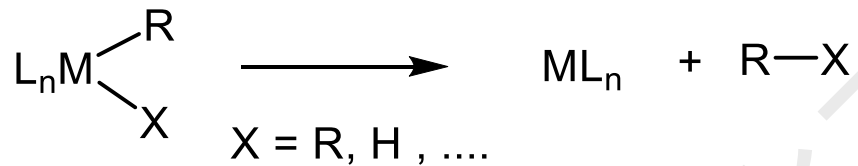


For d^8 complexes, possibly $16 \leftrightarrow 14 \leftrightarrow 16 \dots$



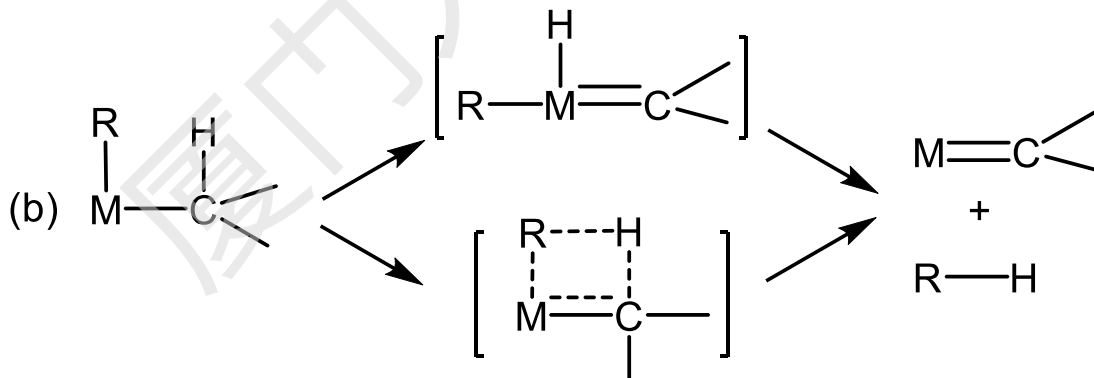
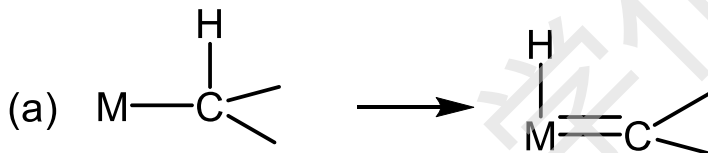
2.3 Chemical properties

B. Reductive elimination



- ✓ Coordination number decreases by 2.
- ✓ Oxidation number decreases by 2.

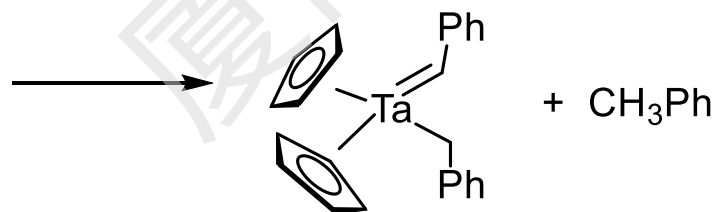
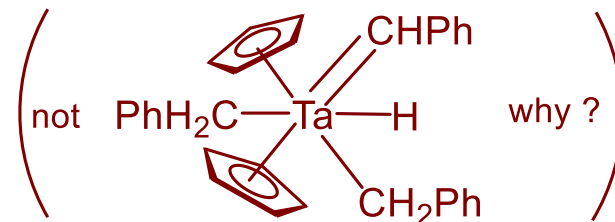
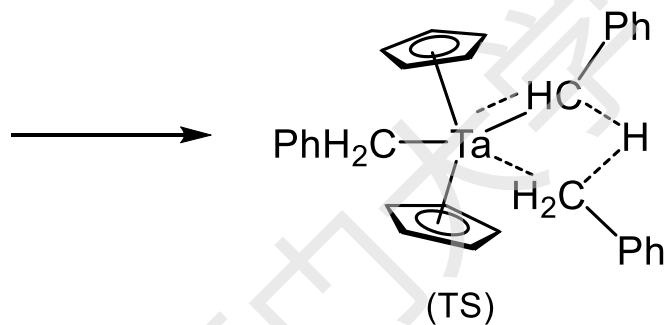
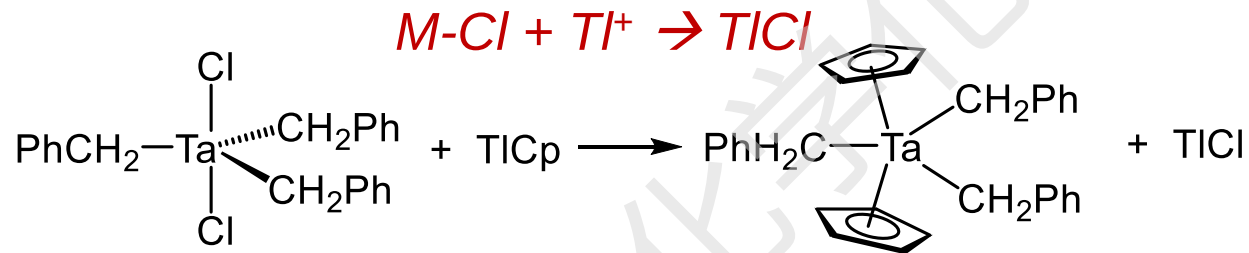
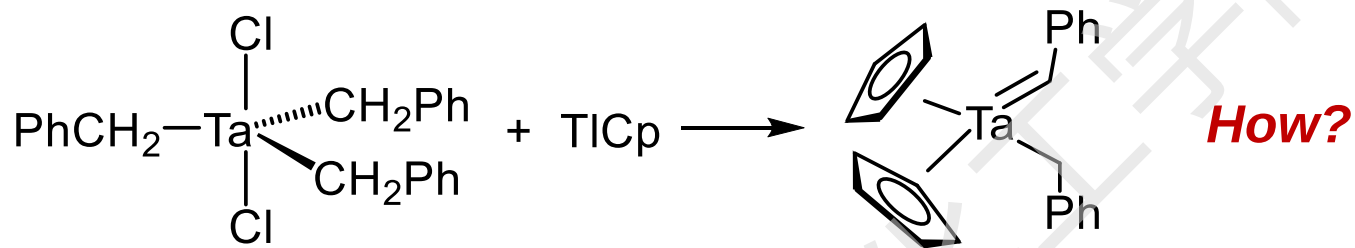
C. α -H elimination





2.3 Chemical properties

D. γ -H elimination



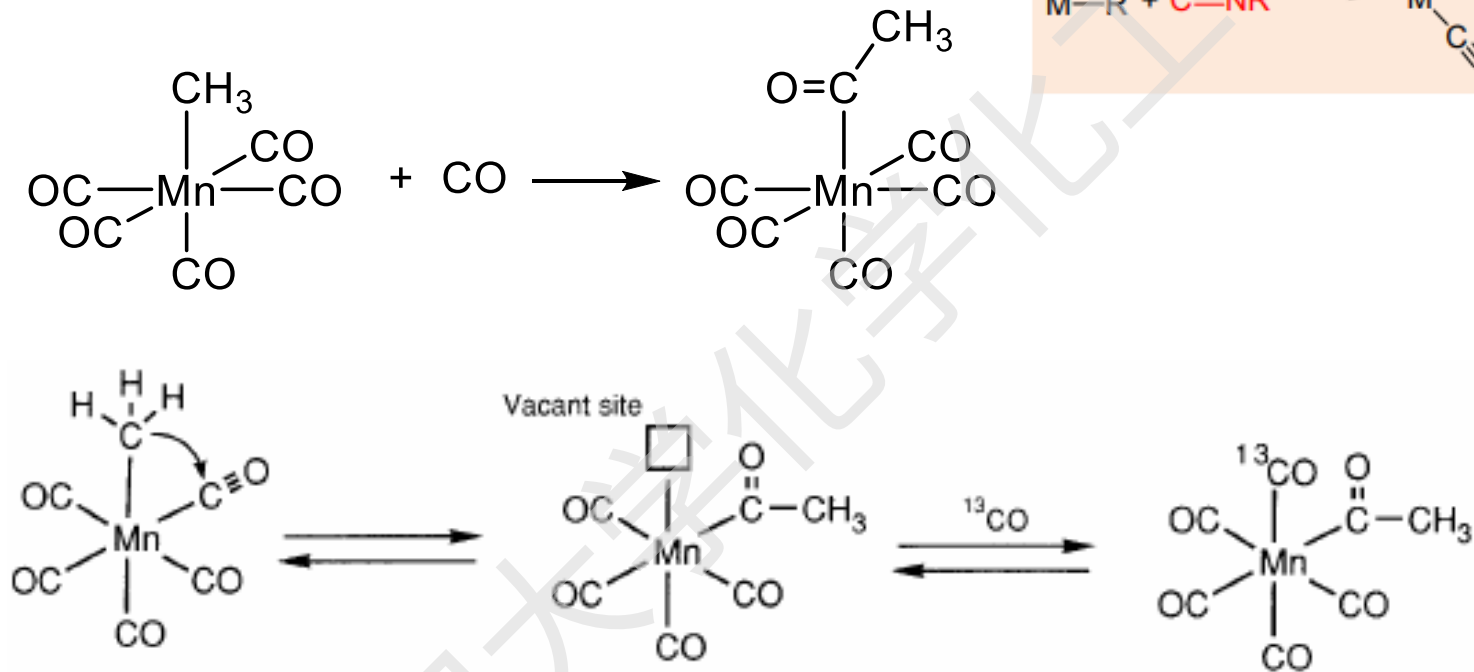
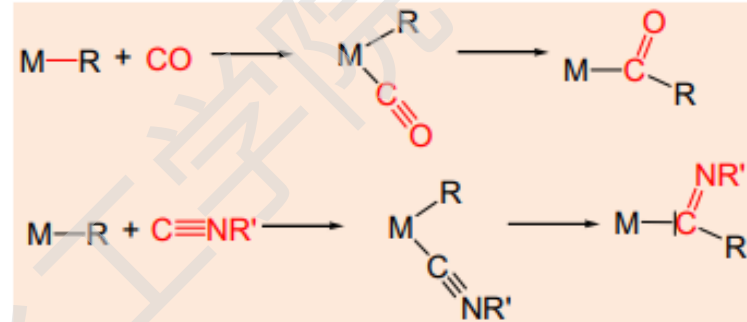
O. S. of Ta is too high



2.3 Chemical properties

E. Insertion Reaction

Examples of CO insertion

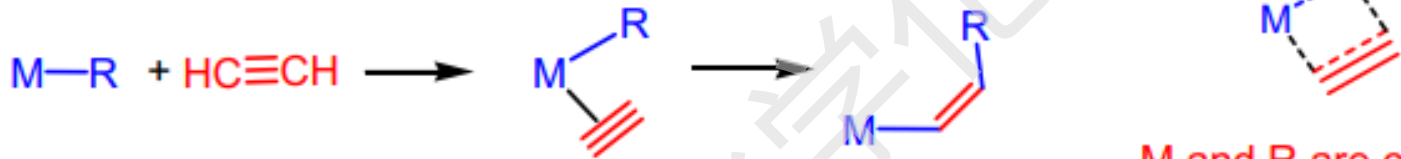
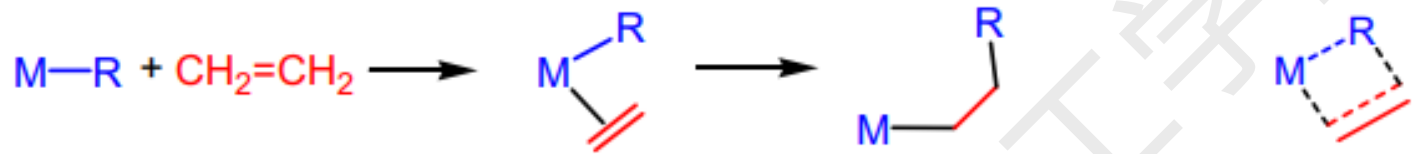


The insertion of CO into an M-C bond results in formation of an acyl ligand. This is one of the industrially important reactions of metal alkyls. Mechanistically, it involves *intramolecular nucleophilic attack* by the alkyl on the carbon of CO. ➔ **More discussion later.**



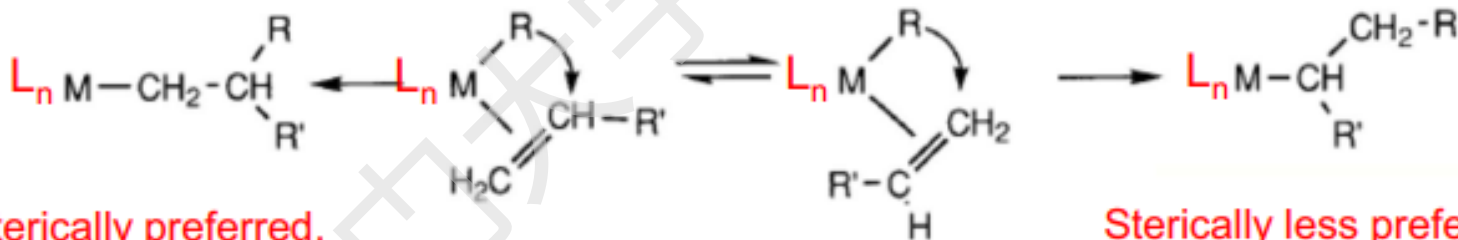
2.3 Chemical properties

Insertion of alkenes and alkynes



M and R are cis to each other!

Regioselectivity:



Sterically preferred.

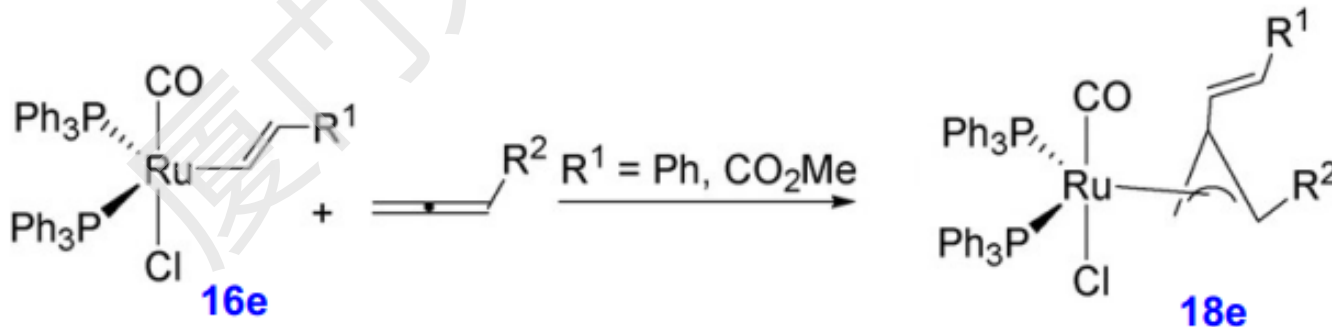
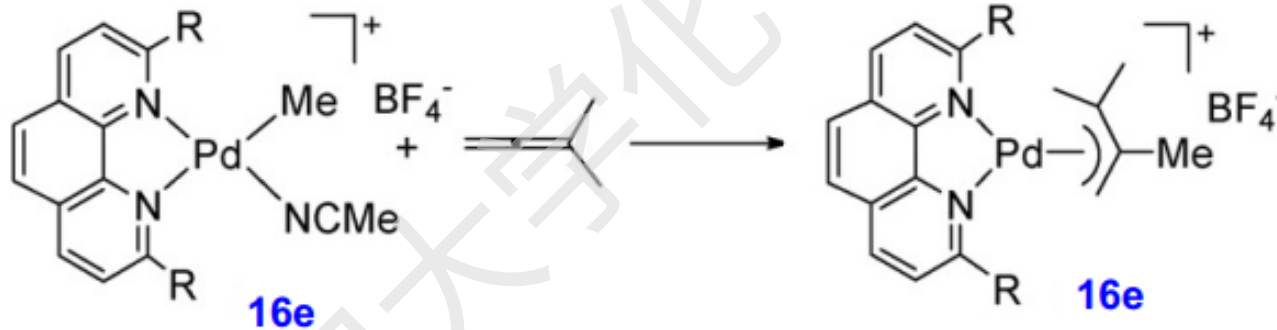
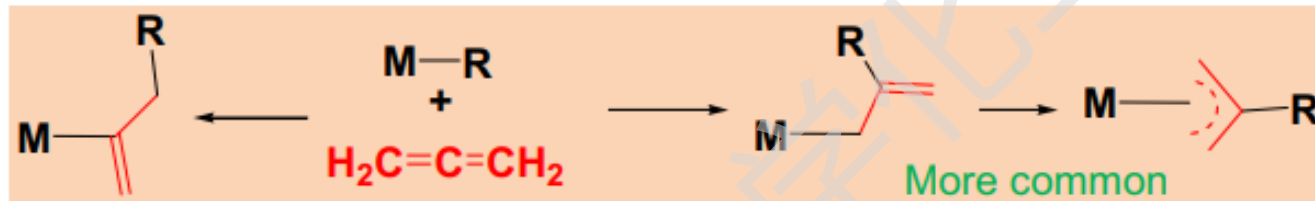
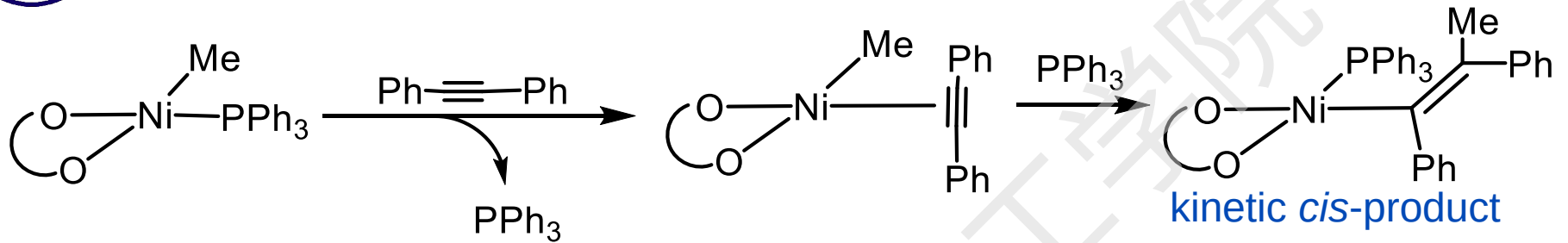
Sterically less preferred.

The preference may be reversed when R' is electron-withdrawing.

➔ More discussion later.



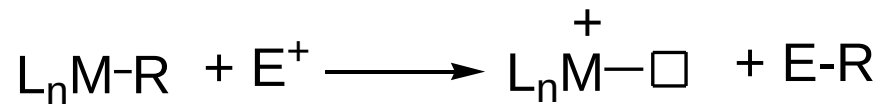
Insertion of **alkynes** and **allenes**





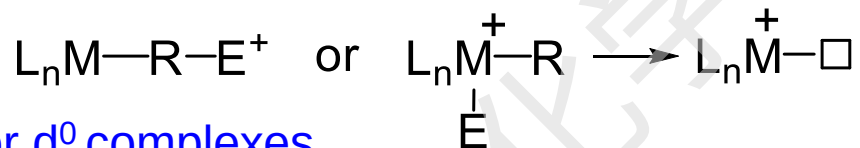
2.3 Chemical properties

F. Electrophilic attack on alkyls



($E^+ = H^+$, Br^+ (from Br_2), $HgCl^+$ (from $HgCl_2$), even $CHCl_3$ )

Two possible pathways:



For d^0 complexes

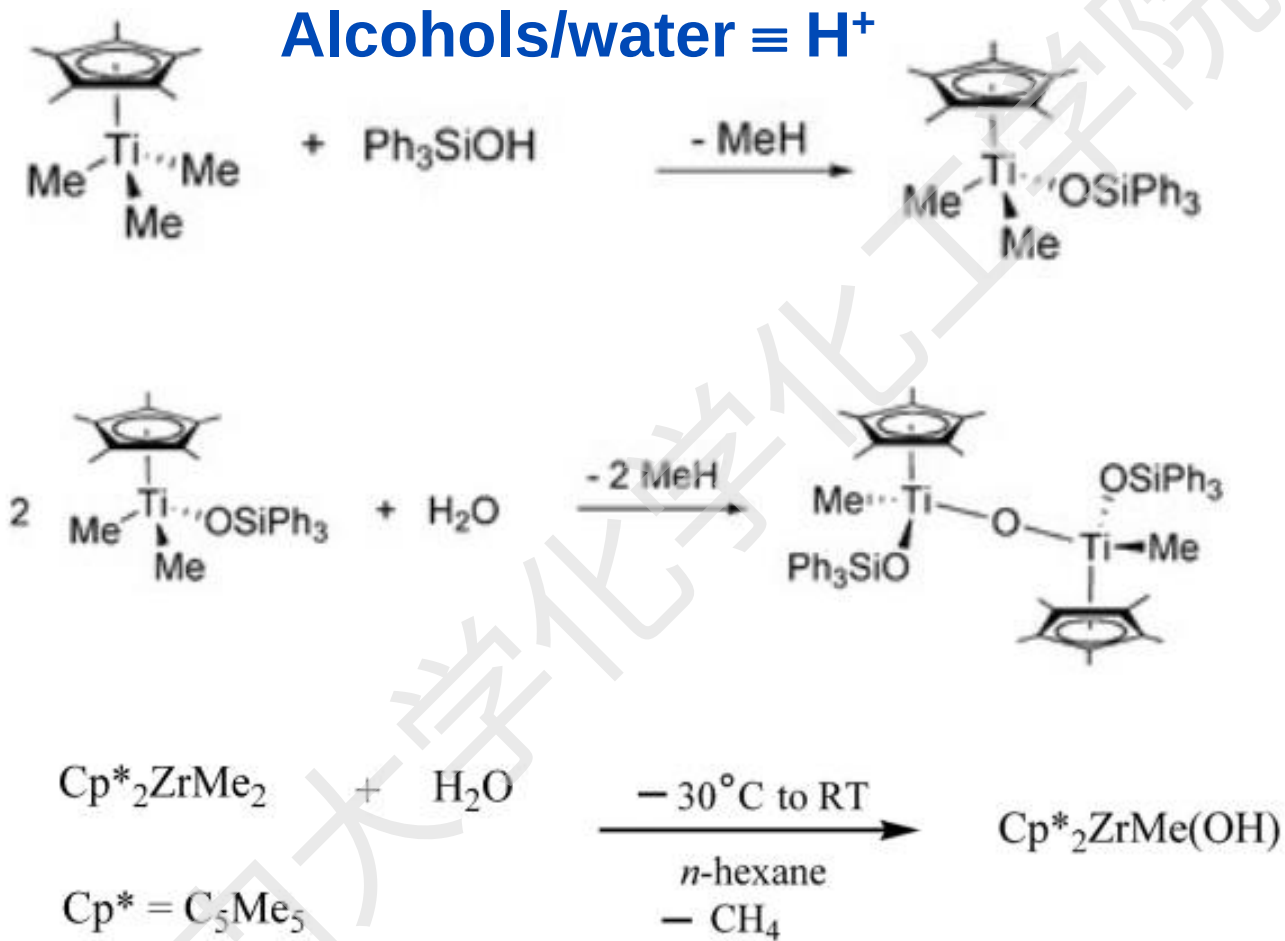
-As the metal's electronegativity increases, the nucleophilic reactivity of the M-C bond decreases.

-Nucleophilic reactivity decreases with more stable carbanions (i.e. $-CH_3 < -C_6H_5 < -C\equiv CR$).

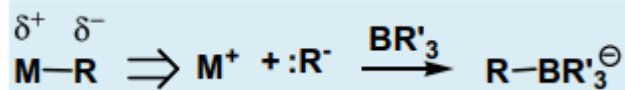
-Early transition metal alkyls are very sensitive to water/alcohol.
Late transition metal alkyls are more stable to water.



Reactions with alcohols and water

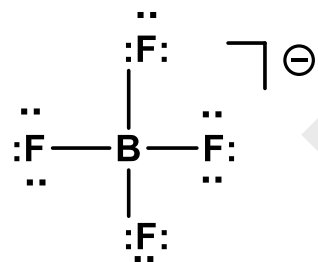
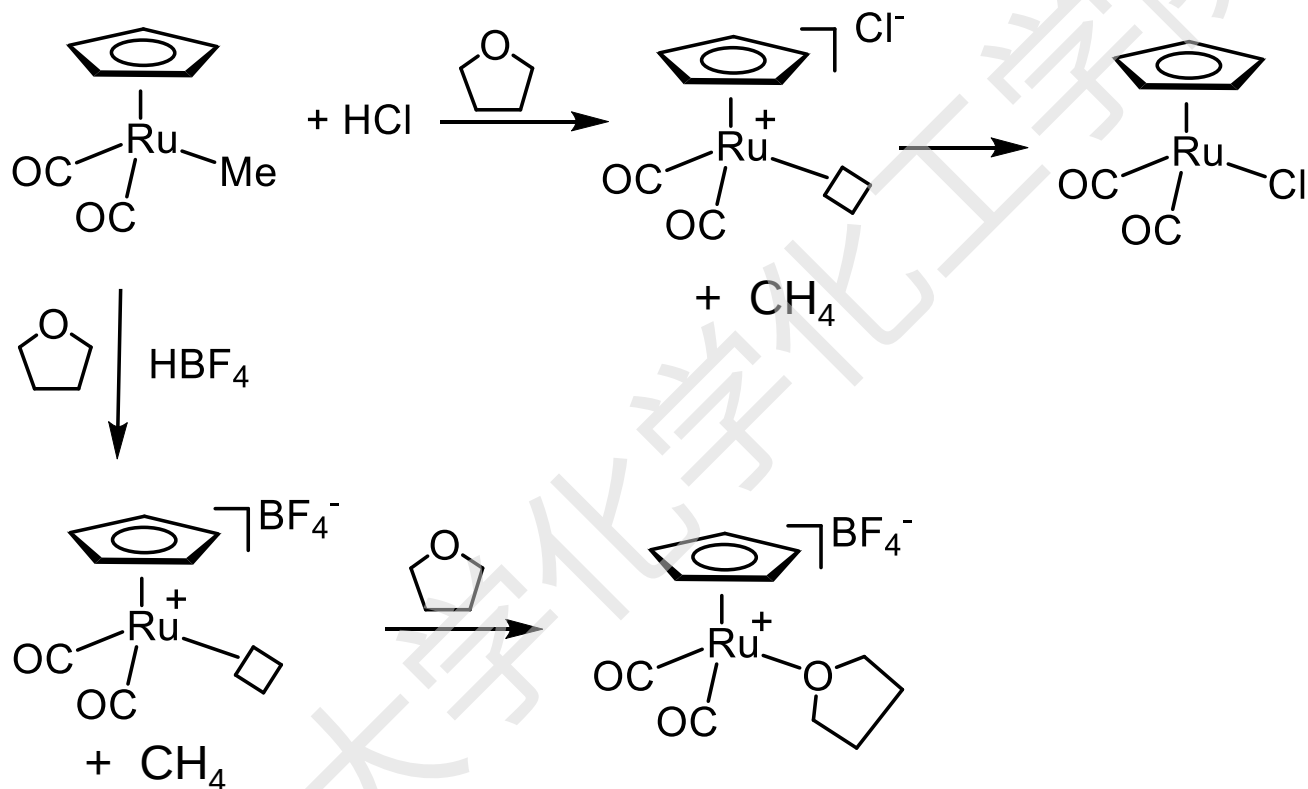


For early transition metal alkyl complexes, the M-C bond can even be cleaved by Lewis acids.





Example



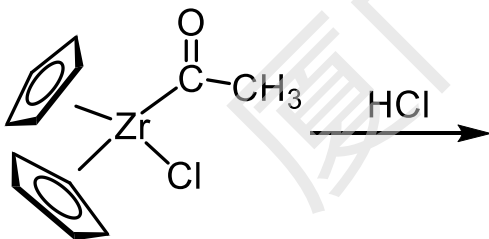
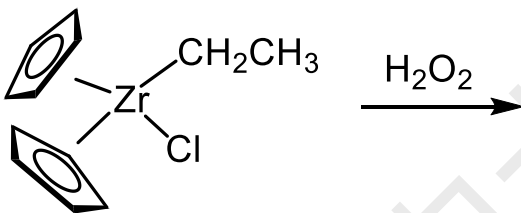
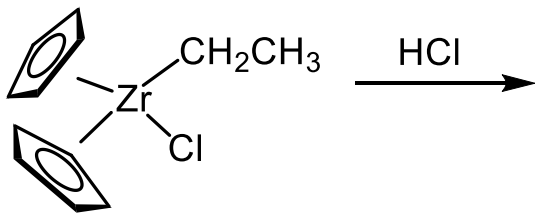
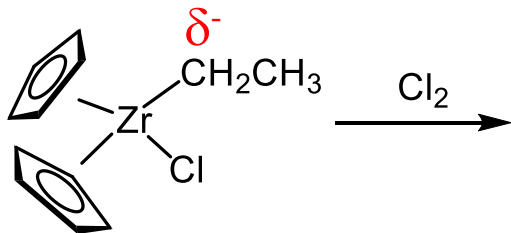
BF_4^- is a weakly coordinate anion.

Other weakly coordinate anions: BPh_4^- , PF_6^- , AsF_6^- , SbF_6^-



Practice Problems

Predict the products of the following reactions.

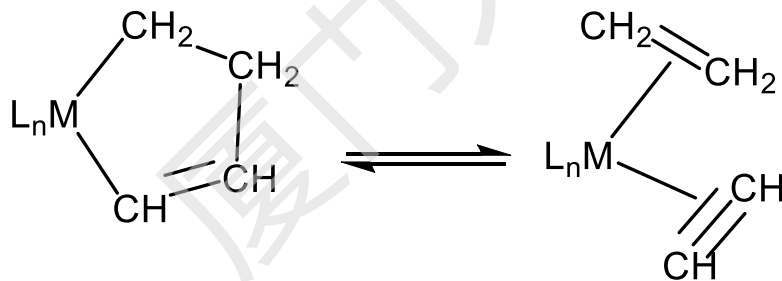
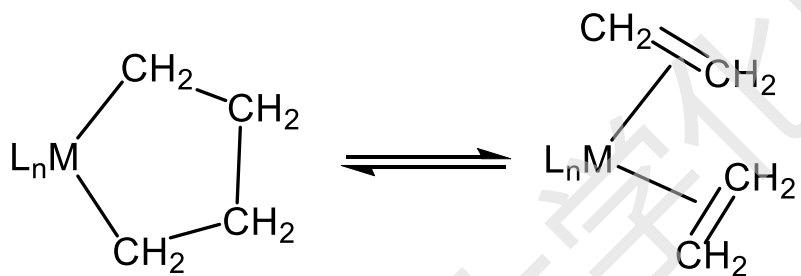
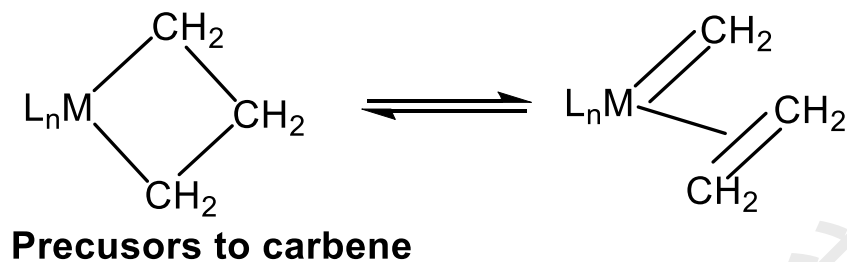




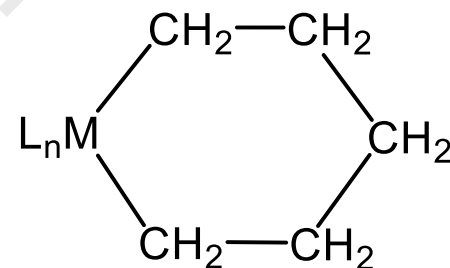
2.3 Chemical properties

G. Special properties of some M-R complexes

- Small metallacycles show interesting rearrangement.



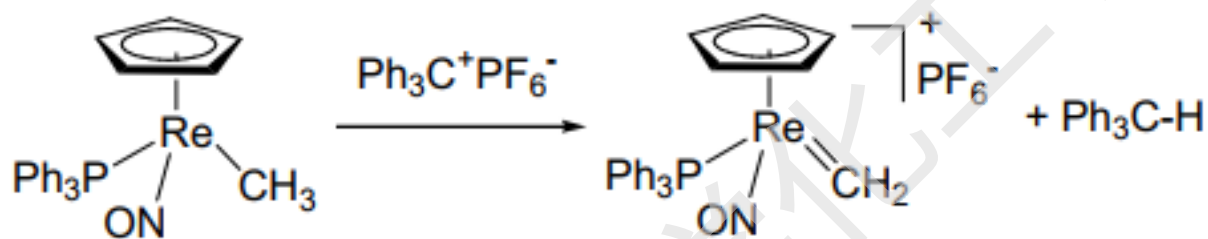
Larger metallacycles show normal properties of alkyls, e.g.





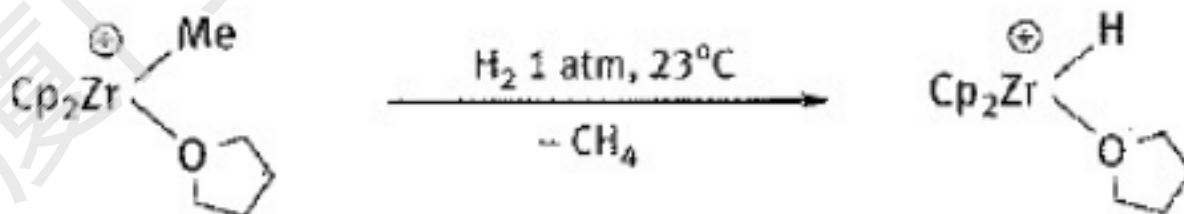
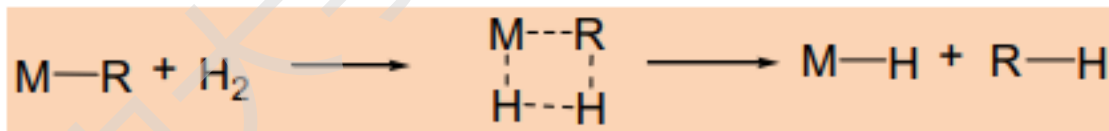
Other reactions

Electrophilic abstraction of α -H. e.g.



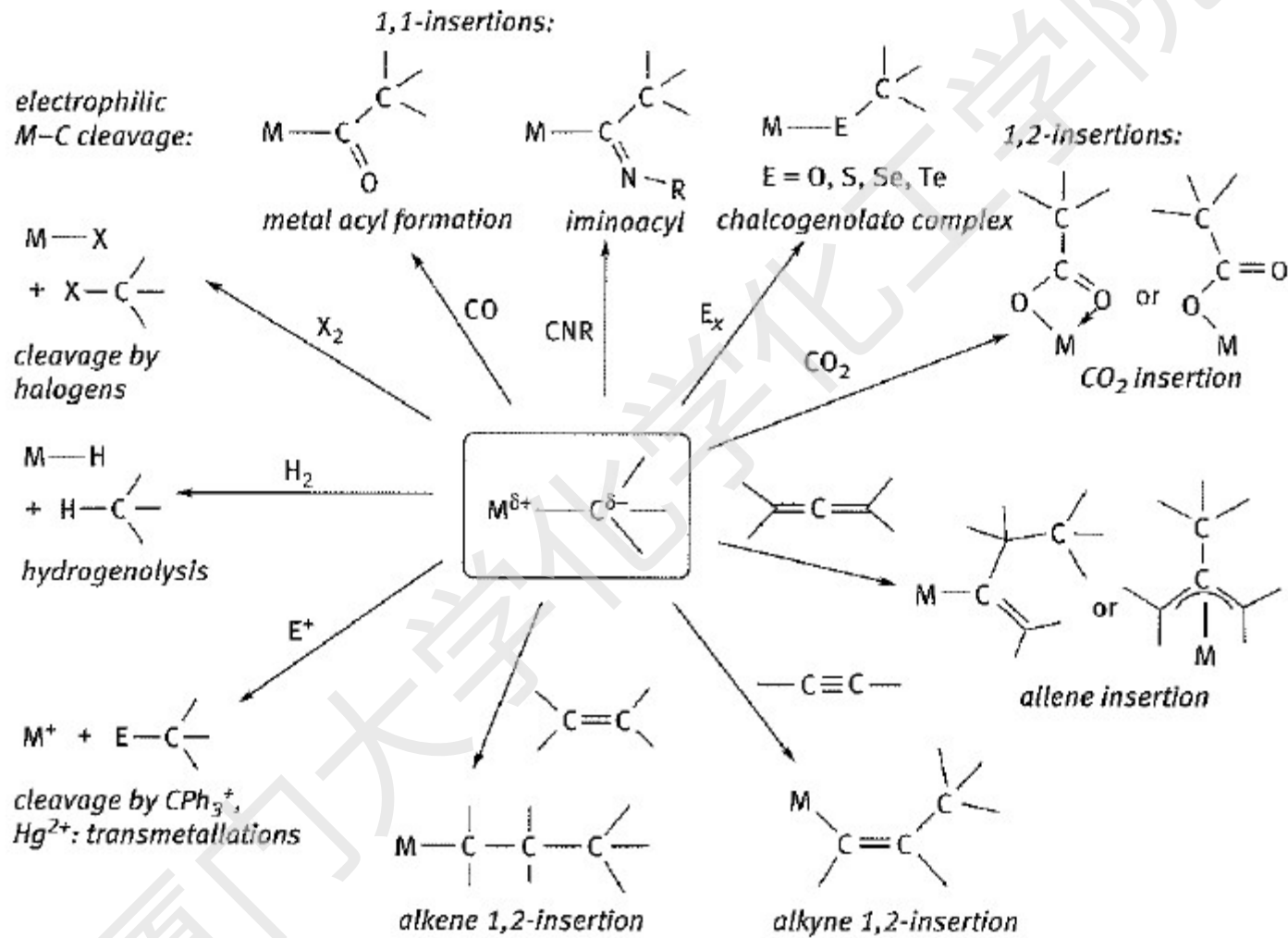
σ -Bond metathesis

✓ d^0 or high valent metal alkyls may undergo such reactions. e.g.





typical reactivity of alkyl complexes





4. Complexes with related σ -bonded ligand complexes

A. $M-AR_3$ (A = Si, Ge, Sn)

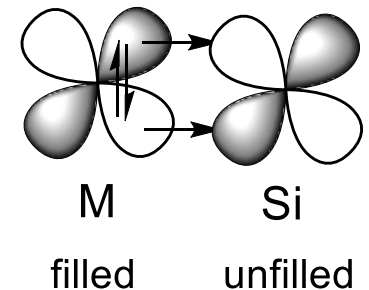
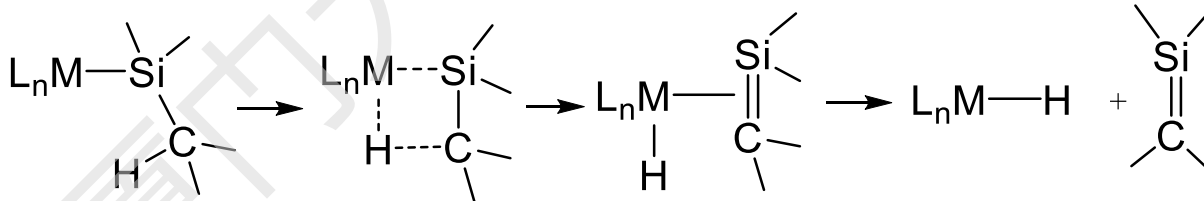
- M-Si bond are longer than M-C
- $BDE(M-SiR_3)$ may be stronger than $BDE(M-CR_3)$

Reasons?

- a) some π -interaction (Si has empty d orbitals)
- b) $M-SiR_3$ is less sterically congested.

AR_3^- (A = Si, Ge, Sn) are π acceptors

- Can $L_nM-SiCH_2R$ undergo β -H elimination?



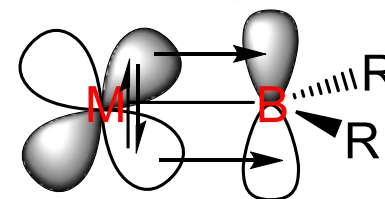
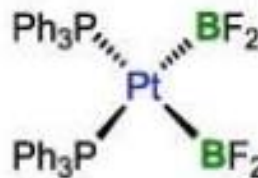
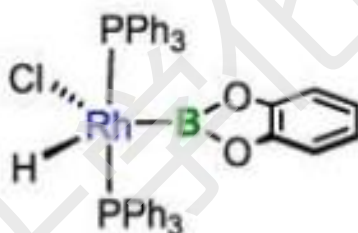
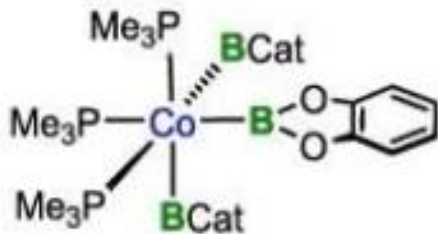
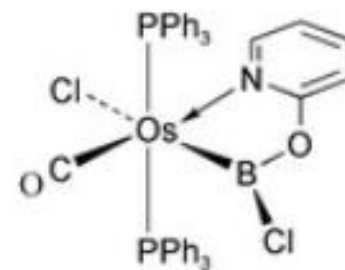
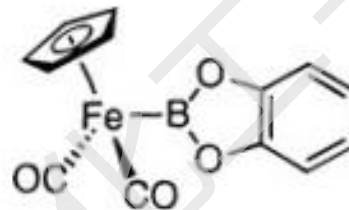
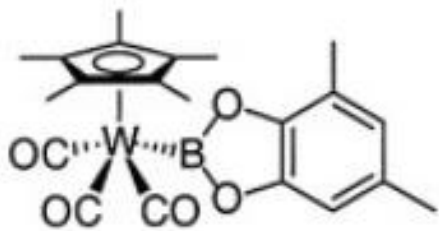
Yes. But more difficult than M-R since $Si=C$ bond is weaker than $C=C$.

In general, $M-SiR_3$ has properties similar to $M-CR_3$.



B. M-BR₂ complexes

- Examples:



filled

unfilled

Is BR₂ a π donor or a π acceptors ?

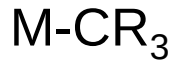
π acceptors

M-BR₂ have properties similar to M-CR₃.

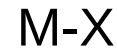


C. M-OR, M-NR₂, M-F, M-Cl, etc.

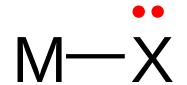
(1) Difference between M-CR₃ and M-X



no $d\pi$ - $p\pi$ interaction
(or weak)

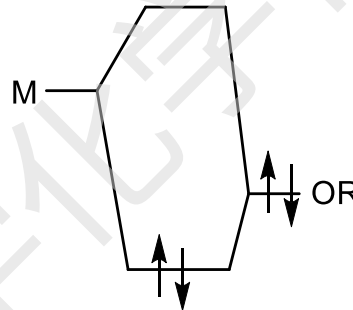
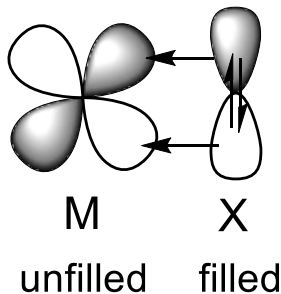


possible $d\pi$ - $p\pi$ interaction



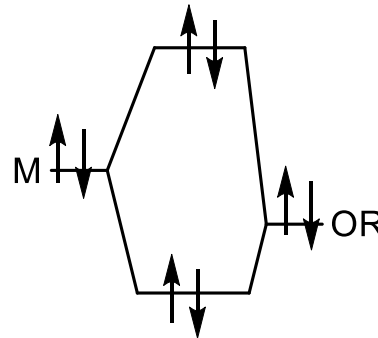
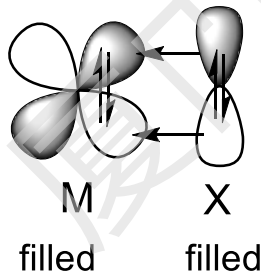
π donors

- M-X bond is **stabilized** for **16e or less** species (early TM)



Early TM forms stronger
bonds with such ligands

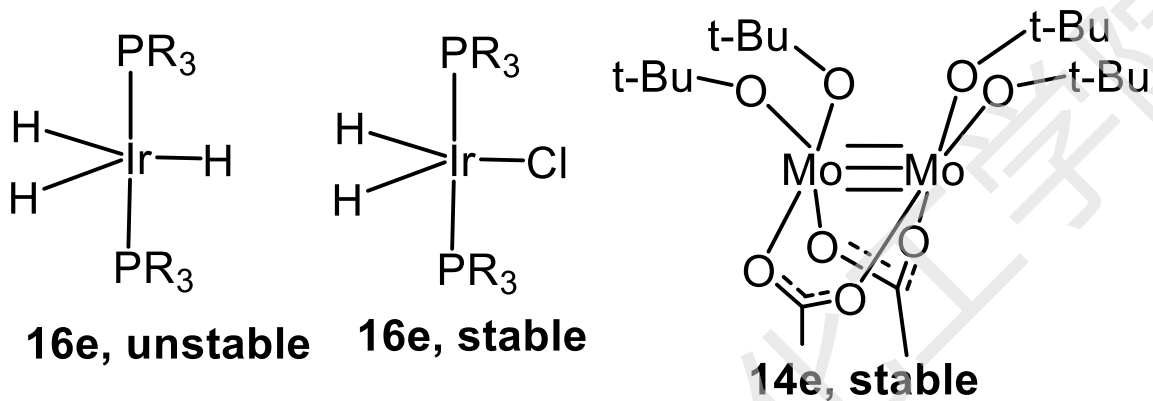
- M-X bond is **destabilized** for **18e** species



Late TM forms weaker
bonds with such ligands

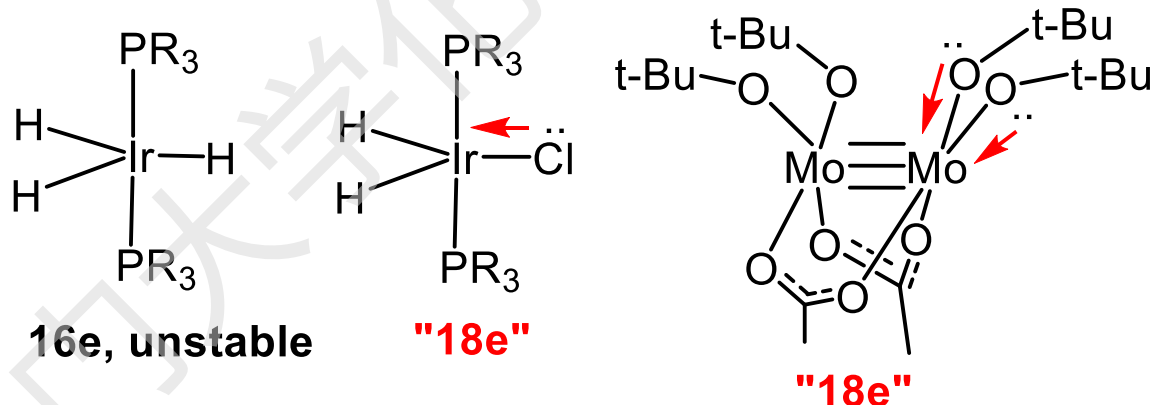


M-X can stabilize complexes with less than 18e



Why?

X (π donors) can stabilize complexes with less than 18e



π bonding can make these complexes to **formally have 18e**.

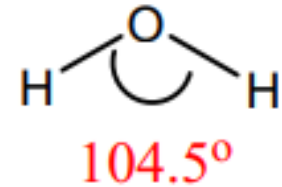
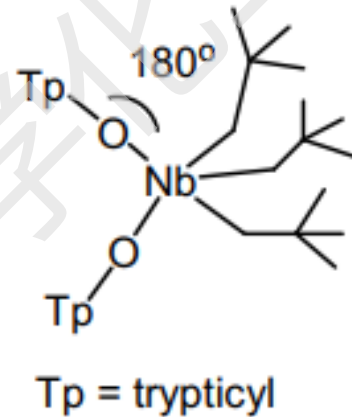
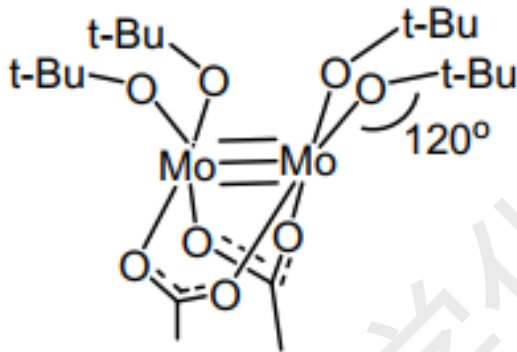
Note: π bonding is normally not included in valence electron counting. 65



Why?

With high valent transition metals,
 π -donation becomes important.

- In H₂O, O atom has sp^3 hybridization, leading to the A-O-B angle being about 104.5 °
- However, M-O-C angles approaching 120 or even 180 ° are common



If no π -bonding: 14e

10e

With π -bonding: 18e

18e

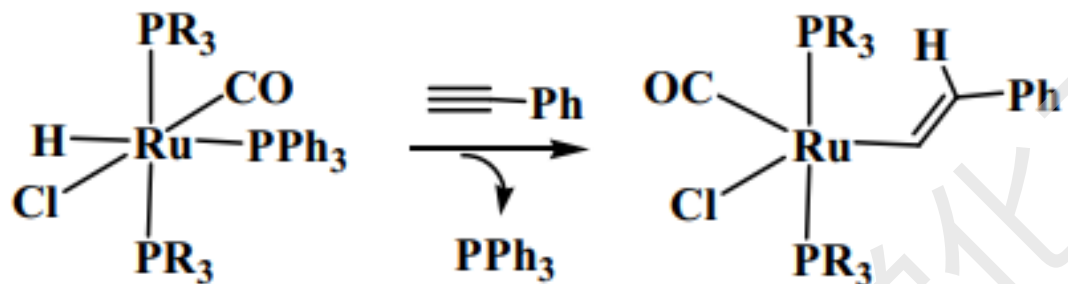
The difference can be explained in term of π bonding

In order to achieve formal 18e configuration,
O takes sp^2 or sp hybridization

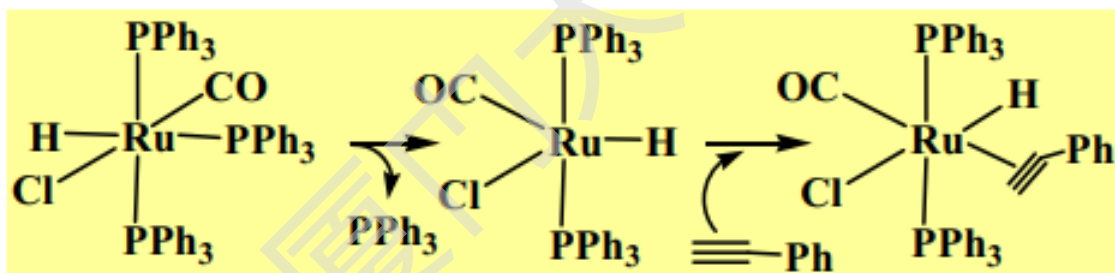
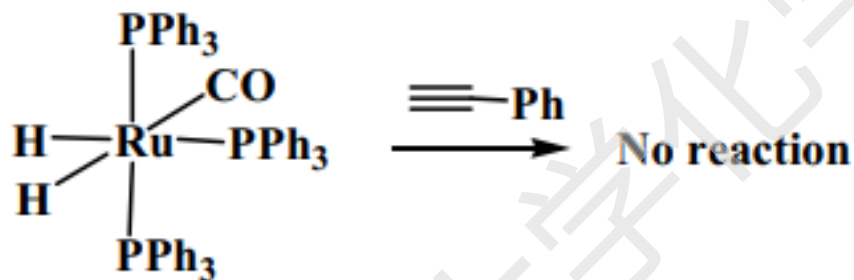


In chemical reactivity

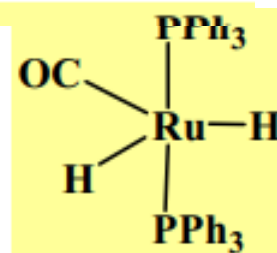
π -donors can help to generate reactive 16e species. e.g.



Why?



relatively easy

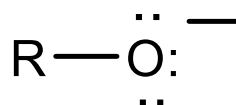
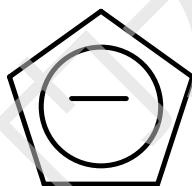
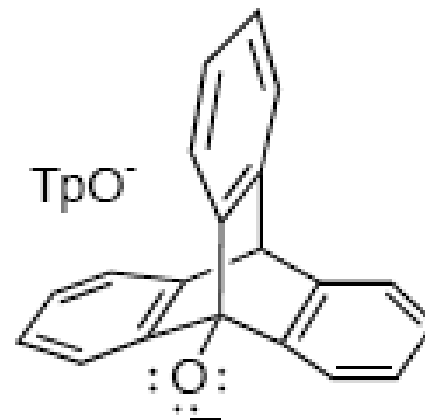
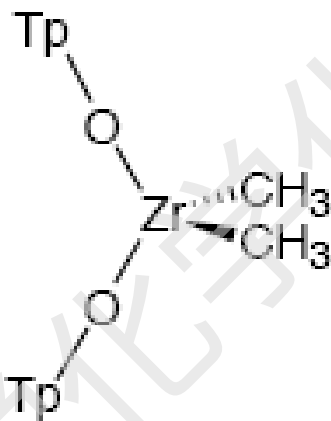
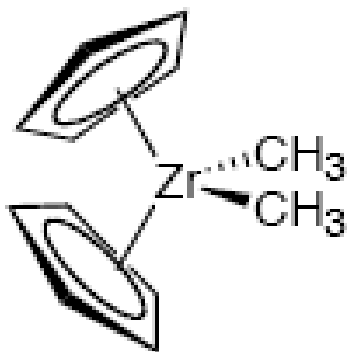


difficult to be generated



Linear alkoxide

Linear alkoxide have sometime be considered to be isoelectronic with the Cp ligand.



Ionic model

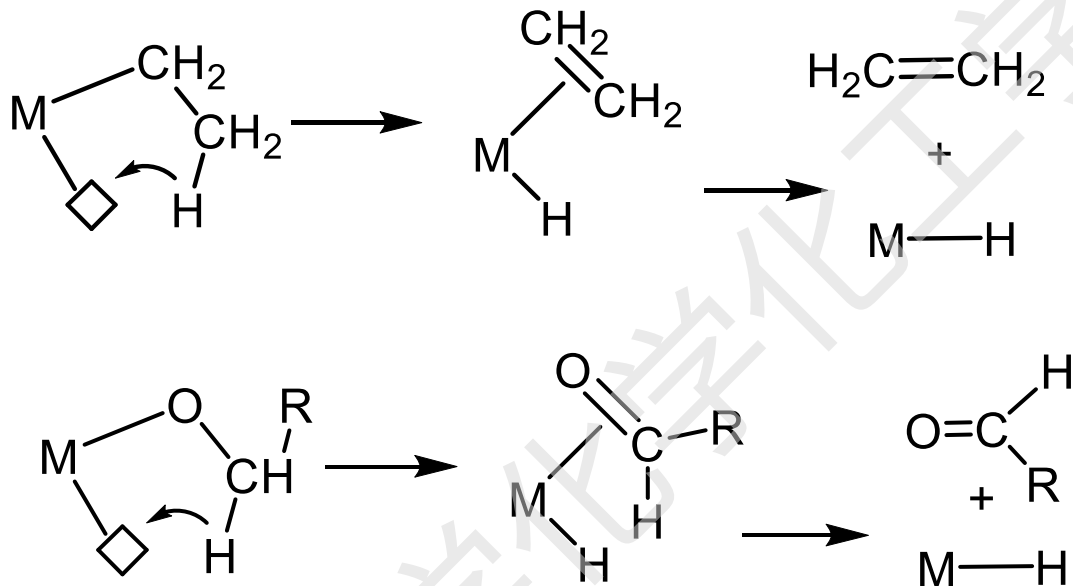
6e-

6e-, if π bonding

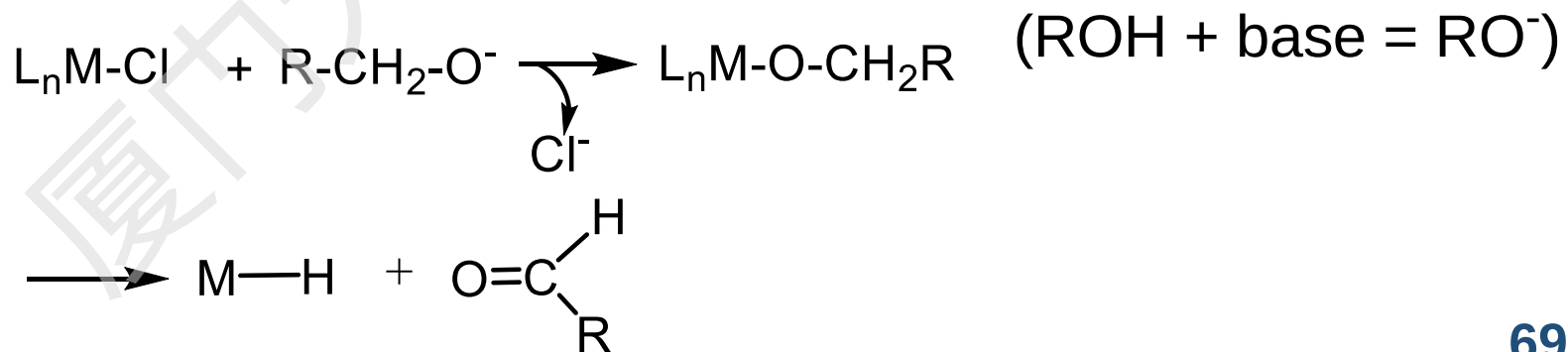


(2) Similarity between $M-CR_3$ and $M-OR$

Both can undergo β -H elimination.



Use of the reaction:





5. Metal hydrides M-H

History

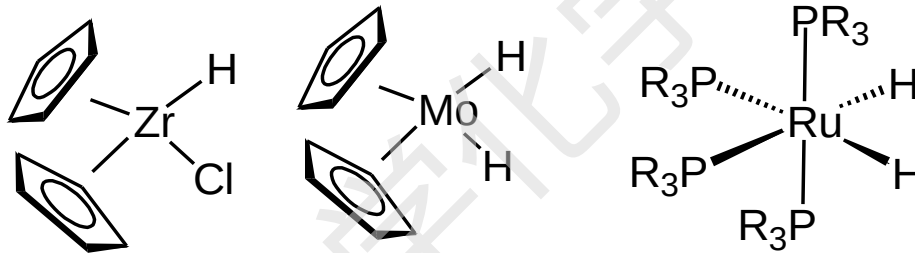
1931: $\text{H}_2\text{Fe}(\text{CO})_4$ prepared by Hieber, although the structure was not accepted initially.

1955-1964: Cp_2ReH , $(\text{PR}_3)_2\text{PtHCl}$, and $\text{K}_2[\text{ReH}_9]$ prepared showing that M-H bonds do exist.

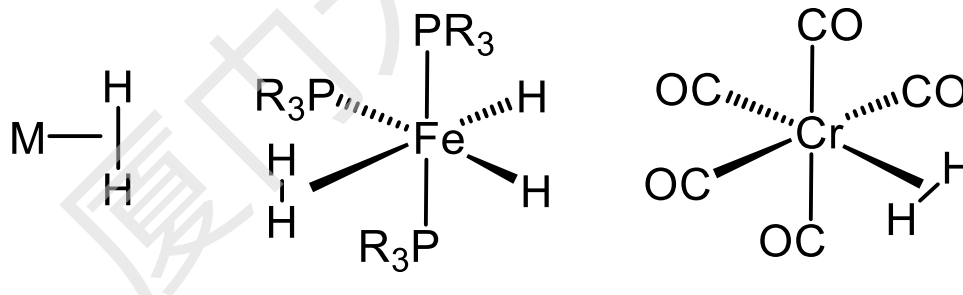
1984: Dihydrogen complexes discovered ($\text{M}-\text{H}_2$).

A. Structural types of metal hydrides.

(1) Terminal hydride or classical hydride, e.g.



(2) Non-classical hydride or molecular hydrogen complexes



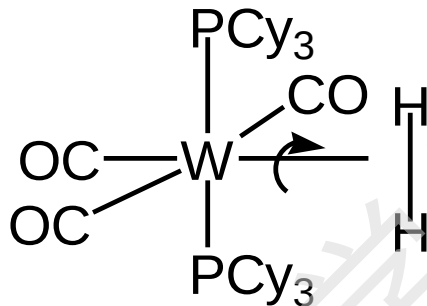
H-H	0.74 Å
M-H	0.8-1.3 Å
M-H ₂	>1.5 Å

Molecular hydrogen complexes are also called $\sigma\text{-H}_2$ complexes. 70

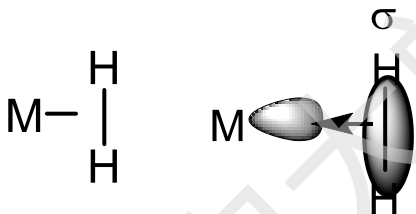


Practice Problems

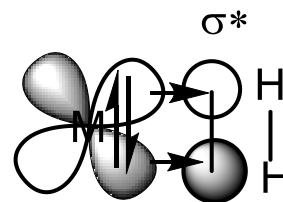
1. Would you expect the H_2 molecule in the following complex can rotate freely? **why?**



Consider the bonding between M and H_2 .



1. e^- from $\sigma(\text{H}_2) \rightarrow \sigma(\text{M})$



2. e^- from $d\pi(\text{M}) \rightarrow \sigma^*(\text{H}_2)$

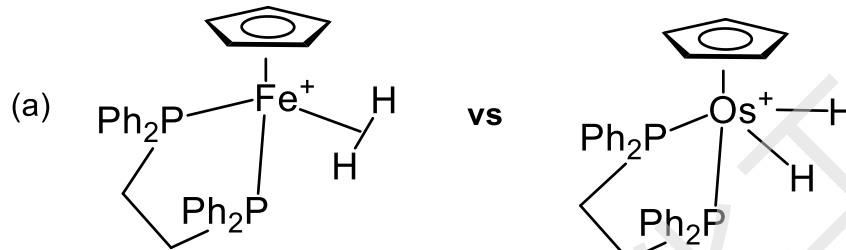
No.

Rotation will break the π bond between M and H_2

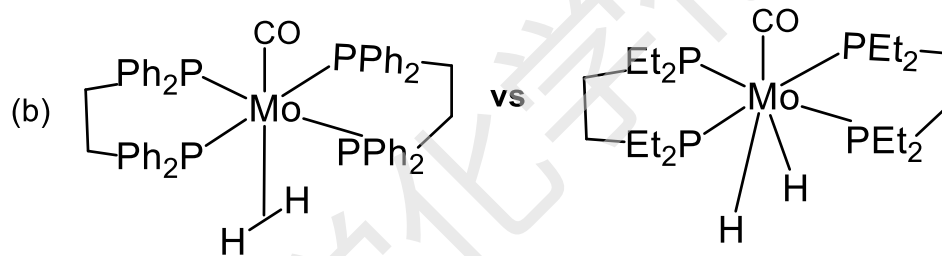


Practice Problems

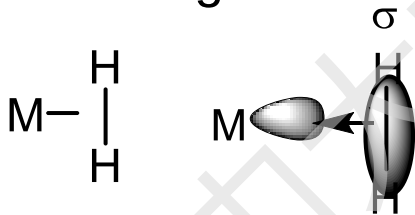
2. L_nMH_2 can have two forms, depending on M and L.



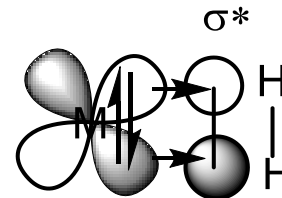
Why?



In terms of bonding:



1. e^- from $\sigma(H_2) \rightarrow \sigma(M)$



2. e^- from $d\pi(M) \rightarrow \sigma^*(H_2)$

H-H bond will break when there is excessive backdonation from M to $\sigma^*(H_2)$.

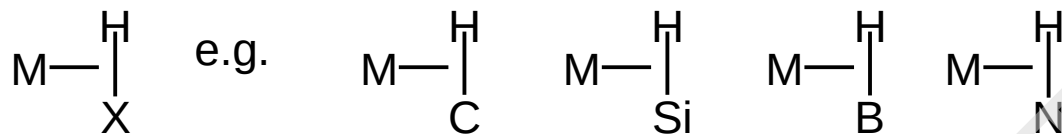
For (a), Os has stronger backdonating ability than Fe

For (b), Et is more electron-releasing than Ph, leading to stronger backdonation.



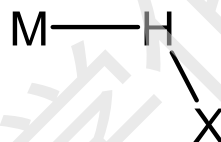
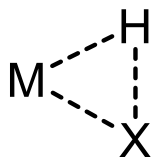
A. Structural types of metal hydrides.

(3) H-X σ complexes

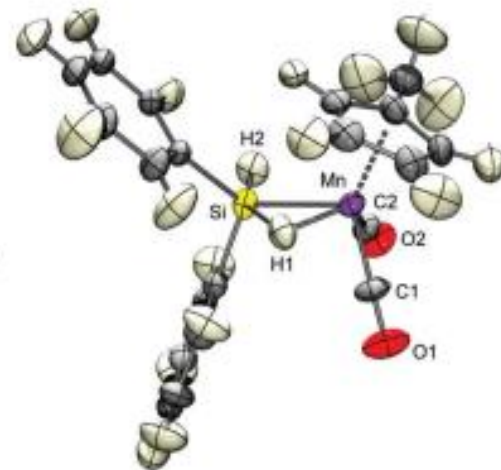
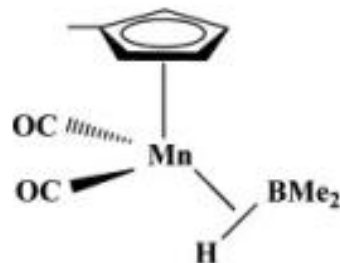
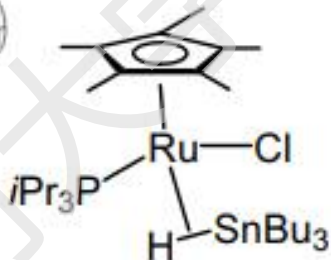
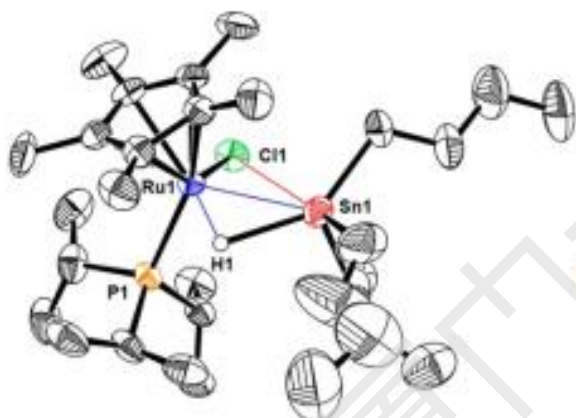


In terms of bonding:
similar to $M(\eta^2-H_2)$

Other representations:



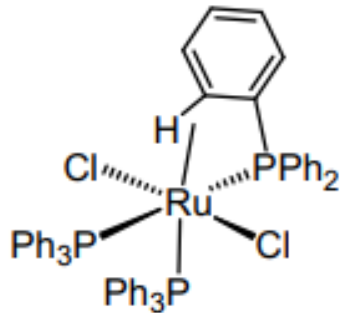
Examples of H-X σ -complexes



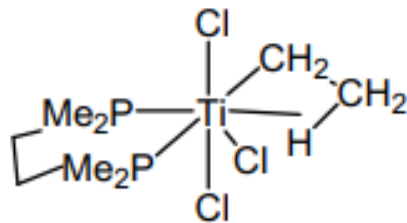
J. Am. Chem. Soc. **2015**, 137, 16, 5506



Additional examples

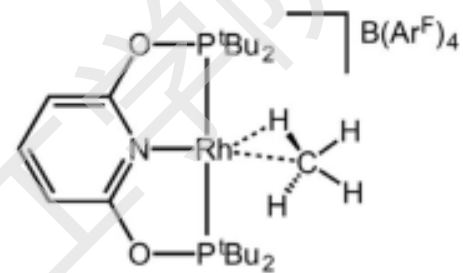


agostic

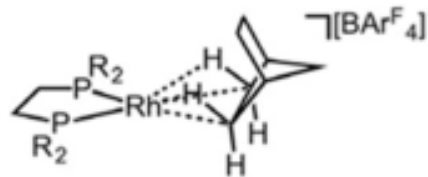


agostic

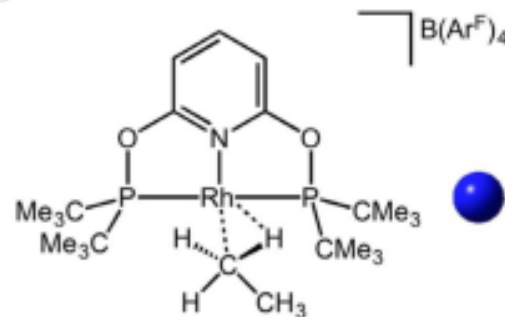
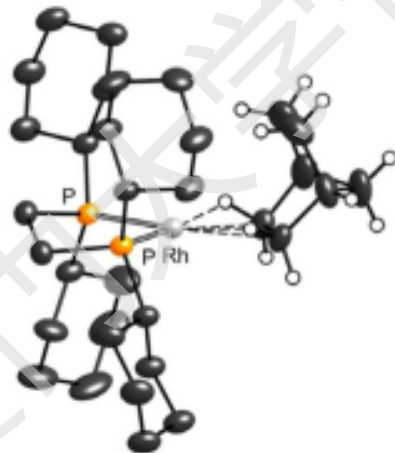
Observed in both solution and solid states.



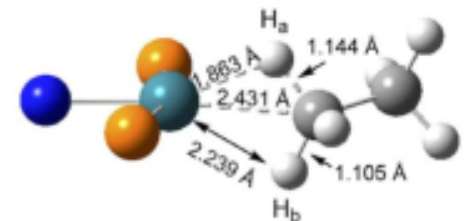
Observed at 163 K in CDFCl_2 by NMR. Science, 2009, 326, 553.



Observed in the solid-state. J. Am. Chem. Soc. 2015, 137, 820.



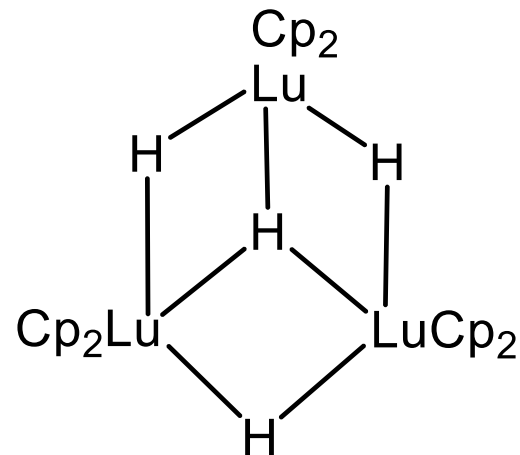
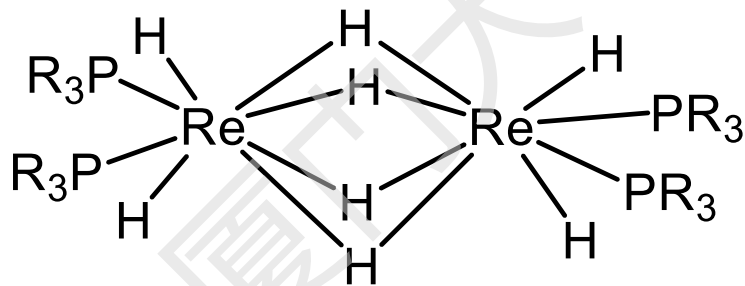
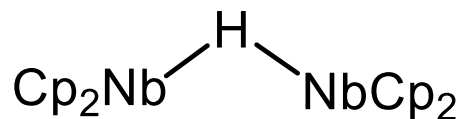
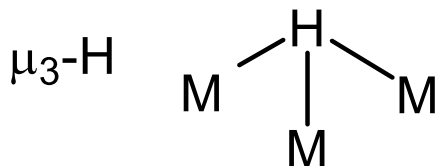
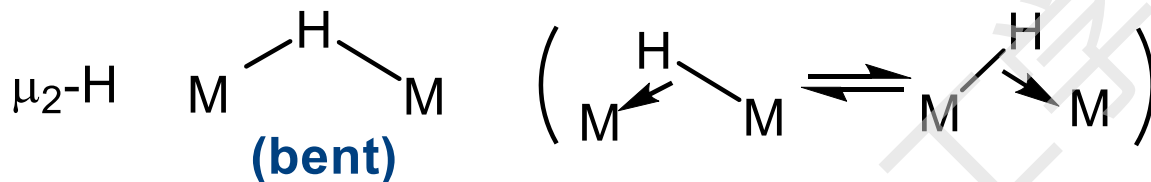
Observed at 123 K in $\text{CD}_2\text{Cl}_2/\text{CDFCl}_2$ by NMR. J. Am. Chem. Soc. 2013, 135, 15933.





A. Structural types of metal hydrides.

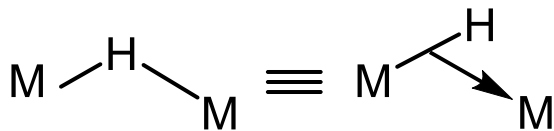
(4) Bridging hydrides



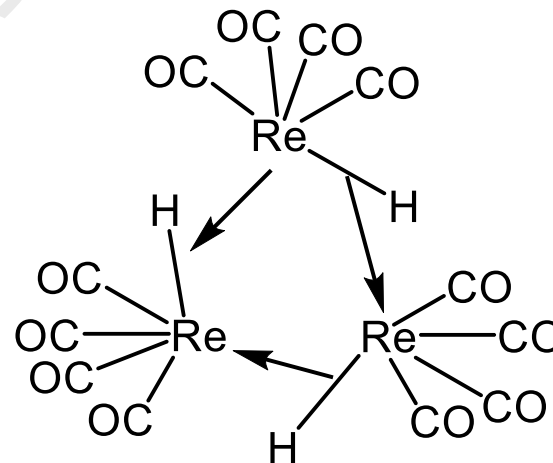
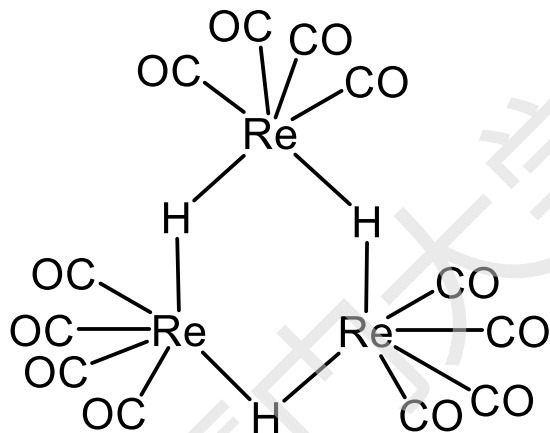


Practice Problems

For electron counting purpose, we can use the resonance structures like:



18e or not?



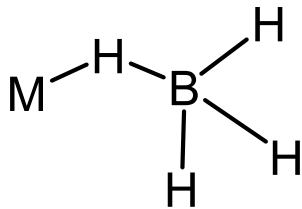
$$7(\text{Re}) + 4 \times 2 (\text{CO}) + 1 (\text{H}) + 2 (\text{Re-H}) = 18$$



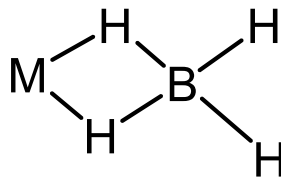
(5) BH_4^- , BH_3L , and AlH_4^- complexes.

A variety structures are possible.

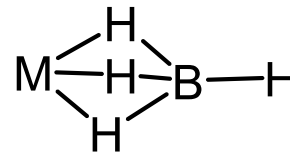
e.g. for BH_4^-



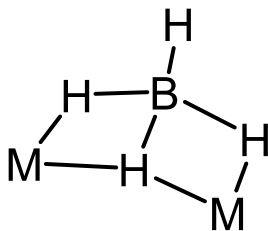
$\eta^1\text{-BH}_4$



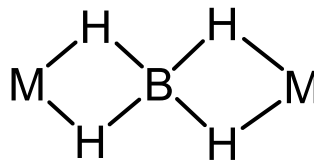
$\eta^2\text{-BH}_4$



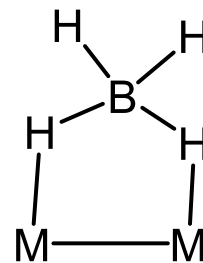
$\eta^3\text{-BH}_4$



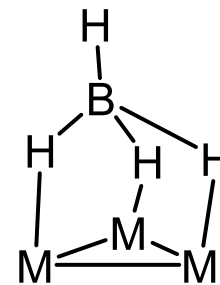
μ_2, η^3



μ_2, η^4



μ_2, η^2



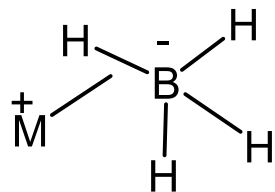
μ_3, η^3



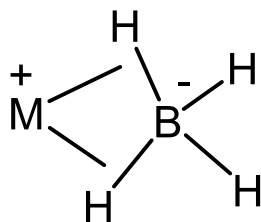
(5) BH_4^- , BH_3L , and AlH_4^- complexes.

For electron counting purpose:

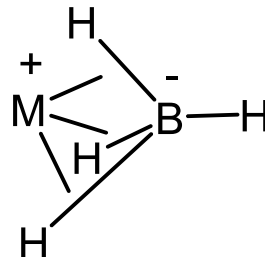
A variety structures are possible



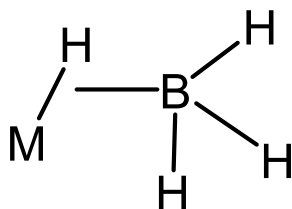
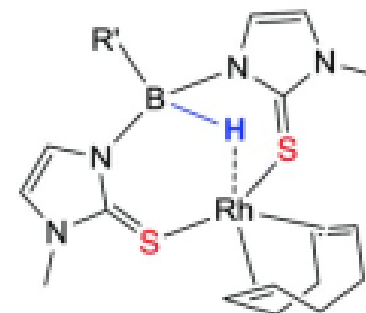
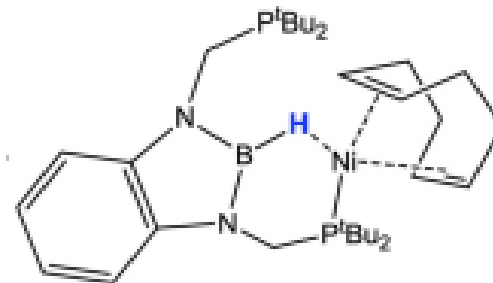
2e



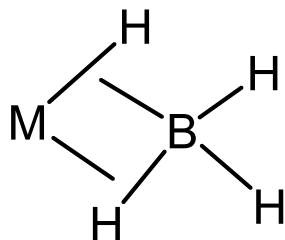
4e



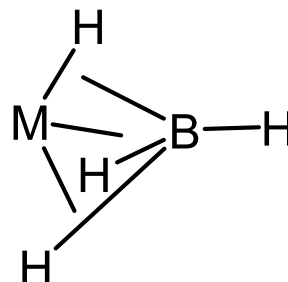
6e



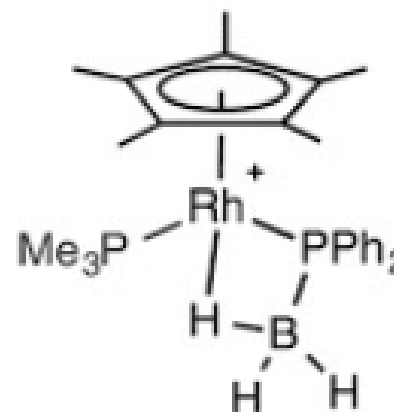
1e



3e



5e

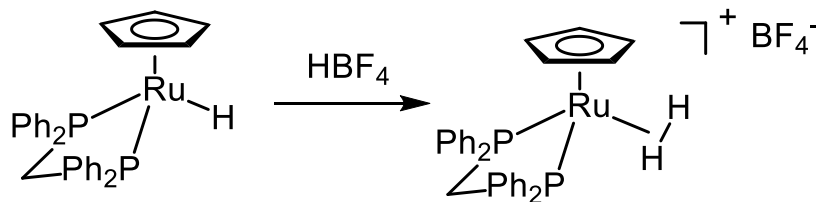
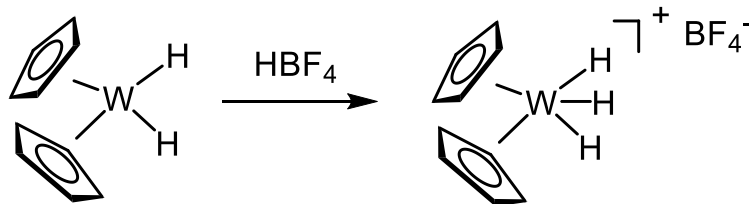
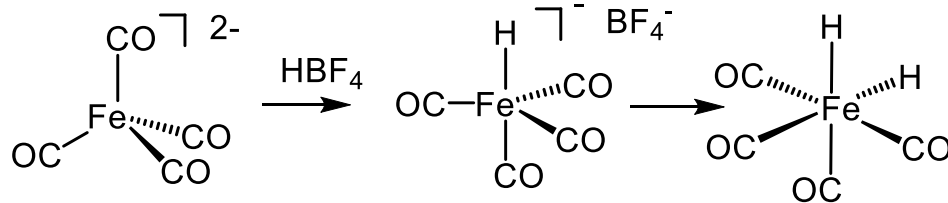
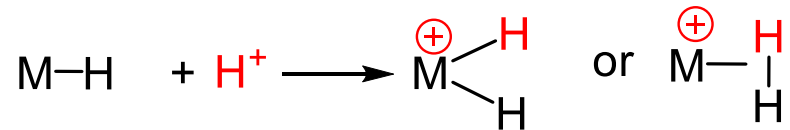
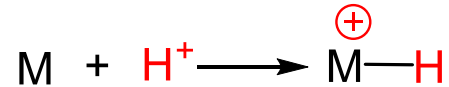




B. Synthesis of metal hydride complexes

(1) By protonation

Sources of H: H^+ , H^- , H_2 , ligands



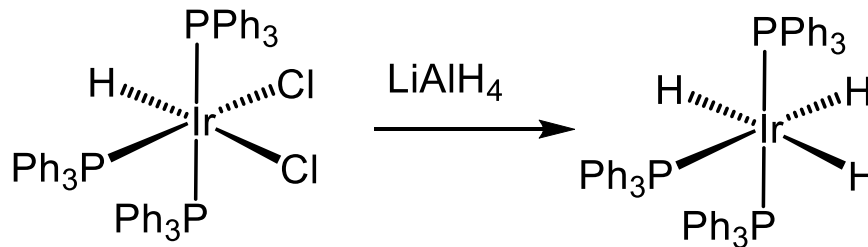


B. Synthesis of metal hydride complexes

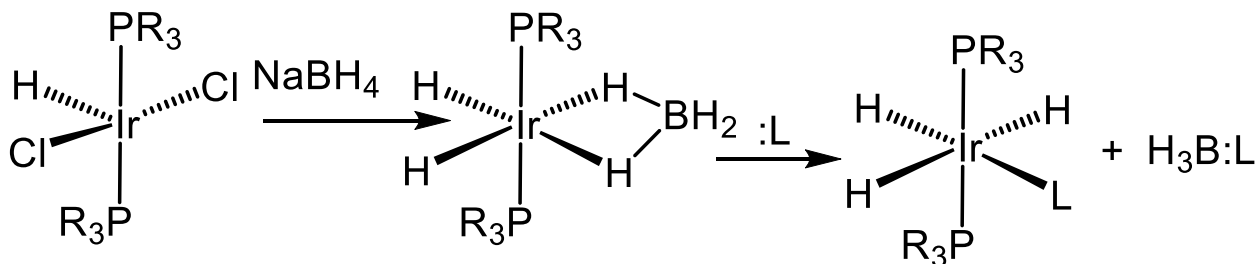
(2) From hydride donors



Typical H^- donors: NaH, $LiAlH_4$, $NaBH_4$, $LiBHEt_3$...



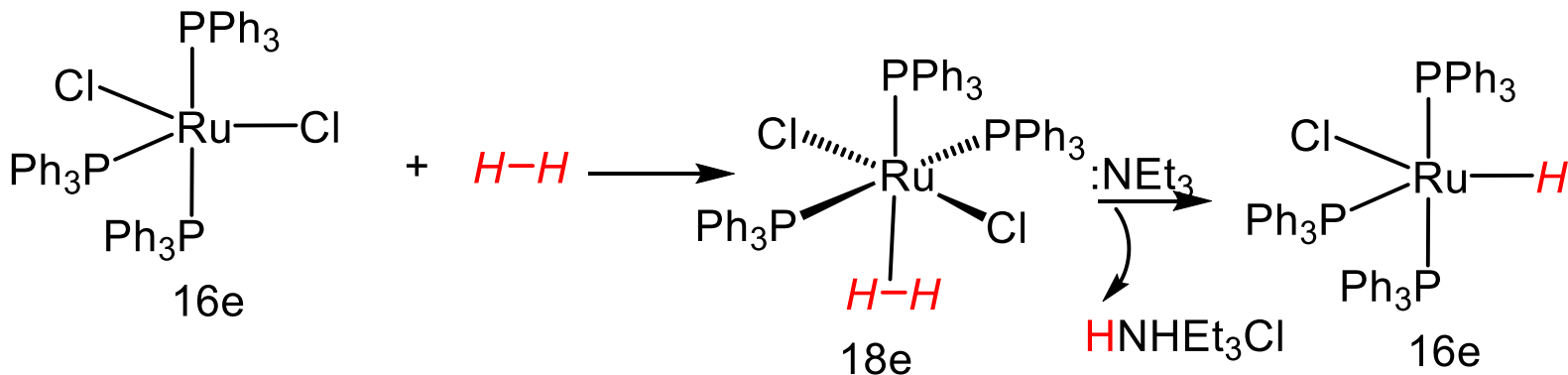
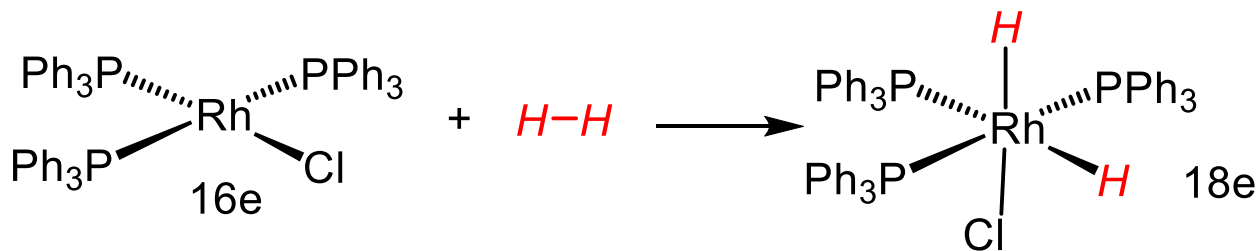
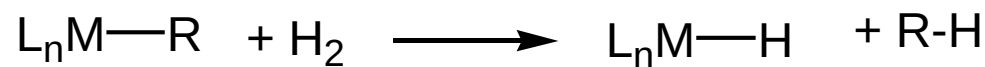
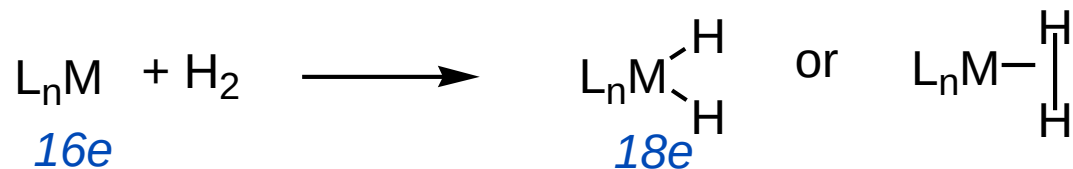
In some cases, complexes with BH_4 or AlH_4 entities may be obtained when $NaBH_4$ or $LiAlH_4$ are used.





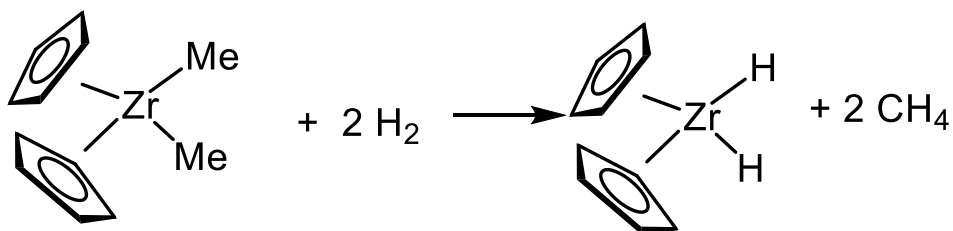
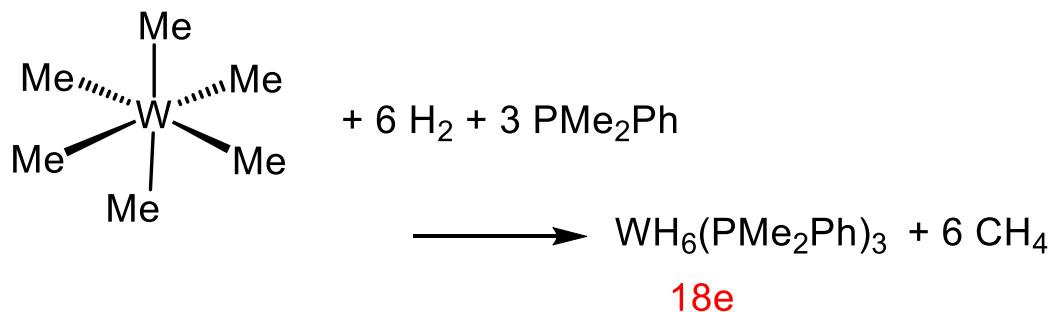
B. Synthesis of metal hydride complexes

(3) From H_2

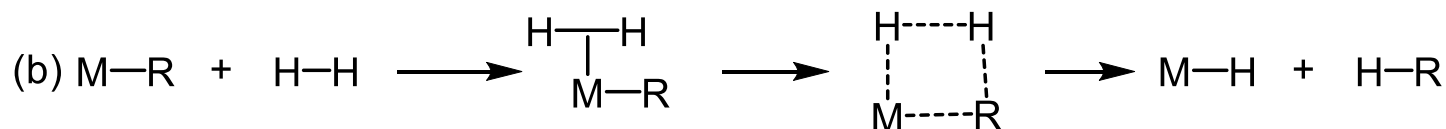
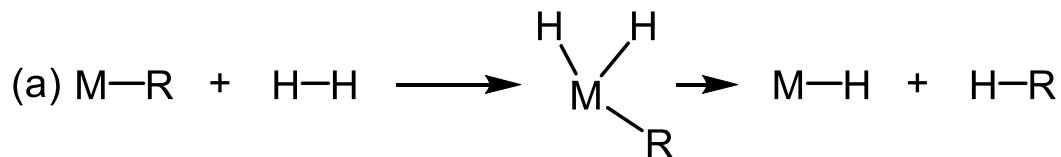




Additional Examples



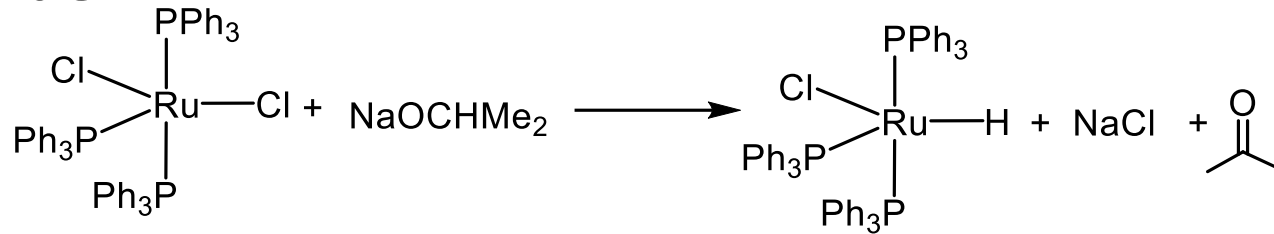
What are the likely intermediates or transition states?





B. Synthesis of metal hydride complexes

(4) From ligands



(a) Mechanism?

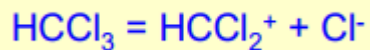
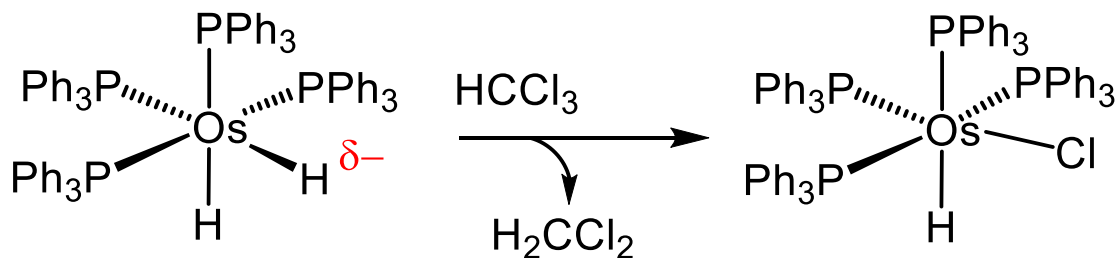
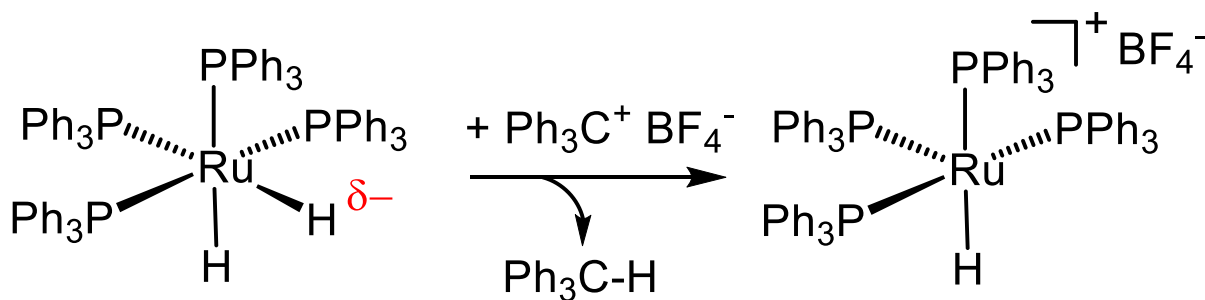
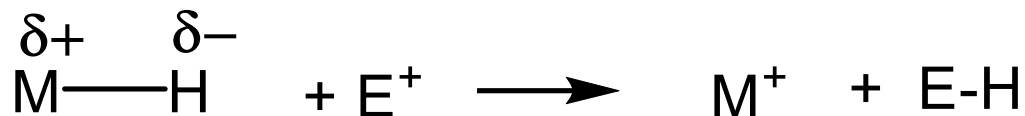
(b) Can you use NaOH, NaOMe, NaOEt?

(c) If NaOCDMe₂, or NaOCH(CD₃)₂ were used, what is the product?



C. Chemical properties of hydride complexes

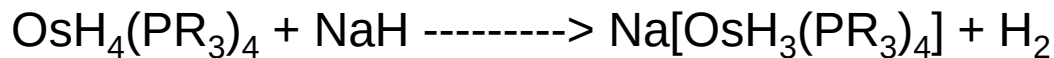
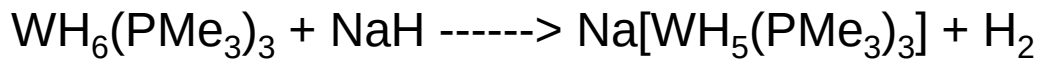
(1) Reactions with electrophiles.



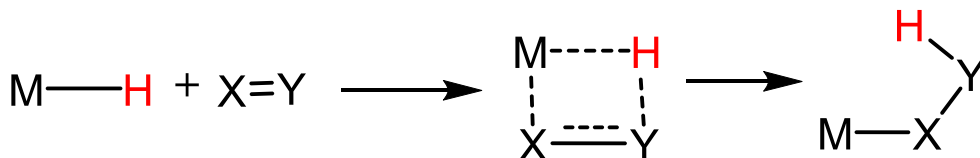
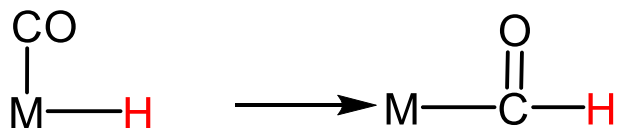
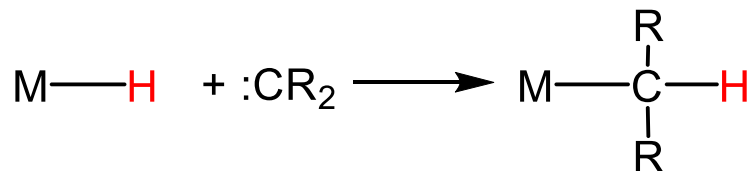


C. Chemical properties of hydride complexes

(2) Reactions with bases

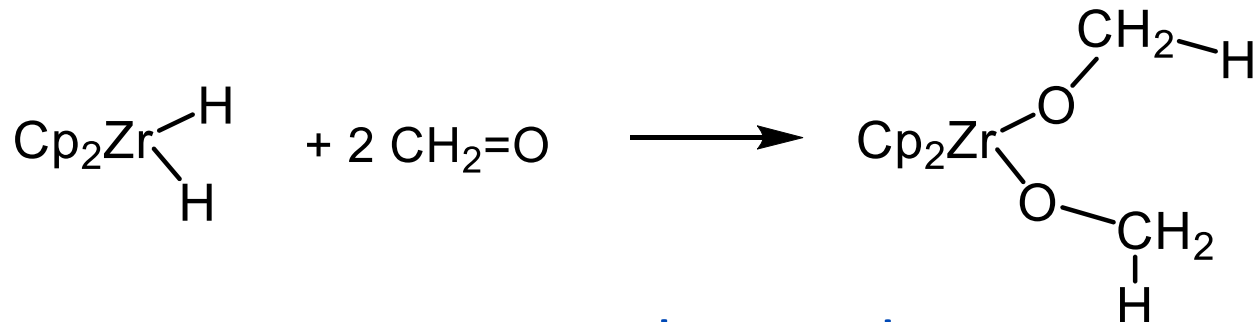
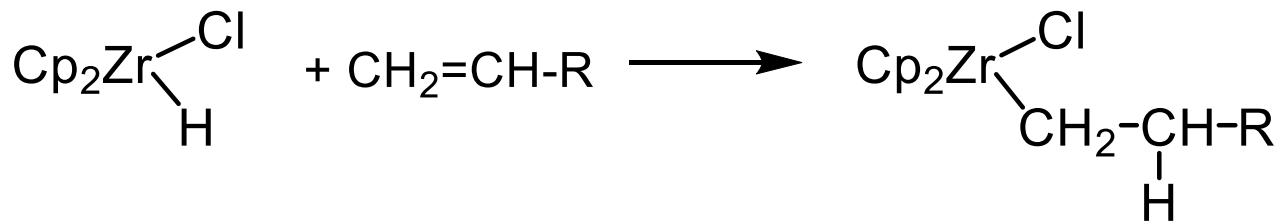
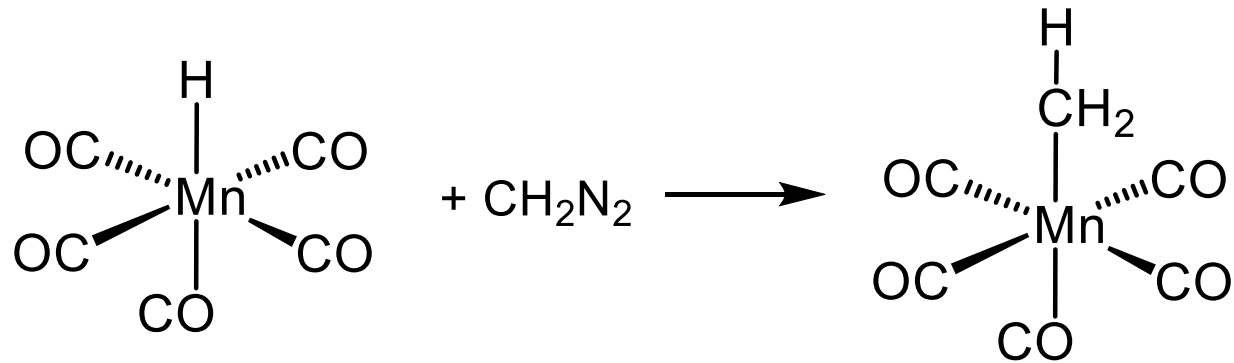


(3) Insertion reactions





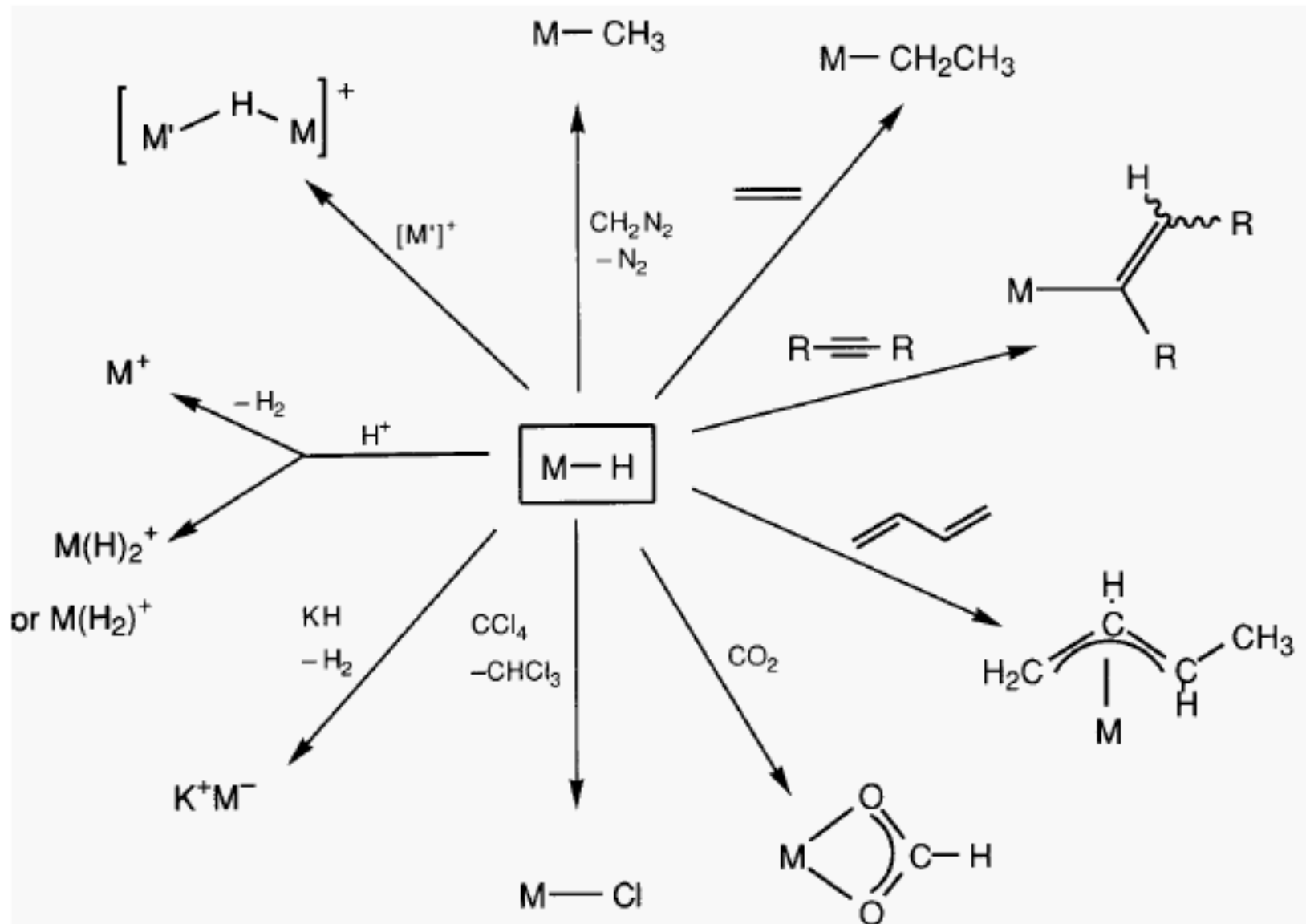
Examples of insertion reaction



Zr-CH₂OH not observed.



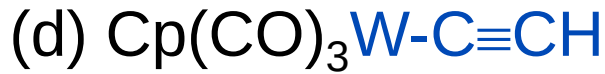
Summary of reactivity of M-H





HOMework (I)

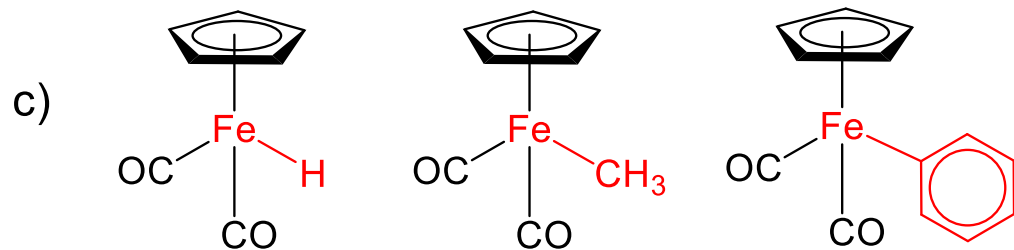
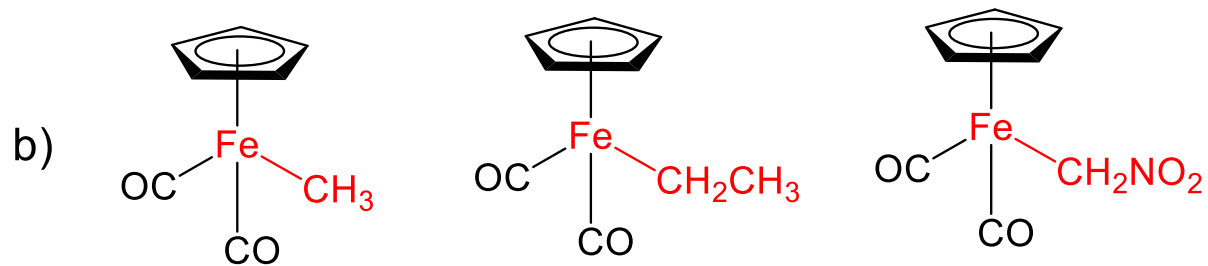
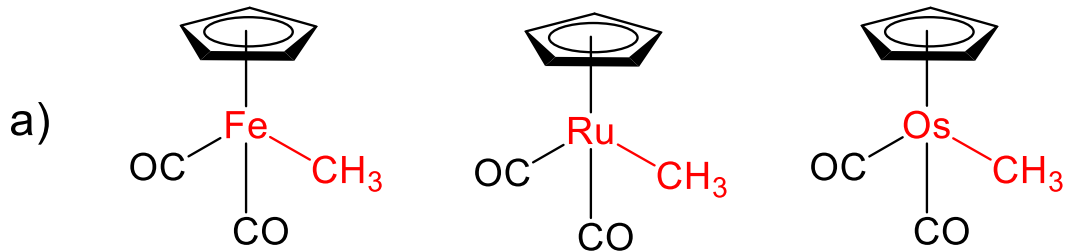
1. Which of the following complexes would you expect to have the strongest metal-alkyl bond?





HOMework (I)

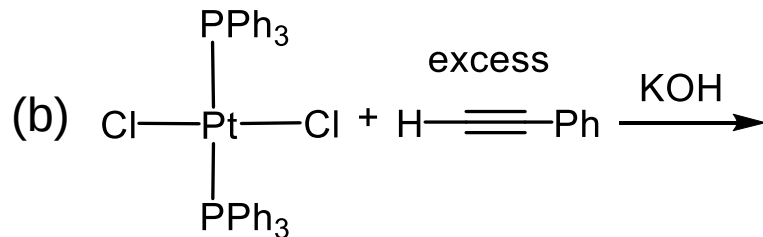
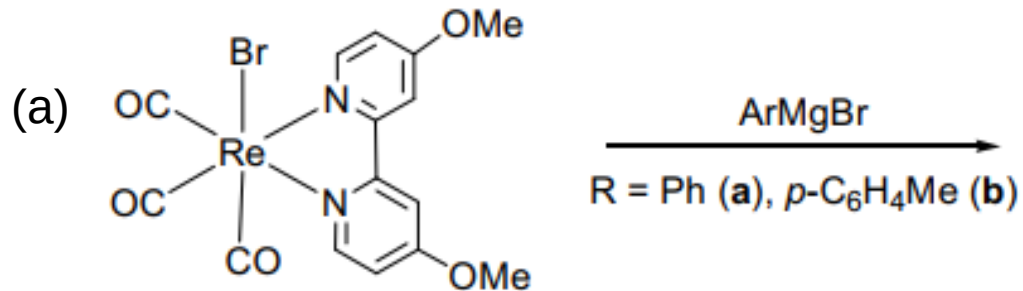
2. Select complex with the strongest **M-X** bond





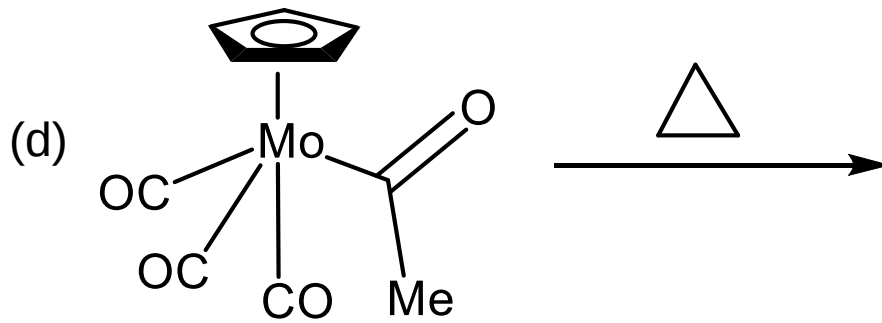
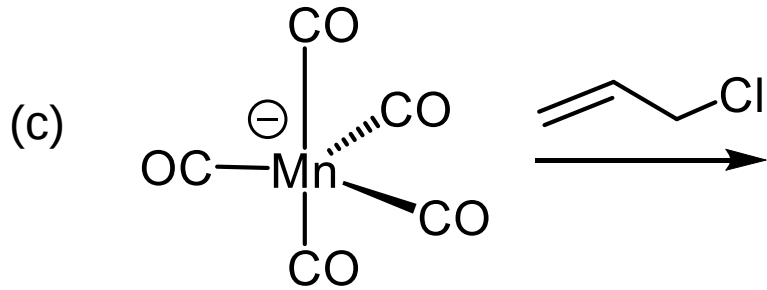
HOMework (I)

3. Predict the metal-containing products of the following reactions





HOMework (I)

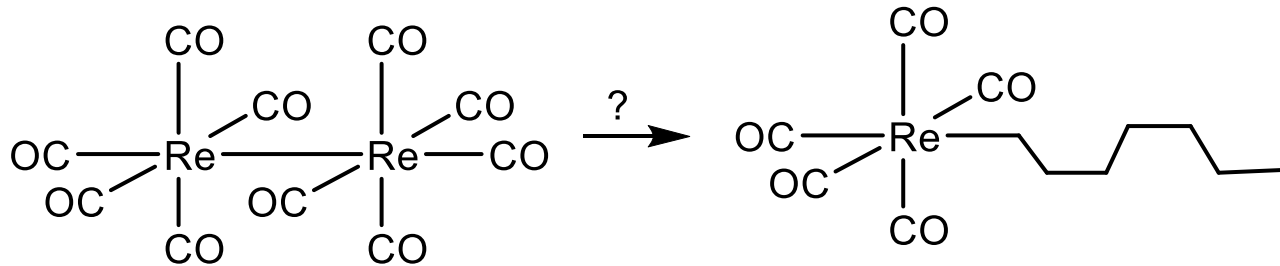




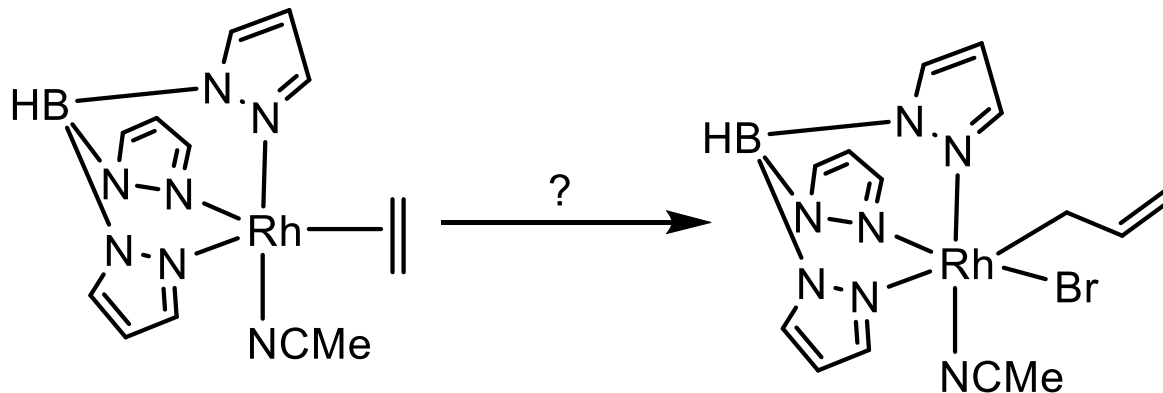
HOMework (I)

4. Provide appropriate reagents for the following transformations

(a)



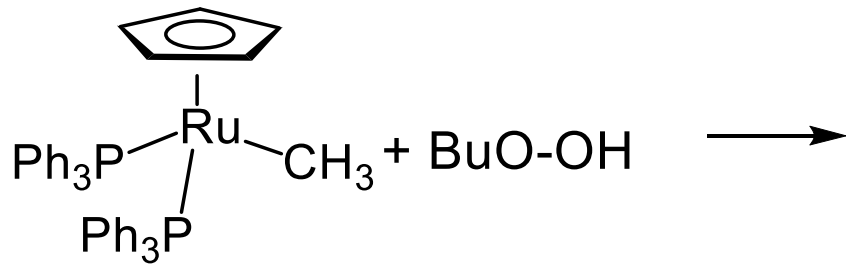
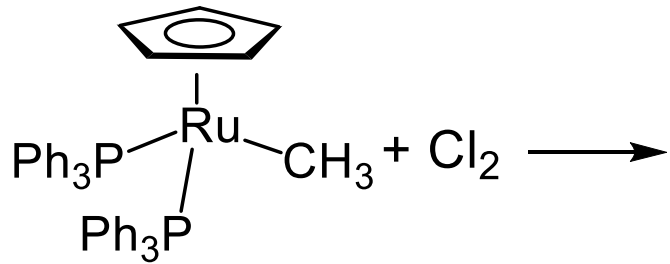
(b)





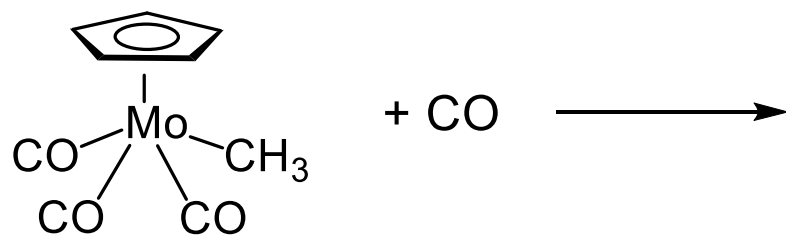
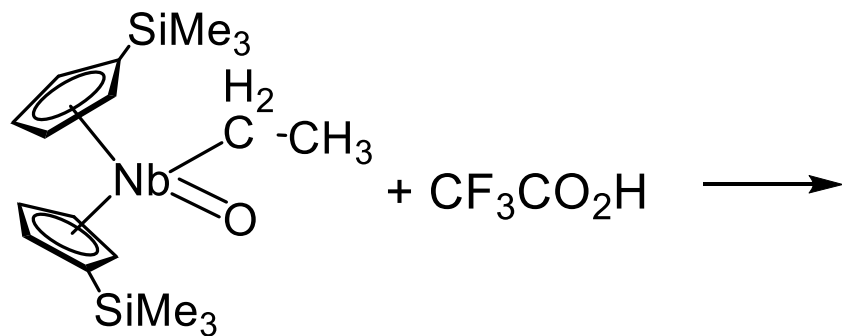
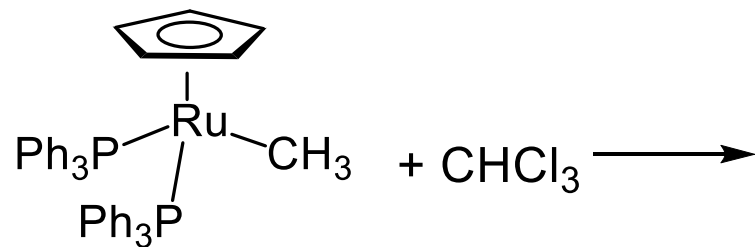
HOMework (II)

1. Predict the metal-containing products of the following reactions





HOMework (II)





Review Question

Predict the metal-containing products of the following reactions

